

Guest Editorial: A Practical Approach to Understanding—and Applying!—the Scholarship of Application

I. INTRODUCTION

AS MANY as 80% of papers published by the IEEE TRANSACTIONS ON EDUCATION fall within an area known as the scholarship of application [1].¹ Recently, the TRANSACTIONS established a new set of criteria to evaluate manuscripts submitted in this area. This guest editorial is intended to demystify these new criteria and to help writers avoid some common mistakes. The guidelines given here suggest systematic ways for researchers to design their studies and present their findings that will increase the likelihood of their papers being accepted by the IEEE TRANSACTIONS ON EDUCATION.

The TRANSACTIONS' "Description and Review Criteria" describe the scholarship of application as follows:

Scholarship of Application: "Contributions within this area of scholarship will often describe how prior research on learning and teaching (either general research, or research in a specific knowledge domain such as engineering) has been applied to create or design educational activities in electrical engineering (EE), computer engineering (CpE), and other fields within the scope of interest of IEEE. These activities include, but are not limited to, courses, course segments, curricula, laboratory experiments, course projects, capstone courses, and outreach activities. Faculty members across the world design these activities for their students, but to be published in the IEEE TRANSACTIONS ON EDUCATION a paper describing this work must (a) demonstrate application of published educational research and (b) provide a cogently articulated rationale for key design decisions" [2].

One way to think about this is that, much as all other forms of scholarship, the scholarship of application often means making a claim, laying out a hypothesis, or setting out a goal to be evaluated. Frequently, the assertion that the application is effective at achieving its stated goals is implicit (although it is better to make this claim explicit). Specific examples of the scholarship of application and their associated claims, taken from recent IEEE

TRANSACTIONS OF EDUCATION manuscripts, might include the following:

- implementation of a new organic electronics laboratory in a graduate curriculum—the explicit claims of this manuscript are that the new laboratory enhances student learning and improves satisfaction [3];
- development of a hands-on lab project with the explicit claims that it improves interest in the course and encourages students to learn the topics required in designing real embedded systems [4];
- integration of hardware and software design instruction using a cryptography application—the explicit claims of this manuscript are that the course teaches both hardware and software design, along with skills in decision-making, presentation skills, teamwork, and design creativity, skills generally overlooked in engineering [5];
- comparison of the effectiveness of a flipped classroom and a traditional classroom in an upper-division engineering course—the explicit claims are that the new approach allows the instructor to cover more material than the traditional classroom and increases students' performance on in-class assessments [6].

In contrast, examples that might be classified as more representative of the scholarship of discovery or of the scholarship of integration include the following:

- study of the factors that influence dissemination in engineering education (scholarship of discovery) [7];
- examination of the effects of anonymity in group discussion on peer interaction and learning achievement (scholarship of discovery) [8];
- review and synthesis of various perspectives on plagiarism (scholarship of integration) [9];
- description of the 10-year evolution of bullfighting robotics to help teach engineering at the Universidad Politecnica de Madrid (scholarship of integration as well as scholarship of application) [10].

II. DIVING DEEPER INTO THE SCHOLARSHIP OF APPLICATION

The TRANSACTIONS' new review criteria for the scholarship of application comprise eight elements [2]:

- 1) the *relevance* of the described efforts;
- 2) the clarity and significance of *intended outcomes*;
- 3) the *context* of the work (within the prior research);
- 4) the *application design* of the project;
- 5) the presentation of the *findings*;
- 6) the contributions as described in the *conclusions*;
- 7) the overall *organization and clarity* of the manuscript;
- 8) the meaningfulness of the *illustrations*.

Digital Object Identifier 10.1109/TE.2014.2313036

¹As background information, Ernest Boyer developed a model for scholarship that was divided into four areas—the scholarship of discovery (obtaining new knowledge), the scholarship of integration (synthesis), the scholarship of application, and the scholarship of teaching. There are obviously a variety of ways to categorize the many types of scholarship, but many have found Boyer's model to be useful. In Boyer's discussion of the scholarship of application, he made clear that he was describing activities where theory and practice interact. Clearly, the areas of scholarship can overlap—development of a novel lab project is an application, but it also involves applying something novel, which is a discovery.

Most of the review criteria form a straightforward checklist that authors can use to ensure their manuscript is properly constructed. The difficulty for many researchers lies in the subtleties of the fourth criterion, *application design*, which the TRANSACTIONS defines as follows:

Application design: “In the scholarship of application, a key focus is on decisions that authors made as they applied existing knowledge and research. This criterion focuses on these decisions. What were key decisions that needed to be made in the process of designing approaches, strategies, tactics, activities, behaviors, and structures to achieve the outcomes? Have the authors identified and explained the key decisions that were needed for their approach? For each decision, what are alternatives drawn from the literature that were, or should have been, considered? To what extent have the authors considered possible alternatives, including alternatives described in prior research? For each decision, what were the criteria used to guide prioritization among the alternatives? To what extent have the criteria that the authors use in making their decisions been clearly and adequately explained? How did the authors prioritize their alternatives, based on their criteria? Did they describe how they arrived at their decisions? Were the descriptions clear and the reasoning solid?”

It is well known that science is subject to considerable biases. A study by Stanford University’s John Ioannidis, for instance, has provided compelling evidence that more than half of all published research findings are false [11]. Another major study found that 88% of 53 “landmark” cancer studies could not be replicated, even despite assistance from those who produced the original study [12]. It is as easy to fall prey to biases and problematic study designs in scholarship of application as it is in other types of research [13]. The application design criterion describes many factors to be considered in order to prevent misdirected and potentially false findings. It also lays out the many considerations underlying a study publishable by the IEEE TRANSACTIONS ON EDUCATION. To clarify these issues, two contrasting approaches to undertaking such a study will be discussed here.

III. SCHOLARSHIP OF APPLICATION EXAMPLE

This section illustrates two very different ways in which a faculty member could approach planning, implementing, and reporting an experiment to “flip” his or her classroom. Study #1, conducted by Dr. Humdrum, is unlikely to be published in the IEEE TRANSACTIONS ON EDUCATION. Study #2, on the other hand, conducted by Dr. Dig-Deeper, has a high potential for publication. For purposes of simplicity, we represent both Dr. Humdrum and Dr. Dig-Deeper as male professors.²

Note that these two fictitious situations are exaggerated to provide insight into good (and bad) ways to approach the scholarship of application. While we all would hope to

make more informed decisions than those of Dr. Humdrum, we cannot all expect to achieve the superhero status of Dr. Dig-Deeper. He applies so many good practices that he may appear intimidating, but there have been many strong application papers published by the IEEE TRANSACTIONS ON EDUCATION that have been more modest in their approach. The key idea is to address all eight elements of the review criteria, designing the study such that the methods for evaluating the claims align with stated goals, presenting a clear rationale for the design decisions while acknowledging the inherent limitations.

A. Study #1

Dr. Humdrum reads an article in *The Chronicle of Higher Education* about the flipped classroom, and he decides to implement the approach in his own class. He is teaching Electric Circuits again this year, as he did last year; he designs an experiment to compare the flipped class with his conventional lecture approach. Last year, he videotaped all of his class sessions for Electric Circuits, so this year he posts his videos online for students to review before class. Then, he uses the class period for students to work homework problems. He writes a final exam for the class that is equivalent to last year’s. He finds no difference in the average exam score for last year’s traditional course and this year’s flipped classroom, and his students complain that the video lectures are dry and dull. He concludes that the flipped approach is a poor substitute for traditional approaches to teaching and writes up his observations and conclusions for the TRANSACTIONS. To his surprise, the manuscript is rejected.

B. Study #2

Dr. Dig-Deeper has heard the buzz about the flipped classroom. He reads an article extolling the benefits of the flipped classroom, and he wants to see if he can replicate the results in his class. He does a thorough review of the research literature—he even reads some recent books and online articles and forums on the phenomenon of flipped classes—and he begins to formulate his project. (Although he did not realize this until much later, his preliminary exploration of the literature applies to the scholarship of application’s review criteria for *relevance* and *context*.)

For the *application design* element of the project, Dr. Dig-Deeper enlists a team of two fellow faculty with complementary backgrounds. His team includes an educational psychologist with a good grasp of statistics and a new mechanical engineering professor who is eager to apply his recent pedagogical training from his teaching assistant work. Early on, Dr. Dig-Deeper sets up a meeting with a member of his Institutional Review Board (IRB) to ensure he has understood the appropriate considerations for collecting and compiling data. Then, he starts to plan a flipped class experiment, actually drafting tentative sections of the manuscript the team intends to submit. He realizes that the process of writing will force him to carefully evaluate and lay out the choices he and his team are making as they conduct the study. (The team addresses the review criteria of *organization and clarity* of the manuscript, presentation of the *findings*, and even the contributions as described in the *conclusions* as they wrestle with getting their preliminary ideas down on paper. Even the element of meaningfulness of the *illustrations* is brought into play at this early stage as the team realizes that certain key factors could ultimately result in a

²We carefully considered the gender options for our two professors, especially given the male-dominated nature of the engineering faculty profession. Because Dr. Humdrum is not a very good model, we did not want to be accused of practicing gender discrimination by making him female. On the other hand, we thought it would be a clear case of reverse discrimination to make the superhuman Dr. Dig-Deeper a female. Making both characters women would have created an unrealistic scenario that would be too unrepresentative of typical engineering professors to be believable. Every alternative had drawbacks, and we finally decided to settle for the least of all evils by making both professors male.

valuable graphical display of the study's contributions.) Writing the literature review proves especially valuable as, during the process of finalizing that section, Dr. Dig-Deeper finds some papers he had missed earlier that give him a better idea of how others have flipped their classrooms. He even contacts some of the authors of the papers to ask a few questions about their findings. As Dr. Dig-Deeper digs deeper, he begins to realize that this study has the potential for significant impact, and that it is going to take several years to complete the work. He begins to look for funding to support his team's work in this area.

Dr. Dig-Deeper's developing manuscript focuses on *application design* decisions in two areas: design decisions made about the pedagogy of the flipped classroom, and design decisions made about the evaluation of the impact of the flipped classroom.

1) *Classroom Design (How Dr. Dig-Deeper Enacts the Flipped Classroom)*.: Dr. Dig-Deeper discovers that the "flipped classroom" is a buzzword used haphazardly to define a myriad of pedagogical innovations. He begins by working with his team to clearly define their approach to the pedagogy (this relates not only to the *application design* element of the project, but also to the element of clarity and significance of *intended outcomes* and the *context* of the work). They opt to use a definition that characterizes a flipped classroom as one in which three key things happen [14]: 1) the first exposure to course content occurs prior to each class; 2) students complete a pre-class assessment to ensure preparation; and 3) class time is used for processing, practice, group learning, and other active learning techniques.

As a way of providing first exposure to course content, Dr. Dig-Deeper decides to use videos and pre-class readings. He has read the research on attention span, and he knows that shorter videos will be far more conducive to learning than standard 2-hour lectures. Therefore, he creates a series of short informational videos to describe some of the more complicated concepts in his class. He then requires his students to watch select videos before each class; he also assigns relevant portions of the course textbook.

Dr. Dig-Deeper hears from some students that they are frustrated and uncomfortable at having to assume responsibility for their own learning by completing the pre-class work. Therefore, he provides support that helps students improve their self-learning ability by detailing clear expectations for what they should know and by providing practical resources on learning more effectively (e.g., [15]) accessible through the course Web site. He also addresses effective self-learning in class.

Consistent with the definition of his pedagogy, Dr. Dig-Deeper designs and administers a series of pre-class quizzes using his university's learning management system. These quizzes contribute a small portion to the students' grade and help him ensure that students prepare for class. In addition, by reviewing student responses before class, Dr. Dig-Deeper is able to ascertain students' level of understanding about key ideas and adjust his class plan each day.

Dr. Dig-Deeper begins each class period by briefly reviewing the topics that, based on the pre-class quizzes, are giving the students the most difficulty. Then, he has students engage in active learning. On a typical day, he has students work individually on an initial problem using the concepts they studied prior to the class. Following that, he has the individual students turn to

a neighbor to compare their answers and refine their solutions. Finally, he has students work together in groups of four to solve a second, more complex problem. Throughout the process, he roams around class offering suggestions and feedback because he knows that personalized guidance, interaction, and encouragement are important ways to foster student success.

2) *Design of the Evaluation (How Dr. Dig-Deeper Evaluates the Flipped Classroom)*.: Dr. Dig-Deeper works with his team to outline three main claims about the flipped classroom study. First, he has read that the flipped class can increase student learning, so he plans the bulk of his study around evaluating learning gains students make in his flipped class and comparing them to those in his traditional course. (Note that this relates to the element of clarity and significance of *intended outcomes*.) His first/primary claim is that: *As a result of engaging in the flipped classroom, students will have greater conceptual understanding of the course objectives*. Dr. Dig-Deeper plans to invest considerable effort in making students responsible for learning some material on their own, and he hopes this will result in improvements in their confidence. Accordingly, his second claim is that: *Students who participate in the flipped class will have increased confidence in their ability to succeed*. Finally, because this is a new endeavor for him, he decides to compare student perspectives about both versions of the class to explore other ways in which the flipped classroom is better (or worse). His third claim is that: *Students will have positive perspectives about their learning in the flipped class*. Dr. Dig-Deeper is wary of his own claims, however, because he knows they reveal his implicit bias, so he continues to work with his team to refine his claims and design a fair evaluation of them.

To study the impact of his flipped classroom experiment, Dr. Dig-Deeper decides to teach the course in the traditional way one year and use the flipped method in the next year. (He realizes that an alternative approach might have been to enlist another professor to teach one of the courses during the same timeframe. However, he considers the tradeoffs and concludes that, in his opinion, having the same professor provides a better control experience.) He chooses to use the same textbook, same weekly homework assignments, and same testing methods for each of the two years—he carefully describes all these considerations in the paper. Sometimes, to his chagrin, he writes up a few paragraphs for the draft manuscript describing his plans, but then the plans change after discussion with his team, and he has to delete and rewrite. Ultimately, he redrafts much in the manuscript as the study develops, but he is grateful that the manuscript provides a continuing means for him and his team to focus their thoughts.

To evaluate the primary claim for his course, *greater conceptual understanding*, Dr. Dig-Deeper decides to compare student performance on two separate instruments: 1) homework problems, and 2) a validated concept inventory. First, for the homework problems, he works to carefully ensure that the assignments used in the two different versions of his course are matched by soliciting two independent reviews. He asks both the teaching assistant for the course and one of his colleagues to help revise the problems. He then pilots the two versions of the problems with some undergraduate researchers in his lab.

Dr. Dig-Deeper also considers relative time in the semester when each problem is assigned—he tries to maintain the same schedule in order to eliminate inadvertent bias. (The team’s psychologist was very helpful in pointing out potential sources of bias.) Second, Dr. Dig-Deeper finds the Electric Circuits Concept Inventory [16], a validated, multiple-choice instrument that measures students’ understanding of DC circuits, and notes that the concepts addressed in the inventory are well matched with the ones he teaches in his Electric Circuits course. Dr. Dig-Deeper decides to administer the instrument on the first day of class and then again as part of the final examination for both versions of his classes.

To evaluate his second claim, *increased student confidence*, Dr. Dig-Deeper and his team review the literature on confidence and self-efficacy and identify several potential self-report survey instruments. After considering the pros and cons of three of these instruments and contacting the developers of each, they decide to use the Longitudinal Assessment of Engineering Self-Efficacy (LAESE), a validated instrument that has been used previously in engineering classes [17]. Dr. Dig-Deeper administers the LAESE in both versions of his class during the first week of class and again at the end.

Finally, to evaluate his third claim, *positive student perspectives*, Dr. Dig-Deeper decides to enlist the assistance of his campus’s center for teaching and learning to collect student feedback about both classes. He has worked with the center’s professional staff before, when they had administered a Small Group Instructional Diagnosis [18] in the middle of the term for his Signals and Systems class; the anonymous student feedback he received allowed him to make small changes to this teaching that resulted in large improvements. An objective consultant visits his flipped class in the middle of the term and then meets with Dr. Dig-Deeper to share student perspectives of the class. This candid student feedback provides Dr. Dig-Deeper with some good insight.

In writing about his efforts, Dr. Dig-Deeper begins by showing that students enrolled in both courses had similar background characteristics and ability prior to taking the courses. He compares the GPAs of entering students for both courses, as well as self-efficacy scores and concept inventory data. He finds and reports minor differences. Then, he and his team conduct careful statistical analyses to show that the flipped group performed statistically better on many problem sets as well as on the concept inventory items. They also present a comparison of student perspectives in both courses, noting both the strengths and weaknesses of each approach.

Dr. Dig-Deeper takes great care to ensure that the methods used to evaluate the goals of the study *align* with what he is evaluating. He is aware that his claim that his students learn more deeply cannot be substantiated by, for example, survey data on the students’ satisfaction with the course. Underlying the design of both his new initiative (the flipped classroom) and his evaluation of it, there is a careful, thorough discussion of the design decisions, reasons, rationales, evidence, justification, and tradeoffs so that readers can clearly understand why the approaches were chosen. In this way, all eight elements of the review criteria for the scholarship of application come into play.

Despite his busy schedule, Dr. Dig-Deeper has taken the time to watch a massive open online course (MOOC) on how to improve his writing for research [19]. Then, he and his team finalize the draft manuscript and send it to a few colleagues from other disciplines for a “pre-review.” When they receive the feedback, they find they can do a final review with fresh eyes. They make several revisions, rewriting one section of their paper to invite less controversy and also responding to the edits of a writer-friendly who is not an academic [20], [21]. They are confident that the final draft they submit to the IEEE TRANSACTIONS IN EDUCATION is their best work, and they wait anxiously for feedback from the editor.

IV. PUTTING IT ALL TOGETHER—POINTERS FOR SUCCESSFUL PUBLICATION

Dr. Dig-Deeper and his team learned a lot in completing their project and writing it up for the IEEE TRANSACTIONS ON EDUCATION. They read the TRANSACTIONS’ review criteria and framed their project using the guidelines. They reflected on the scholarship of application criteria that emphasized the inclusion of both: 1) the application of published research, and 2) a well-articulated rationale for key design decisions. Here is a quick synopsis of ideas used by Dr. Dig-Deeper and his team that may help you toward successful publication in the IEEE TRANSACTIONS ON EDUCATION.

- *Pick an area that is of broad interest.* If you cannot easily imagine colleagues at other institutions being interested in your findings, or if the context of your research is too narrow to allow generalizing the findings to other situations, consider looking for something different to study.
- *Thoroughly review the literature and link your work to appropriate prior research.* You cannot design a fair or appropriate study if you do not understand the current “state of the art,” so take some time to learn about other published research and use that to shape your research plans.
- *Consider forming a multidisciplinary research team.* Partnering with faculty in education, psychology, business, or statistics, for example, or working with professionals from your campus center for teaching and learning can help you explore the research in a more comprehensive way that provides a better chance for successful publication.
- *Identify significant and relevant claims.* The claims that underscore your project should be stated explicitly and framed as testable hypotheses. Central to the idea of any hypothesis is that the outcome is not predetermined.
- *Beware of your potential biases.* Be objective as you plan your work—avoid designing your project in a way that simply confirms your gut feeling. Similarly, be aware that the disciplinary biases inherent in traditional engineering research may require you to design your project and write about your findings in ways that appeal to a variety of audiences.
- *Collect and analyze the data using proven methodology that is aligned with your research questions, goals, or claims.* Keep in mind that a well-designed study might include quantitative data as well as qualitative data—both can yield strong, rich findings. Be aware of over-relying

on student opinion data—students can be predisposed to telling you what they think you want to hear. However, do not dismiss insights you might be able to get from a well-designed self-report survey or from data collected by objective colleagues.

- *Report findings that are clearly supported by your data.* Be sure that your evaluation data matches with your claims and with the study design and that it provides sufficient evidence (e.g., an ample number of students, cohorts, etc.) from which to draw worthwhile conclusions.
- *Clearly articulate the practical implications of your findings.* Helping the reader understand how your findings might extend to other settings can be an important way to increase the impact of your work.
- *Describe the limitations inherent in your work.* Manuscripts that describe the scholarship of application inevitably have limitations. Acknowledge them and discuss their potential impact on results and conclusions.
- *Consider the ethical implications of working with human subjects.* Different institutions in different countries can have very different requirements in relation to institutional review of your research. It is a good idea to establish a relationship with your IRB officials and to interact with them regularly.
- *Get feedback on your draft.* Good writing allows your work to be broadly cited and appreciated. “Pre-reviews” by colleagues from other disciplines, and even from outside academia, can dramatically improve the clarity and quality of your writing.

V. CONCLUSION

It is very likely that the paper you are planning for the IEEE TRANSACTIONS ON EDUCATION falls into the category of the scholarship of application. If that is the case, do your homework. You want to be able to support and describe the many carefully researched decisions you made in laying out a claim and attempting to prove it. Remember that, in some sense, you are working as a detective who has made an important claim and is now trying to prove that claim using all the wit and intelligence at your disposal. This editorial is meant to guide you in your work.

ACKNOWLEDGMENT

The authors would like to thank Rebecca Brent, Jeff Froyd, the Associate Editors, and Kirsty Mills, IEEE TRANSACTIONS ON EDUCATION Editorial Administrator. These individuals reviewed drafts of this editorial and offered insightful comments, suggestions, and criticisms.

BARBARA OAKLEY
Oakland University
Rochester, MI 48309 USA
(e-mail: Oakley@oakland.edu)

CYNTHIA J. FINELLI
University of Michigan
Ann Arbor, MI 48109 USA
(e-mail: cfinelli@umich.edu)

REFERENCES

- [1] E. L. Boyer, *Scholarship Reconsidered: Priorities of the Professoriate*. Princeton, NJ, USA: Carnegie Foundation/Jossey-Bass, 1990.
- [2] IEEE Transactions on Engineering, “The scholarship of application: Description and review criteria,” 2013 [Online]. Available: <http://sites.ieee.org/review-criteria-toe/application/>
- [3] N. Bednar and G. Stojanovic, “An organic electronics laboratory course for graduate students in electrical engineering,” *IEEE Trans. Educ.*, vol. 56, no. 3, pp. 280–286, Aug. 2013.
- [4] L. Sousa, S. Antao, and J. Germano, “A lab project on the design and implementation of programmable and configurable embedded systems,” *IEEE Trans. Educ.*, vol. 56, no. 3, pp. 322–328, Aug. 2013.
- [5] L. Uhsadel, M. Ullrich, A. Das, D. Karakljajic, J. Balasch, I. Verbauwhede, and W. Dehaene, “Teaching HW/SW co-design with a public key cryptography application,” *IEEE Trans. Educ.*, vol. 56, no. 4, pp. 478–483, Nov. 2012.
- [6] G. S. Mason, T. R. Shuman, and K. E. Cook, “Comparing the effectiveness of an inverted classroom to a traditional classroom in an upper-division engineering course,” *IEEE Trans. Educ.*, vol. 56, no. 4, pp. 430–435, Nov. 2013.
- [7] B. T. Hazen, W. Yun, and C. S. Sankar, “Factors that influence dissemination in engineering education,” *IEEE Trans. Educ.*, vol. 55, no. 3, pp. 384–393, Aug. 2012.
- [8] B.-S. Jong, C.-H. Lai, Y.-T. Hsia, and T.-W. Lin, “Effects of anonymity in group discussion on peer interaction and learning achievement,” *IEEE Trans. Educ.*, vol. 56, no. 3, pp. 292–299, Aug. 2012.
- [9] D. Chuda, P. Navrat, B. Kovacova, and P. Humay, “The issue of (software) plagiarism: A student view,” *IEEE Trans. Educ.*, vol. 55, pp. 22–28, 2012.
- [10] M. Hernando, R. Galán, I. Navarro, and D. Rodriguez-Losada, “Ten years of cybertech: The educational benefits of bullfighting robotics,” *IEEE Trans. Educ.*, vol. 54, pp. 569–575, 2011.
- [11] J. P. A. Ioannidis, “Why most published research findings are false,” *PLoS Med.*, vol. 2, p. e124, 2005.
- [12] C. G. Begley and L. M. Ellis, “Drug development: Raise standards for preclinical cancer research,” *Nature*, vol. 483, pp. 531–533, 2012.
- [13] R. M. Felder and R. G. Hadgraft, “Educational practice and educational research in engineering: Partners, antagonists, or ships passing in the night?,” *J. Eng. Educ.*, vol. 102, pp. 339–345, 2013.
- [14] Center for Research on Teaching and Learning, University of Michigan, Ann Arbor, MI, USA, “Flipping your class,” 2014 [Online]. Available: <http://www.crlt.umich.edu/node/58368>
- [15] B. Oakley, *A Mind for Numbers: How to Excel at Math and Science*. New York, NY, USA: Penguin, Jul. 2014.
- [16] T. Ogunfunmi and M. Rahman, “Concept inventory assessment instruments for circuits courses,” in *Proc. Amer. Soc. Eng. Educ.*, Vancouver, BC, Canada, 2011.
- [17] R. M. Marra, K. A. Rodgers, D. Shen, and B. Bogue, “Women engineering students and self-efficacy: A multi-year, multi-institution study of women engineering student self-efficacy,” *J. Eng. Educ.*, vol. 98, pp. 27–38, 2009.
- [18] C. J. Finelli, T. Pinder-Grover, and M. C. Wright, “Consultations on teaching: Using student feedback for instructional improvement,” in *Advancing the Culture of Teaching on Campus: How a Teaching Center Can Make a Difference*, C. E. Cook and M. L. Kaplan, Eds. Sterling, VA, USA: Stylus, 2011, pp. 65–79.
- [19] K. Sainani, “Writing in the Sciences,” Stanford University, Stanford, CA, USA [Online]. Available: <https://www.coursera.org/course/sciwrite>, massive open online course (MOOC), accessed through Coursera.
- [20] T. Becher and P. Trowler, *Academic Tribes and Territories*, 2nd ed. Philadelphia, PA, USA: Open Univ. Press, 2001.
- [21] R. Toor, “Bad writing and bad thinking,” *Chronicle Higher Educ.* Apr. 15, 2010 [Online]. Available: <http://chronicle.com/article/Bad-WritingBad-Thinking/65031/>



Barbara Oakley (M'98–SM'02) received the B.A. degree in Slavic languages and literature and the B.S. degree in electrical engineering from the University of Washington, Seattle, WA, USA, in 1973 and 1986, respectively, and the Ph.D. degree in systems engineering from Oakland University, Rochester, MI, USA, in 1998.

She is currently an Associate Professor of engineering with the Department of Industrial and Systems, Oakland University, and is a licensed Professional Engineer in the State of Michigan. Prior to her academic career, she rose from Private to Captain in the US Army, worked as a Russian translator on Soviet trawlers in the Bering Sea, and served as a Radio Operator with the South Pole Station in Antarctica. This varied background provides unconventional insight to her work. Her most recent book is *A Mind for Numbers: How to Excel at Math and Science (Even If You Flunked Algebra)* (Tarcher-Penguin, 2014). She is interested in translating research findings from neuroscience and cognitive psychology to instructors and students so that these findings can have a practical impact on people's ability to learn effectively. Her foundational research, published in

the *Proceedings of the National Academy of Sciences*, involves pathologies of altruism and altruism bias.



Cynthia J. Finelli earned the B.S.E., M.S.E., and Ph.D. degrees in electrical engineering from the University of Michigan, Ann Arbor, MI, USA, in 1988, 1989, and 1993, respectively.

She currently serves as Director of the Center for Research on Learning and Teaching Engineering and Research Associate Professor with the College of Engineering, University of Michigan. In these roles, she leads faculty development initiatives in engineering, actively pursues research in engineering education, and assists other faculty at the university in their scholarly endeavors. She leads a national initiative to create a taxonomy/keyword outline for the field of engineering education research. Her current research interests include studying faculty motivation to change classroom practices, evaluating methods to improve teaching, and exploring ethical decision-making in undergraduate engineering students.

Dr. Finelli is an American Society of Engineering Education Fellow. She is Past Chair of the Educational Research and Methods Division of the American Society of Engineering Education.

Knowledge Graphs as Context Sources for LLM-Based Explanations of Learning Recommendations^{*}

Hasan A. Rasheed^{1,*}, Christian Weber¹ and Madjid Fathi¹

¹University of Siegen, Siegen, Germany

Abstract

In the era of personalized education, the provision of comprehensible explanations for learning recommendations is of a great value to enhance the learner's understanding and engagement with the recommended learning content. Large language models (LLMs) and generative AI in general have recently opened new doors for generating human-like explanations, for and along learning recommendations. However, their precision is still far away from acceptable in a sensitive field like education. To harness the abilities of LLMs, while still ensuring a high level of precision towards the intent of the learners, this paper proposes an approach to utilize knowledge graphs (KG) as a source of factual context, for LLM prompts, reducing the risk of model hallucinations, and safeguarding against wrong or imprecise information, while maintaining an application-intended learning context. We utilize the semantic relations in the knowledge graph to offer curated knowledge about learning recommendations. With domain-experts in the loop, we design the explanation as a textual template, which is filled and completed by the LLM. Domain experts were integrated in the prompt engineering phase as part of a study, to ensure that explanations include information that is relevant to the learner. We evaluate our approach quantitatively using Rouge-N and Rouge-L measures, as well as qualitatively with experts and learners. Our results show an enhanced recall and precision of the generated explanations compared to those generated solely by the GPT model, with a greatly reduced risk of generating imprecise information in the final learning explanation.

Keywords

Large language models (LLMs), Knowledge graphs, ChatGPT, Generative AI (GenAI), Learning recommendations, explainable AI (XAI)

1. Introduction

In personalized education, the provision of precise and comprehensible explanations for learning recommendations is both important and challenging. Explainability of learning recommendations has been found to enhance the student acceptance of the recommended content [1]. However, the precision of the textual and visual explanations faces several limitations when these are generated semi- or full-automatically. This is due to the level of knowledge, understanding, and reflection, that the learner needs from the generated explanation, which an automatic system may not be capable to offer. The recent research on LLMs opened new possibilities for solutions to generate explanations for learning recommendations on a deeper linguistic and semantic level. The massive datasets used to train such LLMs, and the high number of features considered by the model, enables generating more informative texts for the explainability task. However, they face serious concerns about their ability to generate a sufficiently precise text to reflect the learning intent from the recommendation, especially in sensitive domains. On the one hand, the technology readiness level (TRL) of the majority of LLMs still does not exceed the level TRL-2 [2]. From an ethical perspective, the transparency of LLMs is rarely more than Tier-1 [2], based on the classification of transparency by Chaudhry et al. [3]. This apprehension aligns with the observations of Fullan et al. [4], who also expressed concerns about various adverse effects of ChatGPT in education. They highlighted issues such as the lack of originality in its responses, potentially leading to answers that lack meaning and fail to stimulate

exploration or imagination due to their linearity or flatness [5]. Despite these drawbacks, Hargreaves [6] underscored that utilizing such technology in education exhibits promise, advocating against its outright rejection.

Our research addresses this challenge by combining the strengths of KGs and Generative Pre-Trained Transformers (GPT) models, aiming to enhance the precision, and thus reliability, of the explanations they generate for learning recommendations. We utilize the features of KGs to enhance the querying process of LLMs, through extracting contextual knowledge from the KG and then performing an informed prompt engineering process that is supported by pedagogy experts' inputs. The resulting LLM prompt is then used to guide the model away from generating wrong or irrelevant information and limit the scope of answers it provides to the curated information provided from the knowledge graph. This ensures more precise answers, while still utilizing the model's ability to combine pieces of information and phrase a human-like explanation, which learners can easily comprehend

2. Background

Knowledge Graphs are structured representations of knowledge, composed of entities and the relationships between them [7]. They serve as powerful tools for organizing information, capturing semantic connections, and providing a foundation for contextual understanding. A KG is defined as in (1). In educational contexts, KGs can be constructed from curated educational materials, forming a reliable source of factual knowledge [8].

$$KG = (h, r, t) | h, t \in E, r \in R, \quad (1)$$

Where h is the head entity, t is the tale entity, and r is the relation between h and t . E is the entity group and R is the relation group.

The use of KGs to support LLMs is gaining an increased focus recently [9, 10, 11], which can be traced back to: 1) the new breakthroughs of LLMs, 2) the growing concerns

Accepted in the IEEE Global Engineering Education Conference 2024 (EDUCON24), Kos, Greece

^{*}You can use this document as the template for preparing your publication. We recommend using the latest version of the ceurart style.

^{*}Corresponding author.

✉ hasan.abu.rasheed@uni-siegen.de (H. A. Rasheed);

christian.weber@uni-siegen.de (C. Weber)

✉ 0000-0002-2921-4809 (H. A. Rasheed); 0000-0001-6606-5577

(C. Weber); 0000-0002-7602-9593 (M. Fathi)

© 2024 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

about their risks and limitations [2], and 3) the effective role KGs can play in reducing LLM hallucinations and enhancing their accuracy [12]. Sequeda et al. show that KGs-based querying enhanced the results of a one-shot question answering task in comparison to SQL-querying, when a KG representation of the SQL database was used [12]. KGs have also been well utilized for explainability tasks in the literature. This is due to their potential to provide causal reasoning [13], factual knowledge [11], and interpretable semantic relations [10, 14, 15]. To that end, KGs not only enhance the reasoning of the LLM with facts, but also offer more information, which influences the precision of the automatically generated text. This information is parsed to the LLM in form of a prompt context that is designed during the prompt engineering phase. Depending on the LLM used, contextual parts of the prompt act as guidelines that steer the model to generate more relevant outputs. We build on the concepts in [14] and [9] to utilize KG’s structure and semantic relations for enriching the contextual part of OpenAI’s GPT-4 model prompt [16], thus enhancing the precision of its explanations of a learning recommendation.

3. Methodology

To utilize the GPT-4 potentials while reducing the amount of irrelevant and imprecise text it generates for the learning recommendation, we propose a strategy for constructing the data structures and the LLM prompt to maximize the amount of factual and contextual information that the model receives before generating an explanation for the recommendation, see Fig. 1. Contextual information that KG provides is acquired from the structural relations and the metadata of the learning content connected to the recommended elements. Both types of information are then translated into textual strings that are inserted as prompt context in the LLM query.

3.1. Knowledge Graph Structure

We create the KG from educational materials, aligning with a pre-defined taxonomy, to ensure comprehensive coverage of relevant topics. Our taxonomy includes four levels: 1) Learning goals, 2) Courses, 3) Topics, and 4) Open educational resources (OER). For simplicity, we use the term learning object (LO) as defined in [17] hereafter to refer to any of the last three taxonomy levels, unless they are explicitly named. Learning goals and LOs are represented by graph nodes, who’s properties include their titles and descriptions among other metadata. We focus on textual properties of the LOs to search for semantic relations amongst them. Semantic relations are extracted because the educational content in our database is created by different experts over time. A customized text mining pipeline is utilized to extract the main topics covered each LO and compare them to topics covered by other LOs in the graph [8]. A relation is created between the two LOs if the semantic similarity is above a predefined threshold. Relations extracted between LOs are used to enhance the coverage and composition of context, which is provided to the LLM for generating an explanation of the connected materials in the learning recommendation.

The recommendation algorithm we use is a graph exploration and path weighing algorithm [18], based on Markov decision process (MDP) to identify an optimal learning path for each learner. Recommended learning path is explained

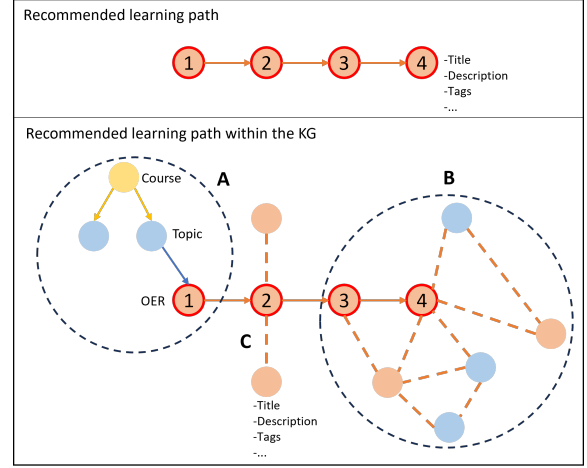


Figure 1: Structural information added to the LLM context from the KG. Top: learning path as an output of the recommendation system. Bottom: recommended path as it appears in the KG. Area (A): hierarchical structure of the learning goal. Area (B) KG community around LO3 and LO4. Connection (C): semantic relation extracted by the relation extraction algorithm.

in terms of its content selection (why each node is recommended), as well as its relation to the learning goal (how a node supports achieving the desired learning goal). To explain the recommended path, we extract four main types of information from the KG, see Fig.1:

1. *the hierarchical structure of the LO based on the pre-defined taxonomy*: this structure informs the LLM about the placement of a recommended LO within its original curriculum, which is defined by the human content-curator of the corresponding learning goal, see Fig. 1 area (A).
2. *Semantic relations to similar LOs in the KG*: which is discovered by the relation extraction algorithm based on the textual content and the semantic similarity between the LOs, see Fig. 1 connection (C).
3. *KG communities around LOs*: which are areas on densely connected LOs in the KG. These communities reflect in our KG fields of application, in which several LOs are usually required together as a group, see Fig. 1 area (B).
4. *Supporting metadata from connected LOs in the KG*: where not only the relations to semantically similar LOs are utilized, but also the textual content of the related LOs’ metadata, which sheds an additional light on the context of the recommended LO.

Using this additional information from the KG, we construct the GPT-4 prompt on the task level, as well as on the context one.

3.2. GPT-4 Prompt and Explanation Construction

The extended explanations for recommended learning paths are generated using the “GPT4-1106-preview” model, which we will refer to here as GPT-4 for simplicity. This model is the most capable model offered by OpenAI at the time of conducting this research, with training data that is updated till April 2023. To use the model, we utilize OpenAI’s API, which allows parsing the prompt to the GPT-4 model with

additional data, information, and instructions, in order to guide the model output generation. We design our prompt to include a main body and a contextual part, see Fig. 2. Prompt’s body represents the direct query from the user. It is constructed as a set of tasks that the model has to perform. This is to enhance the focus of the query and ensure less deviation from the user intent.

The contextual part of the prompt is extracted directly from the KG, based on a search strategy that integrates the four types of information discussed in 3.1, into the prompt’s context. The context is designed to include: 1) The role that the model assumes when answering a user’s query, e.g., answering as a teacher. 2) Required definitions that are needed to understand the terminology, especially in the cases where the domain-specific meaning of a word differs from its general meaning. 3) The information from semantically connected LOs in the KG, which is provided to the model in the form of “supporting content” in the query’s context. The model’s role and terminology definitions are also influenced by a direct input from the domain experts, who can set the pedagogical and domain related guidelines for the model.

The response of the GPT-4 model is used to fill the information gaps in an explanation template. The design of explanations for the learning recommendations is tightly connected to the pedagogical value of the explanation, to ensure an improved learning outcome. For that reason, each template includes specific information spaces or slots, which are filled by the LLM, guided by the data of the KG. It is important to mention here that parts of the explanation template can also be filled with a direct input from the experts. In this paper, however, we limit the scope of information in the explanation to that generated automatically by the GPT-4 model, in order to evaluate its performance against a golden standard, which is extracted from the expert’s input.

3.3. Chatbot-based interaction with the explanations

Once the explanation is complete, it is provided to the user through an interactive interface element, which represents the chat features that LLMs offer. We refer to this approach as “conversational explainability”, which differs from regular textual and visual explanations by its ability to provide the explanation through a multi-step interaction with the user, taking into account previous chat messages to influence the following model output. The ability of GPT models to respond to individual questions, as well as to engage in a multi-step chat with the user, means that explanations can be extended with additional information based on user inputs and requests. This, however, would induce the risk of deviating from the main point of the explanations. Therefore, we limit the chatting feature of GPT-4 in this paper to a first response with confirmation. This approach stands for a one-shot query that is supported by a confirmation step from the model, to ensure that the chatbot understands the user request correctly. With this approach, a learner has the option to ask about the learning material itself, i.e., its content or why it is relevant for their learning goal, as well as the relationships between the learning materials are in the recommended path.

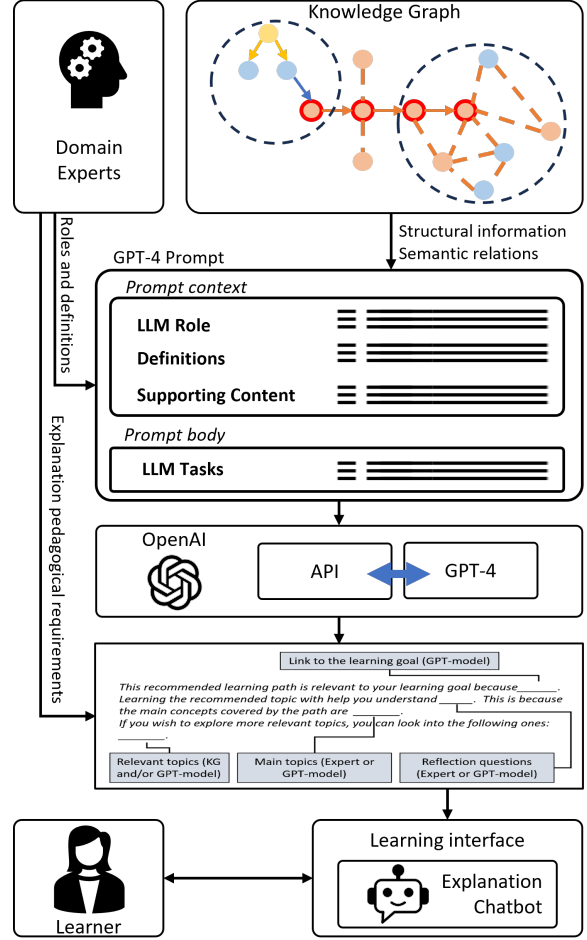


Figure 2: Proposed approach for constructing the GPT-4 prompt, with KG-based contextualization, as well as the Chatbot-based user interaction, and the expert roles in the design for context and explanation-templates.

4. Evaluation and Results

To evaluate our proposed approach, we devise a hybrid quantitative/qualitative evaluation strategy. The quantitative evaluation utilizes Rouge-based measures, namely Rouge-N, Rouge-L and Rouge-Lsum. Qualitative evaluation is conducted through a questionnaire-based user feedback on the explanation approach and its outcomes. It involved domain experts and learners, who were asked to evaluate the generated explanations and their relevance to the learning materials and learning goal.

Through Rouge metric, we aim to measure the amount of text in the generated explanation, which offers precise information, against the amount of irrelevant text, which is either wrong, or simply a filler text that is task-related but has low relevance to the goal of the explanation.

4.1. Experiment Set-up

Rouge (Recall-Oriented Understudy for Gisting Evaluation) [19] is an evaluation metric commonly used in natural language processing (NLP) to assess the quality of automated text summarization and machine translation systems. It employs both recall and precision to compare the overlap of n-grams (contiguous sequences of words) between model-

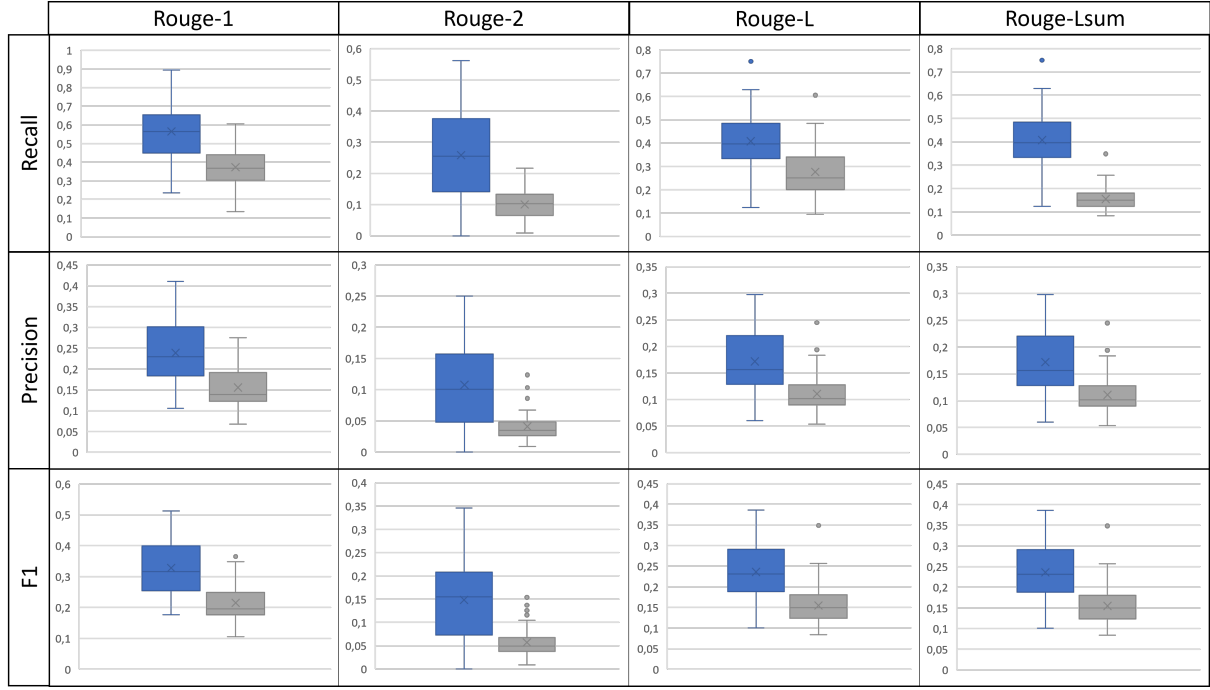


Figure 3: Recall, precision, and f1-measure values of the Rouge metric, for both explanation types: 1) with KG-based contextualization (blue), and 2) without contextualization (gray).

generated summaries and human-generated ones. We utilize this metric to measure the overlap between reference explanations from a human text and each of the explanations generated automatically by the GPT-4 model, i.e., with a KG-based contextualization, and without it. To ensure a fair comparison between the two explanation approaches, we fix the length of the text generated by the GPT-4 model. This is because Rouge measures take into consideration the total amount of n-grams in the model generated text.

The reference text is composed from human defined descriptions and reflection information in the LO’s metadata. In this paper, we do not account for the phrasing of the explanation, i.e., sentence structures and writing styles, since there is no certain phrasing or writing style that can be considered as “correct” or “referential”. Instead, we focus on the word choice and word patterns that appear in the generated text and match the patterns in the reference text.

4.2. Dataset

To calculate Rouge values, we construct a reference data set of reference explanations. 52 samples of human-generated texts were collected from clarifications of the recommended content of 10 different learning-path recommendations. For each reference explanation, we generate two candidate explanations, one through the proposed KG-based contextualization approach, and one without the context.

4.3. Results and Discussion

We calculate the recall, precision, and f1-measure from four Rouge measures: Rouge-1, which is the measure used to for the overlap between single words, i.e., unigrams, Rouge-2 that accounts for two-word patterns, i.e., bigrams, Rouge-L that takes into account the longest matching patterns between the reference text and the candidate one, and Rouge-

Lsum which is a variant of Rouge-L that calculates the overlap on the sentence level, not the complete text. Calculated values of the different measures, see Fig. 3, show a clear enhancement of the precision, recall, and thus f1-measure of the explanations generated with KG-based context. The results also show that recall scores are best with Rouge-Lsum measure, while the precision and f1-measure values are better with Rouge-1. With a domain expert support, we evaluate that none of the automatic explanation samples was wrong or misleading, which leads to the conclusion that the amount of less irrelevant text in the non-contextualized explanations was considerably higher than the one in the contextualized explanations, since both have comparable lengths.

Qualitatively, the explanations were provided in a user study to a group of eight learners and five domain experts. Participants included two PhD holders, four PhD candidates and seven graduate students. We surveyed the participants for their perception on the explanation quality, and the overall impressions, remarks, and limitations of the two explanation approaches. The survey evaluation revealed an enhanced acceptance of the explanations when it is generated with KG-based context. On a Likert scale, the quality of the contextualized explanation reached 4.7/5. The amount of irrelevant text, which was correct in general but not related to the user intent from the explanation, was also reported to be less in the contextualized explanations, which aligns with quantitative results. Participants also highlighted important limitations that the LLM revealed when generation the explanation. Three out of five domain experts pointed out that not only the word selection, but also the phrasing of the explanation plays an important role in determining its meaning. Four experts also emphasized that the explanation is not only a task of answering a user question, but also includes high-level reflection information, which enables the learner to understand how the learning content can

affect their own context, e.g., their daily work. This level of reflection exceeds the ability of the LLM and requires a human mentor to provide it based on their understanding of the learner's context.

This feedback shows a limitation in our valuation approach, which should also include the effect of the explanation phrasing to the comparison with reference explanations. The nature of LLMs and the complexity of evaluating what a correct phrasing is, challenges this evaluation, and defines at the same time several prospects for continuing this ongoing research. Other limitations that we address in our study include: 1) The limited sample sizes in the experiment, which is used to provide a proof-of-concept and set the starting point for a larger scale test. 2) The comparison between GPT-4 and other LLMs is required to measure LLM-based performance deviations. 3) We do not include user-specific data in our contextualization approach, to comply with the GDPR, since GPT-4 is provided by a third party. A local LLM is intended to be used for further evaluating the role of user data in enhancing the relevance of the explanation to user contexts, as well as materials contexts.

5. Conclusion

In this paper, we proposed and evaluated an approach for utilizing KGs as a source of contextual information that supports LLMs in generating more relevant explanations of learning recommendations. Our approach extracts different types of information from the KG and constructs a contextual part of a GPT-4 model's prompt accordingly. The design of the prompt's context is conducted with experts-in-the-loop to ensure the fulfillment of the pedagogical requirements from the explanation. Experts also participated in the design of the final explanation shape and content, which was offered to the learners as a textual template filled by the GPT-4 model. A chatbot-based interaction with the user is used to provide the explanations as answers to user questions. We evaluate the proposed approach qualitatively and quantitatively using Rouge measures. Evaluation results provided a proof of the proposed concept, showing a reduced amount of less-relevant text in the generated explanation. The results of the evaluation also pointed out limitations of the LLM's performance which defines a starting point for future work to further evaluate the role of the explanation phrasing and the use of user-data for further personalization of the explanation text.

References

- [1] J. Ooge, S. Kato, K. Verbert, Explaining Recommendations in E-Learning: Effects on Adolescents' Trust, in: 27th International Conference on Intelligent User Interfaces, ACM, Helsinki Finland, 2022, pp. 93–105. doi:10.1145/3490099.3511140.
- [2] L. Yan, L. Sha, L. Zhao, Y. Li, R. Martinez-Maldonado, G. Chen, X. Li, Y. Jin, D. Gašević, Practical and Ethical Challenges of Large Language Models in Education: A Systematic Scoping Review, *British Journal of Educational Technology* (2023) bjet.13370. doi:10.1111/bjet.13370, arXiv:2303.13379 [cs].
- [3] M. A. Chaudhry, M. Cukurova, R. Luckin, A Transparency Index Framework for AI in Education, in: M. M. Rodrigo, N. Matsuda, A. I. Cristea, V. Dimitrova (Eds.), *Artificial Intelligence in Education. Posters and Late Breaking Results, Workshops and Tutorials, Industry and Innovation Tracks, Practitioners' and Doctoral Consortium*, volume 13356, Springer International Publishing, Cham, 2022, pp. 195–198. doi:10.1007/978-3-031-11647-6_33, series Title: Lecture Notes in Computer Science.
- [4] M. Fullan, C. Azorín, A. Harris, M. Jones, Artificial intelligence and school leadership: challenges, opportunities and implications, *School Leadership & Management* (2023) 1–8. doi:10.1080/13632434.2023.2246856.
- [5] K. Dana, H. Harouni, ChatGPT Is Unoriginal—and Exactly What Humans Need, 2023. URL: <https://www.wired.com/story/chatgpt-education-originality/>.
- [6] S. Hargreaves, 'Words Are Flowing Out Like Endless Rain Into a Paper Cup': ChatGPT & Law School Assessments, *SSRN Electronic Journal* (2023). doi:10.2139/ssrn.4359407.
- [7] D. Fensel, U. Şimşek, K. Angele, E. Huaman, E. Kärle, O. Panasiuk, I. Toma, J. Umbrich, A. Wahler, Introduction: What Is a Knowledge Graph?, in: *Knowledge Graphs*, Springer International Publishing, Cham, 2020, pp. 1–10. doi:10.1007/978-3-030-37439-6_1.
- [8] H. Abu-Rasheed, M. Dornhöfer, C. Weber, G. Kismihók, U. Buchmann, M. Fathi, Building Contextual Knowledge Graphs for Personalized Learning Recommendations Using Text Mining and Semantic Graph Completion, in: 2023 IEEE International Conference on Advanced Learning Technologies (ICALT), IEEE, Orem, UT, USA, 2023, pp. 36–40. doi:10.1109/ICALT58122.2023.00016.
- [9] S. Pan, L. Luo, Y. Wang, C. Chen, J. Wang, X. Wu, Unifying Large Language Models and Knowledge Graphs: A Roadmap, 2023. ArXiv:2306.08302 [cs].
- [10] L. Luo, Y.-F. Li, G. Haffari, S. Pan, Reasoning on Graphs: Faithful and Interpretable Large Language Model Reasoning, 2023. URL: <http://arxiv.org/abs/2310.01061>, arXiv:2310.01061 [cs].
- [11] R. Logan, N. F. Liu, M. E. Peters, M. Gardner, S. Singh, Barack's Wife Hillary: Using Knowledge Graphs for Fact-Aware Language Modeling, in: *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, Association for Computational Linguistics, Florence, Italy, 2019, pp. 5962–5971. doi:10.18653/v1/P19-1598.
- [12] J. Sequeda, D. Allemang, B. Jacob, A Benchmark to Understand the Role of Knowledge Graphs on Large Language Model's Accuracy for Question Answering on Enterprise SQL Databases, 2023. ArXiv:2311.07509 [cs].
- [13] M. Munch, J. Dibie-Barthélemy, P.-H. Willemin, C. Manfredotti, Interactive Causal Discovery in Knowledge Graphs, in: E. Demidova, S. Dietze, J. Breslin, S. Gottschalk, P. Cimiano, B. Ell, A. Lawrynowicz, L. Moss, A.-C. N. Ngomo (Eds.), *Joint Proceedings of the 6th International Workshop on Dataset PROFILING and Search & the 1st Workshop on Semantic Explainability*, volume 2465 of *CEUR Workshop Proceedings*, CEUR, Auckland, New Zealand, 2019, pp. 78–93. ISSN: 1613-0073.
- [14] H. Abu-Rasheed, C. Weber, J. Zenkert, M. Dornhöfer, M. Fathi, Transferrable Framework Based on Knowledge Graphs for Generating Explainable Results in Domain-Specific, Intelligent Information

- Retrieval, Informatics 9 (2022) 6. doi:10.3390/informatics9010006, number: 1 Publisher: Multidisciplinary Digital Publishing Institute.
- [15] E. Rajabi, K. Etminani, Knowledge-graph-based explainable AI: A systematic review, Journal of Information Science (2022) 01655515221112844. doi:10.1177/01655515221112844, publisher: SAGE Publications Ltd.
 - [16] OpenAI, GPT-4 Technical Report, 2023. doi:10.48550/arXiv.2303.08774, arXiv:2303.08774 [cs].
 - [17] IEEE Standard for Learning Object Metadata, IEEE Std 1484.12.1-2020 (2020) 1–50. doi:10.1109/IEEESTD.2020.9262118.
 - [18] H. Abu-Rasheed, C. Weber, M. Dornhöfer, M. Fathi, Pedagogically-Informed Implementation of Reinforcement Learning on Knowledge Graphs for Context-Aware Learning Recommendations, in: O. Viberg, I. Jivet, P. Muñoz-Merino, M. Perifanou, T. Papathoma (Eds.), Responsive and Sustainable Educational Futures, volume 14200, Springer Nature Switzerland, Cham, 2023, pp. 518–523. doi:10.1007/978-3-031-42682-7_35, series Title: Lecture Notes in Computer Science.
 - [19] C.-Y. Lin, ROUGE: A Package for Automatic Evaluation of Summaries, in: ACL 2004, 2004.

LLM-Assisted Knowledge Graph Completion for Curriculum and Domain Modelling in Personalized Higher Education Recommendations

Hasan Abu-Rasheed^{1,*}, Constance Jumbo², Rashed Al Amin², Christian Weber^{1,3}, Veit Wiese², Roman Obermaisser² and Madjid Fathi¹

¹*Institute for Knowledge-Based Systems and Knowledge Management, University of Siegen, Siegen, Germany*

²*Institute for Embedded Systems, University of Siegen, Siegen, Germany*

³*Department of Digital Health Sciences and Biomedicine, University of Siegen, Siegen, Germany*

Abstract

While learning personalization offers great potential for learners, modern practices in higher education require a deeper consideration of domain models and learning contexts, to develop effective personalization algorithms. This paper introduces an innovative approach to higher education curriculum modelling that utilizes large language models (LLMs) for knowledge graph (KG) completion, with the goal of creating personalized learning-path recommendations. Our research focuses on modelling university subjects and linking their topics to corresponding domain models, enabling the integration of learning modules from different faculties and institutions in the student's learning path. Central to our approach is a collaborative process, where LLMs assist human experts in extracting high-quality, fine-grained topics from lecture materials. We develop a domain, curriculum, and user models for university modules and stakeholders. We implement this model to create the KG from two study modules: Embedded Systems and Development of Embedded Systems Using FPGA. The resulting KG structures the curriculum and links it to the domain models. We evaluate our approach through qualitative expert feedback and quantitative graph quality metrics. Domain experts validated the relevance and accuracy of the model, while the graph quality metrics measured the structural properties of our KG. Our results show that the LLM-assisted graph completion approach enhances the ability to connect related courses across disciplines to personalize the learning experience. Expert feedback also showed high acceptance of the proposed collaborative approach for concept extraction and classification.

Keywords

Higher education, Knowledge graph (KG), Large language models (LLM), curriculum model, domain model, Recommender systems

1. Introduction

Higher education is a structured learning setting, where curriculums are designed to meet predefined learning goals for each study program. This leaves a small margin to meet the individual goals of different students in the same program, especially in programs where students have different study backgrounds, or come from various countries, such as international study programs. In these scenarios, the likelihood of a student repeating a course or a topic they already studied before becomes higher. In order to increase the efficiency of the full learning path that a student takes within their study, it is essential to consider their level of knowledge, the courses and modules they already studied, as well as their career and professional goals, to tailor their learning path to those individual needs, goals, and requirements.

However, the learning setting in higher education poses several challenges to accomplishing this personalization. On the one hand, teachers are occupied with many tasks [1] and lack sufficient time to analyze the background knowledge of individual students, as well as to analyze the content of other courses in the university, which might be used to construct a personalized learning path. On the other hand, there is a lack of standard representation of the courses and study modules in different study

Accepted in the IEEE Global Engineering Education Conference (EDUCON2025), London, UK, 22-25 April, 2025

*Corresponding author.

✉ hasan.abu.rasheed@uni-siegen.de (H. Abu-Rasheed)



© 2025 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

programs, sometimes even inside the same university. To enhance the interoperability of different study programs, and thus the potential to create an efficient learning-path recommendation, we aim in this research to provide a foundation for creating a comprehensive and homogenic pool of learning materials, through a flexible and sufficiently-detailed representation of curriculums in higher education.

We propose a human-AI collaboration approach to support teachers with intelligent, LLM-assisted algorithms that extract and analyze their educational materials. Automatic extraction is then evaluated and controlled by the human teacher to ensure its high-quality. To ensure the comparability of the learning materials amongst different study programs and different universities, we propose an ontological foundation that is designed specifically for higher education, covering the curriculum model, domain model, and student model. Our main research questions in this paper are therefore:

1. How to develop a higher-education ecosystem, where LLM supports teachers in creating a knowledge graph for higher education?
2. What are the sufficient classes of a higher education ontology, to integrate domain, curriculum, and user models?

To answer those questions, we developed the collaboration process based on a KG, which serves as the network data structure that represents the three models and their educational content. The KG also forms the homogeneous database that allows a recommender system to find and recommend personalize learning paths.

2. Background

The recent developments of LLMs opened several doors to support teachers and experts in a faster extraction and analysis of the wide range of educational content, whether online or within universities and higher education institutions. In the following, we highlight some of the systems and approaches that have similar goals to our research.

2.1. LLMs Support in Creating KGs

LLMs are powerful tools for generating and enhancing KGs [2]. They are designed to understand and generate natural language and are capable of a wide range of natural language processing tasks [3]. In the context of KGs, LLMs enhance KG embedding, completion, and construction, as well as supporting text generation and question-answering tasks [4]. KGs are valuable means for representing the interconnected nature of information, especially in educational settings[5]. They serve as a solid data structure for organizing and storing knowledge [6]. Kommineni’s study shows that integrating LLMs into KG creation can enhance entity recognition and relationship extraction streamlining KG creation[7]. This is an essential tool in higher education where vast information must be organized cohesively.

2.2. LLM-Enhanced KG Creation for Higher Education

Enhancing KG creation using LLMs in higher education introduces a robust method for modelling course content and structure [8]. The traditional process of constructing KGs is largely dependent on human experts and this has proven to be more complex when constructing large-scale and high-quality KGs in domains that evolve constantly [1]. Jhaji [8] investigated the use of LLM in creating and enhancing educational knowledge graphs by analyzing three KGs. The human evaluators rated the graph created from LLM and contextual data (course syllabus) higher regarding its relevance and educational value. Together with course-specific information, LLM can extract information and generate nodes that have high semantic similarity with course materials enabling a more efficient semi-automated KG creation process in higher education. The fusion of KG and LLMs is synergistic, particularly in higher education. KGs can be used as factual context sources for LLM prompts to enhance learning recommendations [9]. Evaluating the approach both qualitatively and quantitatively, results showed improvement in

the quality of the generated text. Despite these use cases, LLM support for KG creation in higher education remains under-explored, particularly in comprehensive multi-dimensional models, linking users, curriculums, and domain elements.

2.3. A Comprehensive Ontology for Higher Education

Ontologies are formal representations of data, which in our use case provide a holistic view of higher education requirements, incorporating curriculum, user, and domain models. Recently, ontologies have been commonly used in higher education [10]. However, traditional ontologies focus on user and educational models [11], which represent the subject matter within the educational environment. This underscores the need for providing a detailed and comprehensive representation of relevant knowledge and concepts in real-world domains. Including a domain model provides contextual depth, which is important for recommending courses that align with user backgrounds, goals, and career orientations. LLM-support for curriculum and domain modelling offers a solid foundation for personalized learning recommendations in higher education.

3. Methodology

To accomplish creating a graph data structure for modelling content in higher education and supporting teachers to represent their existing materials in this model, we propose a four-step implementation strategy: Graph ontology definition, automated topic extraction, human-AI collaboration for validation task, and KG construction.

3.1. Defining a Higher Education Ontology

To develop a graph structure that accommodates learning content from different study programs and universities, we developed the content representation models as interoperable ones. This meant selecting ontological classes that apply to a wider range of course structures, enabling teachers to adapt their learning content to that model easily. To that end, we select the class “Topic” as the core taxonomical element that is used to break down the content of each session within the study module, see Figure 1. As topics in the learning content differ in their level of detail, we add a “Sub-Topic” class to account for more granularity in the educational content. In this sense, a Topic is defined as an abstract concept, which is being taught in the learning session, while a Sub-Topic is defined as the fine-grained content, which is described through a tangible means, such as lecture slides. A Session is then the iterative instance of a Lecture, which takes place, e.g., on a weekly basis. The proposed ontology is composed of three models:

1. The curriculum model: which includes the classes representing educational content as defined by the university.
2. The domain model: which is used to model a higher abstraction level of the domain of knowledge. The domain model includes domains to which the learning content belongs, such as Automotive, Embedded Systems, or Communication domains. In each domain, more granular concepts are represented as Sub-Domains, such as Self-Driving Cars being a sub-domain of the Automotive domain.
3. The user model: which is meant to represent the stakeholders in the educational setting, focusing on students and the factors that can be used for learning personalization. These include the background knowledge, learning goals, preferences, as well as academic parameters, as adopted from [12], which represent all other academic aspects related to the student’s learning process, such as their exam performance.

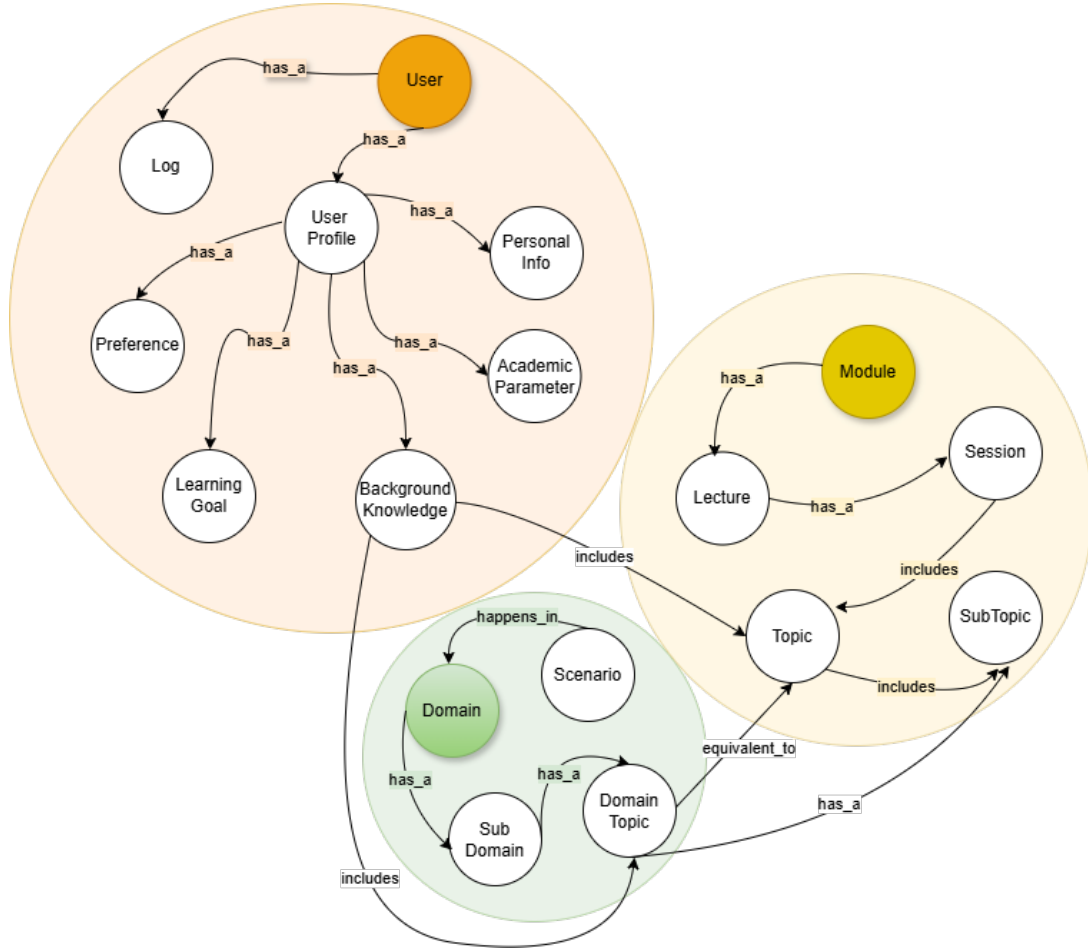


Figure 1: Proposed ontological structure of the knowledge graph, with domain, curriculum, and user models.

3.2. Automatic Content Extraction and Classification

In order for teachers to classify the content of their lectures into topics and sub-topics, a great amount of time and effort needs to be invested, which is rarely available for teachers due to their busy schedules. Therefore, we developed an automated extraction and classification pipeline, to support teachers with this task and considerably reduce the time and effort requirements on their side.

The pipeline, see Figure 2, begins with the extraction of textual content from the learning materials, including lecture slides, manuscripts, and lecture video recordings. While the extraction of the textual data from lecture slides and manuscripts is relatively straightforward, the extraction of the text from lecture videos requires transcribing these videos. Here, we assume that the figures, images, and equations in the slides are annotated or described textually. We conduct this transcription automatically utilizing the advanced capabilities of OpenAI’s Whisper model [13], which is a speech recognition model with high accuracy in zero-shot recognition tasks. Video transcripts are then fed, alongside the other textual content materials to an LLM for the topic extraction and classification tasks.

In the topic extraction step, we utilize OpenAI’s GPT-4o model. The LLM is prompted to extract the main concepts that each session includes. A concept represents here a key topic that is being taught in the session. Extracted concepts are then classified into topics and sub-topics based on the definitions of these two classes in our ontology. A topic is defined in the LLM prompt as an abstract concept, which is taught within the session. A sub-topic is defined as the concept that is being explained, in detail, through learning materials (e.g. lecture slides). The LLM is also instructed to consider the hierarchical structure of the two classes, where each topic includes one or more sub-topics. A topic is then an overarching concept, whose sub-topics are the actual content of the lecture. It is important to point out

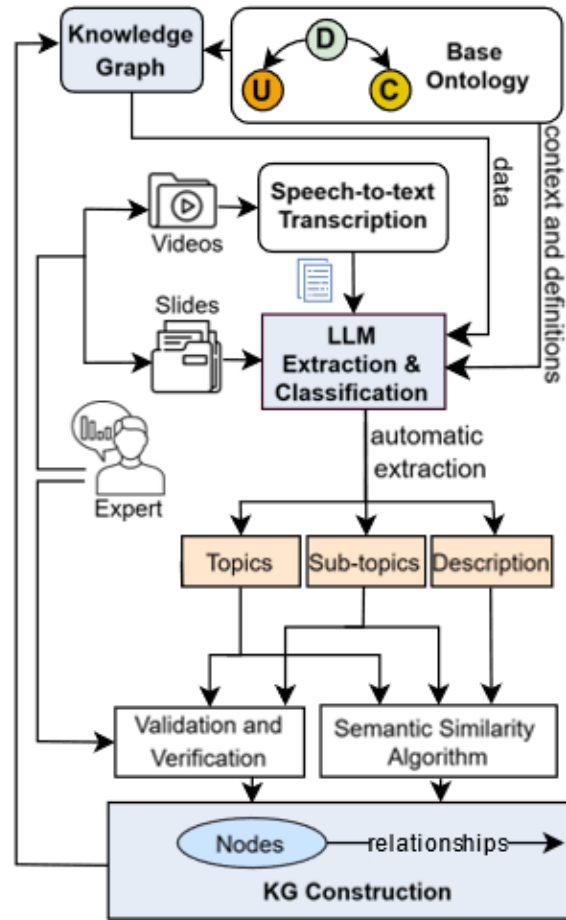


Figure 2: Pipeline components for the transcription, extraction, classification, and KG construction, based on a human-AI collaborative approach.

here that this ontology-based contextualization of the LLM prompt is imperative to extract an accurate representation of the learning content, ensure a sufficient level of interoperability, and enable finding meaningful similarities between the topics and sub-topics among lectures from different programs or universities.

In addition to the topics and sub-topics, the detailed content of lecture slides and videos provides a solid source for generating high-quality descriptions, especially on the sub-topic level, since it is the concept explained in detail in the session. We use those descriptions, in addition to the titles of the topics and sub-topics, to search for semantic similarities between different study programs.

3.3. Human Validation of the Automated Extraction

The risks of LLMs in automatic extraction, classification, and generation tasks, are high in a sensitive field like education. Therefore, we ensure the high-quality of topic extraction and classification through a human-in-the-loop approach, in which teachers validate the predictions of the intelligent algorithm, and make the final decision on extracted concepts and their classification as topics or sub-topics. The preliminary extraction and classification by the LLM serves here as an advanced starting point for the teachers and human evaluators, which saves them a considerable amount of time when analyzing the educational content. This also applies to the automated descriptions of topics and sub-topics, since the phrasing of the description is based on the session content itself and not the general knowledge of the model, i.e., in its training data. The final decision on the topics, sub-topics, and their descriptions is made by the teachers themselves, to ensure a high-quality content curation of the KG, which is necessary to enable generating high-quality learning-path recommendations for the students. The role of the

teachers in our system is not limited to validating the last prediction of the intelligent models. Teachers are also integrated in the contextualization of the LLM prompts by providing input on the definitions, limitations, and rules that the model should consider when analyzing, extracting, and classifying the topics of their learning materials. An example of those definitions is the specific meaning of terminology that is used in the lecture, which may differ, slightly or considerably, among different domains.

3.4. Knowledge Graph Construction

Based on the validated content from the teachers, the KG construction process is then conducted through two steps:

1. Creating graph nodes: according to the base ontology, and populating them with the data from the learning content, domain, and users.
2. Creating graph relations: which is conducted on two levels: the hierarchical structure from the base ontology, and the semantic similarity between the topics and sub-topics of different lectures.

To calculate the semantic similarities, we first build on the natural language processing pipeline in [5], where the title and description properties of the learning content are utilized to find similar content. This approach fits our extraction process of the topics, sub-topics, and their descriptions. Secondly, we utilize the LLM itself to analyze the extracted content and search for similar content from other lectures.

We use a semi-supervised approach for the validation of semantic similarities, where the human teacher validates a random sample of the predicted relations in the KG. This approach is adopted due to the large amount of semantic relations that can be identified when the KG grows to include large-scale educational content. Through teacher validation, hints are collected to adjust the parameters and prompts of the relation extraction pipeline and the LLM, respectively, sustaining the collaborative approach in our overall solution.

Once the KG is constructed, it also becomes a source of data that is fed to the LLM, to provide contextual information for the extraction and classification tasks. We found this method particularly effective when the topic extraction and classification are done to a lecture that is contextually similar to the existing content of the KG, such as topics from the same domain, or the same field of application.

4. Evaluation and Results

To evaluate the proposed pipeline, we construct the knowledge graph from the content of two university modules, namely Embedded Systems and Development of Embedded Systems Using FPGA. The two courses are taught in the University of Siegen, Germany, for master- and bachelor-level students. These courses were chosen because they are taught separately, despite the potential overlap between them. Each course is composed of 9 sessions, spanning the lecture period of one academic semester. Each weekly session covers a diverse number of topics, which differ in their level of detail.

4.1. Expert Evaluation of the Ontology and Pipeline

We conduct the evaluation with module lecturers, to ensure high-quality feedback based on their experience with the module's content and learning goals. Evaluators included one professor and two PhD assistants, working in the Embedded Systems Institute in the same university. We collect qualitative feedback from the lecturers about the base ontology, and the overall acceptance of our approach for human-AI collaboration. Feedback is collected through group meetings with the lecturers, in which the ontology was presented and evaluated from the perspective of use cases they face in their teaching.

Moreover, we collect quantitative input from the lecturers regarding the accuracy of extraction and classification of topics, sub-topics, and descriptions through the pipeline. To achieve this quantitative feedback, each lecturer checked a comprehensive list of extracted topics, their sub-topics, and the sub-topic descriptions, and evaluated the following for each sample:

1. The correctness of extraction.

Table 1

TABLE I. PRECISION (P), RECALL (R), AND F1 MEASURE VALUES FOR TOPICS AND SUB-TOPICS OF THE EVALUATION MODULES.

	Topic			Sub-Topic			Description
	P	R	F1	P	R	F1	P
ESa	0,99	0,94	0,96	1	0,97	0,98	1
FPGA	0,97	1	0,98	0,89	0,99	0,94	0,89

2. The hierarchical allocation of sub-topics within the topic.

Lecturers also evaluated the structural extraction and verified whether the extracted sub-topics were correctly positioned within the corresponding topics. A total of 1197 extraction samples were evaluated by the experts (173 Topics, 512 Sub-Topics, and 512 Descriptions). Table 1 shows the results of the Precision, Recall, and F1 measure for Topic and Sub-Topic classes in both evaluation modules. For the Description class, we calculate the Precision value to verify whether the long description text that was generated automatically fits the corresponding sub-topic.

The proposed pipeline and LLM-supported extraction of topics and sub-topics has shown a high capability of detecting the topics and their corresponding sub-topics from the lecture materials. A slight difference can be noticed between the two modules, which we trace back to the level and style of structuring that each lecturer uses to organize their lecture recordings. While some recordings only included the lecturer's explanation of the session's content, other recordings were simply made of the live, online session, including the discussions with students, their questions, and the lecturer's answers. The difference between these two approaches had a considerable influence on the concept extraction from transcribed videos in the first iteration of LLM extraction and classification. The model extracted concepts from the side discussions, questions, and answers, as well as from the lecturer's highlights about topics that will be covered in the course. Those topics, however, were not in fact discussed in the corresponding session. To account for this effect, we followed a prompt engineering approach to set explicit rules and limitations for the way each video transcript is handled. The prompt engineering process also defined the ontology classes to the LLM, to clarify the contextual meaning of the Topic and Sub-Topic classes.

It is important to highlight here that the differences among lecturers in the ways they record their lectures are vast, and therefore cannot be only accounted for in the prompt engineering steps. To that end, diversifying the input of the model, e.g., by supporting the video transcriptions with structured lecture slides or references, will be important to ensure a good quality extraction. Moreover, the KG itself can play the role of a supporting data source, e.g., in a retrieval augmented generation (RAG) approach, to extend the context of the LLM's prompt and increase the relevance and accuracy of the extraction and classification tasks.

4.2. KG-Structural Evaluation

To evaluate the resulting knowledge graph structure, we use the graph quality measures: Average Degree Centrality (ADC), and Graph Modularity. These measures validate our research goal of discovering connections between learning contents from multiple modules. Average degree centrality evaluates the connectedness of a graph node through its incoming and outgoing relations. In our use case, ADC highlights the role that the semantic similarity algorithm plays in increasing the density of relations among graph nodes. Graph modularity, on the other hand, evaluates the potential to separate learning contents in the graph from each other. Therefore, a lower modularity score is preferred in our case, since it reflects stronger connections between similar contents.

In the resulting KG, see Figure 3, we calculate an increase of the ADC value from 0.9 to 1.03, as well as a slight modularity decrease from 0.769 to 0.767. This slight change can be traced back to the small number of semantic relations added between the two modules, due to the relatively small size of the sample data. Moreover, this is also a result of the high similarity thresholds that we have set for

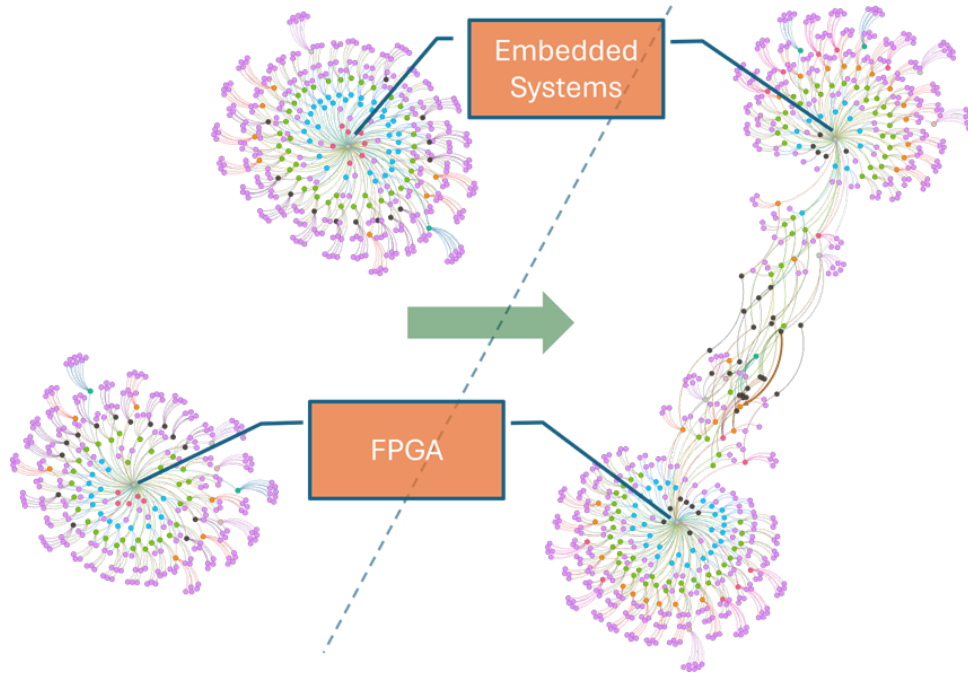


Figure 3: The structure of the KG before (left) and after (right) connecting both evaluation modules through semantic Topic and Sub-Topic relations.

the semantic relation extraction algorithm, to ensure high-quality relations between the topics and sub-topics.

Despite this relatively small improvement on the KG structurally, qualitative expert feedback pointed out that the automatic topic extraction and the possibility to semantically connect topics from different courses is a considerable help for the lecturers in re-structuring their course contents. This is accomplished by being aware of similar content in other courses and modules, which might be overlapping, or can be re-used. The evaluators pointed out that the insights they received from the pipeline’s output did not only support a better structuring of their courses and modules, but also the personalization of the learning experience for individual students, who have different levels of knowledge, scientific backgrounds, and career goals.

5. Conclusion

In this paper, we proposed an approach for modelling higher education curriculums and linking them to domain and user models. We use an LLM-assisted pipeline to conduct a KG completion task, which is accomplished through a collaboration between the intelligent model and domain experts. Our goal was to enable a seamless integration of the existing learning content into the curriculum model, to support learning personalization based on the KG.

We implemented our pipeline on a sample of two university modules within embedded systems and FPGA courses. The proposed base ontology, pipeline, and the extraction and classification models were evaluated with the lecturers of the selected evaluation modules. Our evaluation results showed the effectiveness of the proposed approach for the collaboration between human experts and the AI system. It served as an assistant, providing the human expert with an advanced starting point for the structuring of the learning content, and for enabling better personalization of the learning experience, considering the similarities between module contents and their connection to the domain and user models.

References

- [1] R. Meissner, L. Köbis, Annotated Knowledge Graphs for Teaching in Higher Education: Supporting Mentors and Mentees by Digital Systems, in: M. Bielikova, T. Mikkonen, C. Pautasso (Eds.), *Web Engineering*, volume 12128, Springer International Publishing, Cham, 2020, pp. 551–555. URL: http://link.springer.com/10.1007/978-3-030-50578-3_43. doi:10.1007/978-3-030-50578-3_43, series Title: *Lecture Notes in Computer Science*.
- [2] N. Ibrahim, S. Aboulela, A. Ibrahim, R. Kashef, A survey on augmenting knowledge graphs (KGs) with large language models (LLMs): models, evaluation metrics, benchmarks, and challenges, *Discover Artificial Intelligence* 4 (2024) 76. URL: <https://link.springer.com/10.1007/s44163-024-00175-8>. doi:10.1007/s44163-024-00175-8.
- [3] H. Naveed, A. U. Khan, S. Qiu, M. Saqib, S. Anwar, M. Usman, N. Akhtar, N. Barnes, A. Mian, A Comprehensive Overview of Large Language Models, 2023. URL: <https://arxiv.org/abs/2307.06435>. doi:10.48550/ARXIV.2307.06435, version Number: 10.
- [4] S. Pan, L. Luo, Y. Wang, C. Chen, J. Wang, X. Wu, Unifying Large Language Models and Knowledge Graphs: A Roadmap (2023). URL: <https://arxiv.org/abs/2306.08302>. doi:10.48550/ARXIV.2306.08302, publisher: arXiv Version Number: 3.
- [5] H. Abu-Rasheed, M. Dornhöfer, C. Weber, G. Kismihók, U. Buchmann, M. Fathi, Building Contextual Knowledge Graphs for Personalized Learning Recommendations Using Text Mining and Semantic Graph Completion, in: *2023 IEEE International Conference on Advanced Learning Technologies (ICALT)*, IEEE, Orem, UT, USA, 2023, pp. 36–40. URL: <https://ieeexplore.ieee.org/document/10260850/>. doi:10.1109/ICALT58122.2023.00016.
- [6] M. Kejriwal, Knowledge Graphs: A Practical Review of the Research Landscape, *Information* 13 (2022) 161. URL: <https://www.mdpi.com/2078-2489/13/4/161>. doi:10.3390/info13040161.
- [7] V. K. Kommineni, B. König-Ries, S. Samuel, From human experts to machines: An LLM supported approach to ontology and knowledge graph construction, 2024. URL: <https://arxiv.org/abs/2403.08345>. doi:10.48550/ARXIV.2403.08345, version Number: 1.
- [8] G. Jhaji, X. Zhang, J. R. Gustafson, F. Lin, M. P.-C. Lin, Educational Knowledge Graph Creation and Augmentation via LLMs, in: A. Sifaleras, F. Lin (Eds.), *Generative Intelligence and Intelligent Tutoring Systems*, volume 14799, Springer Nature Switzerland, Cham, 2024, pp. 292–304. URL: https://link.springer.com/10.1007/978-3-031-63031-6_25. doi:10.1007/978-3-031-63031-6_25, series Title: *Lecture Notes in Computer Science*.
- [9] H. Abu-Rasheed, C. Weber, M. Fathi, Knowledge Graphs as Context Sources for LLM-Based Explanations of Learning Recommendations, in: *2024 IEEE Global Engineering Education Conference (EDUCON)*, IEEE, Kos Island, Greece, 2024, pp. 1–5. URL: <https://ieeexplore.ieee.org/document/10578654/>. doi:10.1109/EDUCON60312.2024.10578654.
- [10] K. Stancin, P. Posic, D. Jaksic, Ontologies in education – state of the art, *Education and Information Technologies* 25 (2020) 5301–5320. URL: <https://link.springer.com/10.1007/s10639-020-10226-z>. doi:10.1007/s10639-020-10226-z.
- [11] E. Katis, H. Kondylakis, G. Agathangelos, K. Vassilakis, Developing an Ontology for Curriculum and Syllabus, in: A. Gangemi, A. L. Gentile, A. G. Nuzzolese, S. Rudolph, M. Maleshkova, H. Paulheim, J. Z. Pan, M. Alam (Eds.), *The Semantic Web: ESWC 2018 Satellite Events*, volume 11155, Springer International Publishing, Cham, 2018, pp. 55–59. URL: https://link.springer.com/10.1007/978-3-319-98192-5_11. doi:10.1007/978-3-319-98192-5_11, series Title: *Lecture Notes in Computer Science*.
- [12] E. Ilkou, H. Abu-Rasheed, M. Tavakoli, S. Hakimov, G. Kismihók, S. Auer, W. Nejdl, EduCOR: An Educational and Career-Oriented Recommendation Ontology, in: A. Hotho, E. Blomqvist, S. Dietze, A. Fokoue, Y. Ding, P. Barnaghi, A. Haller, M. Dragoni, H. Alani (Eds.), *The Semantic Web – ISWC 2021*, volume 12922, Springer International Publishing, Cham, 2021, pp. 546–562. URL: https://link.springer.com/10.1007/978-3-030-88361-4_32. doi:10.1007/978-3-030-88361-4_32, series Title: *Lecture Notes in Computer Science*.
- [13] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, I. Sutskever, Robust Speech Recognition

via Large-Scale Weak Supervision, 2022. URL: <https://arxiv.org/abs/2212.04356>. doi:10.48550/ARXIV.2212.04356, version Number: 1.

Analytical Techniques for Developing Argumentative Writing in STEM: A Pilot Study

Patricia Marybelle Davies¹, Member, IEEE, Rebecca Jane Passonneau², Smaranda Muresan³,
and Yanjun Gao⁴

Abstract—Contribution: Demonstrates how to use experiential learning (EL) to improve argumentative writing. Presents the design and development of a natural language processing (NLP) application for aiding instructors in providing feedback on student essays. Discusses how EL combined with automated support provides an analytical approach to improving written-communication skills.

Background: High-quality, timely, feedback is an effective way to improve students' writing. However, large class sizes and limited instructor backgrounds often make formative feedback impossible. Recent trends, including lowering entry requirements, have added to these challenges. Assistive technologies for implementing inclusive education provide viable solutions.

Research Questions: 1) How and why can EL be used to develop argumentative writing skills in university STEM students? 2) How can technologies be developed to support using EL in teaching writing? and 3) How might the holistic impact of using such analytic techniques be evaluated?

Methodology: Participants in an EL project were assigned two essays in sequence. They were given instructions on making good arguments and shown how to use an analytic rubric to maximize their scores. The essays were hand scored by tutors who provided scores for each dimension of the rubric. Subsequently, the content and argumentation of the essays were analyzed using NLP techniques to obtain independent scores. Qualitative data were also collected.

Findings: The project produced transformative writing experiences for the participants. It showed how analytical techniques help improve writing skills and how relevant automated instructor assistance can be developed using NLP technologies.

Index Terms—Argumentation, content annotation, experiential learning (EL), higher education (HE), natural language processing (NLP), rubric reliability, STEM, written communication.

Manuscript received 7 February 2021; revised 2 August 2021; accepted 16 September 2021. Date of publication 11 October 2021; date of current version 15 August 2022. This work was supported by the National Science Foundation under Award 1847853 and Award 1847842. (Corresponding author: Patricia Marybelle Davies.)

Patricia Marybelle Davies is with the College of Sciences and Human Studies, Prince Mohammad Bin Fahd University, Al Khobar 31952, Saudi Arabia (e-mail: pdavies@pmu.edu.sa).

Rebecca Jane Passonneau and Yanjun Gao are with the Department of Computer Science and Engineering, Penn State University, Pennsylvania, PA 16802 USA (e-mail: rjp49@cse.psu.edu; yug125@cse.psu.edu).

Smaranda Muresan is with the Department of Computer Science, Columbia University, New York, NY 10027 USA (e-mail: smara@columbia.edu).

Digital Object Identifier 10.1109/TE.2021.3116202

I. INTRODUCTION

GOOD writing, especially making effective arguments, demonstrates excellent critical thinking skills [1]. Writing as a means of communicating knowledge is a necessity in higher education (HE). Yet students enrolled in STEM programs worldwide often have little opportunity to develop and practice writing during their college years. Several studies have shown that graduates in computer science and engineering seldom have adequate writing skills for work in a professional setting [2], [3]. In fact, the problem starts earlier. As Gibbs [4] argues, many students in the U.K. leave secondary school without proficiency in reading, writing, and communication. In the U.S., student achievement in reading and writing from the National Assessment of Educational Progress points to a serious crisis in writing instruction: most students do not achieve grade proficiency [5]. Furthermore, even those secondary graduates with good language skills may run a risk of diminishing them once at university because there are too few opportunities to write essays and for getting good-quality feedback on writing assignments in STEM courses.

The research reported here demonstrates how analytical techniques could be used to help STEM students develop an understanding of and the skills for writing good arguments. An experiential learning (EL) project, which involved designing, delivering, and assessing two argumentative essays for first-year undergraduates, is presented. Three researchers, including the course instructor, collaborated on the development of a natural language processing (NLP) application designed to assist instructors with providing students with prompt, reliable feedback on argumentative writing assignments. As part of the preliminary investigation, a pilot study involving 141 computer science and engineering first-year students at a U.K. university was conducted. For these students, there is sometimes a lack of contextual coherence due to the dearth of opportunities for systematic inquiry in writing assignments. This is indicative of the fact that traditional approaches to teaching writing differ from those used in STEM courses with laboratory work in which students have the opportunity to investigate hypotheses through actions and activities. Such explorations enable the development of knowledge through internal and external discourses; for example, by watching videos or through discussions with their peers.

EL involves immersing students in educational activities and then encouraging them to reflect on the experience and develop new ways of thinking. It is a constructivist process that includes cycles of inquiry and reflection, a typical approach in STEM fields. Students can work individually, as part of a group, and with or without the guidance of a facilitator. The ways in which HE institutions organize curriculum, integrate technology, and infuse other resources to improve student outcomes have garnered scrutiny in recent decades [6]. Now that universities are being held accountable for the quality of their teaching through instruments such as the U.K. National Student Survey [7], the multidimensional nature of student achievement has assumed a new urgency. Central to students' developing good writing skills is constructive commentary that they can use as a scaffold. Consequently, of concern is the availability of technologies for assisting instructors, especially those with large classes, to provide high-quality, timely, and consistent feedback to guide students as they experiment with writing to develop their written communication skills. The EL project involved providing students with two tasks, each asking them to analyze source material critically prior to writing an argumentative essay.

This article hopes to provide guidance for practitioners seeking to integrate EL in university writing courses for STEM students, as well as information for researchers concerned with using NLP for building applications to support writing instruction. The EL project was set up to explore the following.

- 1) How and why can EL be used to develop argumentative writing skills of students enrolled in university STEM programs?
- 2) How can technologies that help promote EL in argumentative writing be developed?
- 3) How can the holistic impact of using such an approach be evaluated?

Following this introduction—Section I, the remainder of the article is organized as follows. Section II provides background information on EL and previous work on technology for aiding writing instruction. The methodology reported in Section III is divided into two main parts. First, there is the design of the teaching aspects of the EL project and related rubrics, plus the delivery and assessment of the essays. The second part discusses development and use of the NLP application for supporting writing instruction. Section IV discusses the impact the project had on students and what might constitute a full appraisal of the benefits of learning technologies that promote experiential education. Finally, Section V concludes the report and offers ideas on future directions.

II. BACKGROUND

A. Experiential Learning Contexts

The construct of EL or learning-by-doing refers to the work expounded by Dewey [8]. For him, the ultimate purpose of an experience is to reawaken the senses, to uncover ideas that might have been missed, and to validate what is being studied. More recently, Vygotsky [9] have been credited for providing the foundations for EL. His claims are that thinking, understanding, and knowing happen within a context based on

social and cultural factors. The American scholar Kolb [10] expands on these ideas to produce an EL model involving a cycle of observing, formulating, testing, and experiencing. In other words, do something, experience its consequences, take action in response to these, and then repeat the process, this time with a more developed understanding of what the process involves. EL can, therefore, be seen as an analytical process that allows students to develop enduring understandings.

HE institutions worldwide use EL to develop in students 21st century skills and competencies—empathy, resilience, and collaboration [11]—needed to prepare them better for an unpredictable world. Through volunteering, internships, and field studies involving local and global communities, some online, students are expected to harness communal traits that enable them to become more integrated and better connected as human beings. However, such activities are often viewed as extracurricular or co-curricular, separate yet in conjunction with what is presented in textbooks. For a variety of reasons, EL is seldom used in lecture rooms. The EL project is distinctive in that it is situated within the lectures of a university course.

B. Higher Education Contexts

Seeking to tap into the potential of HE to transform individuals, local communities, and society at large, the U.K. Government introduced a *Widening Participation in HE* agenda in 2014 [12]. One measure of an HE institution's performance in this scheme is the proportion of students from disadvantaged backgrounds they admit each year. Ensuring this metric is kept high has given rise to a variety of nontraditional approaches to recruiting students. These include running on-campus Bridge Programs and Taster Days, University Ambassadors visiting schools, and lowering entry requirements. Widening participation initiatives generally focus on the aspirations of applicants rather than their academic preparedness. Additionally, the migration toward a hybrid part-public, part-private funding system in the U.K. has prompted many universities to intensify their campaign for even more students [13]. Once admitted, these students become the responsibility of those who teach. It is instructors, often with no additional training, who must ensure that these students progress to graduation.

A primary concern is assessing these students. The move away from HE students being assessed solely on final examinations happened over the past five decades [14]. Nowadays, overall student assessment is determined by their performance on a series of tasks spread out over the term. Assignments are given in almost every field through projects, dissertations, and theses. The increase in student numbers, added to a lack of STEM instructors' knowledge of scoring essays, makes providing formative feedback on writing tasks challenging. As a result, students miss out on receiving information that could be integral to improving their writing skills. Scholars have argued that being able to receive comments and suggestions that help students revise their writing deepens understandings [15]; when received in a timely and systematic manner, formative feedback could have a lasting impact. Concerns

about the inadequacies of feedback on HE assessment tasks are well documented via the U.K. National Student Survey and the Australian Course Experience Questionnaire [16]. Assistive technologies, including NLP applications for implementing inclusive education by providing prompt feedback on essay drafts, offer new opportunities for improving students' writing performance.

C. Experiential Learning Project Context

A large proportion of the participants in the pilot study were admitted under a widening participation initiative. The essays assigned were two of the five tasks they had to complete in a writing course. These were to be done individually, whereas the other three assignments involved working in teams. The challenges posed by grading individual student work are amplified by reductions in per capita funding. A cohort of 215 students enrolled in academic skills is typically assigned one course leader and six tutors. Previously, each student had his or her essay marked by the tutor responsible for their section of the course. Grading was done by hand and without a rubric to facilitate consistency. The lack of a congruous process often resulted in wide variations in the scores across sections. One repercussion of this approach was that many papers had to be regraded by the course leader as part of the moderation process. Such concerns have prompted many instructors to explore automated scoring as an alternative.

D. Previous Work on Technology for Writing Instruction

A comprehensive review of research on instruction indicates that writing skills develop best given a formative assessment cycle. This involves successive stages of instruction to target specific learning goals, followed by assessments for which instructors provide feedback to help students scaffold their learning. Reliable and valid assessment is seen to be important as part of instruction. There is little work, however, on whether reliability of assessment can be achieved in classroom settings.

Existing literature discussed above together with the present study provide evidence that reliable classroom assessment is difficult to achieve. This section highlights the promise of combining human and automated assessment to ease instructors' assessment burden and improve reliability.

The time involved in traditional methods for assessing writing and the difficulties discussed in the previous section provide strong motivation for technological support for writing instruction. While there is little work on reliability of analytic rubrics in classroom settings, there is research on the reliability of so-called constructed response questions (open-ended prompts), where students provide short answers in a few sentences. Studies of rater cognition and validity for constructed response questions find that automated methods tend to be more consistent than human raters, and that the variability across humans observed in both TAs and raters is difficult to overcome [17]. A similar case has been made in a study that compares human and machine scoring of constructed responses to assess science teachers' pedagogical content knowledge [18].

Reviews of recent papers summarizing automated writing assessment tools and studies indicate that these tools can improve student engagement and support for peer collaboration [19], and provide feedback to generate students' revision. The main drawback noted in [19] is that instructors found it challenging to integrate technology in the writing curriculum. A review of 44 tools for supporting academic writing instruction showed that most tools rely on automated writing evaluation (AWE) techniques used across a variety of subject domains and genres [20]. AWE tools have been defined as consisting of two components, one for scoring and one for feedback [21]. Apart from pointing to the need to address languages other than English, the authors of the tool review found promising results in feedback linked to writing goals and genres, and to strategy instruction, meaning techniques for planning and revising text in general, or specific kinds of text such as persuasive writing. AWE builds on many distinct automated technologies and tools that can support analytic or holistic rubrics, including Coh-Metrix [22], C-rater-ML [23], G-rubric [24], Coh-Viz [25], and PEG [26].

AWE technology to support revision based on providing feedback that automatically applies an analytic rubric shows promise across multiple subject areas. These include second language learners' persuasive essays [27], college students' physics lab reports [28], and middle school students in english language arts (ELA) classes [29]. In [27], a comparison of teacher versus automated feedback for 104 students found that the automated feedback led to the same kinds of improvements between first and second drafts on four of seven classes. Park and Cho [28] investigated the ability to predict peer reviews of lab reports in a study with 41 students, where eight Coh-Metrix indices had modest but significant correlations with the human scores. However, the other work that attempted to replicate findings from the use of Coh-Metrix found a different set of Coh-Metrix features to be predictive from those identified in the previous work [30]. The PEG tool provides scores for six dimensions of writing quality (e.g., idea development, style, and word choice), each on a 5-point scale, based on NLP and machine learning techniques. A few recent studies of PEG for writing revision in ELA classes found PEG to reduce teacher effort, and to lead to improved scores on standardized tests [29], [31]. Other machine learning methods for predicting scores on analytic rubrics have also been investigated for college level essays from second language (L2) learners [32] and middle school argument writing [33].

Rubric-free methods for AWE have also been investigated. G-rubric [24], [34] is a modification of latent semantic analysis (LSA) [35], a method to create numeric vector representations of the meanings of words, where the number of vector dimensions is up to the investigator. G-Rubric converts the LSA vector space with latent dimensions of meaning to a new vector space with semantic grounding i (e.g., 300), a fixed number of relevant concepts. It has been used to give college students iterative feedback during revision of source-based summaries [34], and with business students in a MOOC [36].

Concept maps are another rubric-free feedback method. Concept maps, a visualization tool that has been used in education for decades [37], are graphs that depict explanatory

knowledge, where nodes represent concepts and edges represent relations between them. Sung *et al.* [38] compared four conditions of feedback for sixth graders in the U.S. writing summaries over six weeks: 1) no feedback; 2) LSA-based visualization; 3) concept maps; and 4) LSA plus concept maps. Students who received feedback all improved between pretest and posttest, and students with concept-map feedback outperformed students with the other two conditions. The Coh-Viz tool automatically creates concept maps for individual sentences, similar to subject-predicate-object graphs [39], and has been tested with students studying education in a German university. Students' revisions based on concept map conditions showed significantly greater improvements in cohesion over a baseline.

To summarize, studies show automated analysis can support formative assessment during writing instruction by helping the instructor to provide prompt feedback to students [27], [40], [41] and that feedback can support student revision [31], [34], [36], which can, in turn, lead to improved writing skills [29], [38]. Machine learning methods as used in PEG, C-rater-ML, G-Rubric, and the method reported in [27] generalize better than Coh-Metrix alone, although Coh-Metrix provides useful features for the machine learning approach used in [27]. Lancho *et al.* [36] suggested that automated support could also be integrated with human grading to improve the consistency and reliability of summative assessment.

III. METHODOLOGY

A. Experiential Learning Teaching Approach

1) *Setting—Participants and Assignments*: Participants in the project comprise around 200 first-year computer science and engineering students at a public university in the U.K. They are required to complete an academic skills module in their first year. The semester-long course aims to develop in students the proficiency in writing and communication skills necessary for success in college and future employment. The university attracts students from surrounding towns and cities. Participants come from a wide range of socioeconomic and academic backgrounds. Most study full time, but a small number are part-time students. The learning outcomes of the course are drawn from the UNESCO [42] definition of literacy, which centers on ensuring that students are able to “identify, understand, interpret, create, communicate, and compute, using printed and written materials, as well as ... to solve problems in an increasingly technological and information-rich environment.” The course leader is supported by six tutors. Classes are scheduled over 6 h per week, with an hour-long lecture followed by a 2-h hands-on workshop on each of two days. A highly interactive workshop design was used to help deepen students' understanding of the lectures.

The EL project included two argumentative essays described below. Each student chose one of the following three topics for the first essay. First, they had to analyze critically reading material provided and summarize it in 150–250 words. Next, they were asked to write a short argumentative essay (300–500 words), based on their reading, addressing one of the following prompts.

TABLE I
RUBRIC DIMENSIONS, WEIGHTS, AND SUBDIMENSIONS

Dimensions	Weights	Sub-Dimensions (with points assigned)
Content	3/7	quality, coverage, coherence (0 - 5 each)
Argument	2/7	claims, support, counterargument (0 - 10)
Conventions	1/7	lexis and grammar (0 - 5)
Referencing	1/7	sources and citations (0 - 5)

- 1) *Autonomous Vehicles (AV)*: Will these change how we travel today?
- 2) *Cryptocurrencies (Crypto)*: Are they the currencies of the future?
- 3) *Cybercrime (Cyber)*: Will education and investment provide the solution?

The areas of specialization of the participants include cybersecurity, information technology, and computer engineering. Thus, for the first essay they had the opportunity to choose a topic they were already familiar with or interested in.

For the second essay, all students were asked to discuss the same question—*Should artificial intelligence be used in teaching and learning?* They were limited to making arguments based only on material from two articles they were given. The two essays were designed to be developmental exercises with the second assignment extending the writing skills the students had developed in the first essay.

2) *Rubric—Motivation and Design*: Writing scales arose in the early 20th century to compare performance of schools and teachers [43], and only later were they developed within classroom contexts to provide guidance for students. It has been shown that analytic rubrics, where scores are assigned to distinct dimensions, have greater reliability than holistic rubrics [44]. Still, many studies highlight as problematic inconsistency among raters and scoring professionals [45] when applying analytic rubrics, due to the lack of training or familiarity. Despite these debates, analytic rubrics can serve as an instructional tool to improve students' writing quality [46]. An analytic rubric was developed and used both for instruction and grading.

The rubric was designed through a collaborative process by the three researchers working on the project. The instructor, who is one of the researchers, has a background in educational technology whereas the other two specialize in applying NLP to educational data. Part of the investigation involved understanding how the rubric supports instruction in argumentative writing. The rubric contained explicit descriptions of performance characteristics, each corresponding to a point on a rating scale. Table I shows the four dimensions of the rubric, the dimension weights, and subdimensions with the points assigned. Design of the rubric was guided by Ferretti's well-known argument rubric [47] and the source-based argument scoring attributes (AWC) [48]. The research has shown that the range of a rubric scale is important because it affects reliability and ability to make meaningful distinctions; more than seven levels lead to cognitive difficulty, and fewer levels produce sharper classification.

Timely instructor feedback has long been advocated in the assessment literature as a means of supporting student

learning. It provides a learner-centered approach in which, from a social-constructivist standpoint, students can scaffold their learning. A rubric that provides descriptive feedback is a key component of authentic assessment. It can also be used for self-assessment as a criterion of written work. It was, therefore, important that the students were given the rubric together with the first assignment. The rubric was also used to provide formative feedback on the first essay before the second essay was assigned.

3) *Essays—Assigning and Grading*: A research-based Universal Design for Learning framework [49] guided the formulation of the assignments. The framework provides a structure for developing curriculum—learning outcomes, instructional methods, and assessment. It is composed of three main ideas: 1) providing engagement; 2) encouraging alternate forms of representation; and 3) facilitating action and expression. Students were first introduced to the four elements of writing argumentative essays: 1) engaging with the prompt; 2) formulating a claim; 3) developing arguments and counter-arguments; and 4) concluding the essay [50]. Students spent the tutorial following the lecture exploring these components. Black [50] advocated that argumentative writing should be considered as an aesthetically pleasing art form and that, on completion of the work, authors should have the satisfaction of knowing that they have created something.

The rubric became an instructional tool used to explain the purpose of the assignment. Six model essays on each of the three topics were written by sophomores who were more experienced writers and had previously done well in the course. These exemplars were used to highlight how students could maximize their scores. The first assignment was scored by three course tutors, with each person scoring essays with the same title; the number of students per topic was capped for even distribution. All tutors were trained to use the rubric consistently.

Once scored, examples from the first essays were used to point out avoidable mistakes. As a more learner-centered approach, the students were encouraged to reflect on their own scored essays. Taking this first attempt at argumentative writing as a point of departure, the second essay was assigned. The project participants had experienced the consequences of their first attempt, reflected on the feedback received, and now went through the writing process again.

4) *Reliability of the Rubric*: The use of rubrics for writing has engendered debate about their reliability and purpose. Educational intervention studies apply rubrics whose reliability is usually quite good. For example, Graham and Perin [51], in a metaanalysis of educational interventions, excluded interventions with reliability lower than 0.60. Yet a large body of research, including [52], has documented how trained raters can exhibit different levels of severity on analytic rubric categories. There has also been skepticism about applying rubrics to classroom grading, due to subjectivity in interpreting rubric criteria and overreliance by teachers on the rubric as authority. Turley and Gallagher [43] argued that the debate should not be about whether rubrics are good or bad, but about how to use them. They discuss how the interpretation of a rubric depends, in part, on developing a community of users who

TABLE II
SEVEN-WEEK RATER TRAINING

Week	Activity
1	Virtual meeting to review argument writing: assignment #1 and rubric #1
2	Raters write individual essays: one on AV, and one on Crypto or Cyber; then each rater applies rubric to essays written by the other rater
3	Virtual meeting to review raters' essays and assessments
4	Both raters assess the same three Crypto essays
5	Virtual meeting on their first round of assessment centering on discussion between raters and with all three researchers
6	Each rater assesses the same three additional Crypto essays
7	Feedback on the second round of assessment, with some discussion on assignment #2 and rubric #2

understand the language of the rubric criteria. However, very little work has been done on comparing how rubrics are used for instruction with how they are used in scoring, or to examine the difference between their reliable use and inconsistent classroom use. The present work investigates these problems.

Although having an analytic rubric for both instruction and grading is beneficial for students, it is difficult to apply an analytic rubric reliably in the context of large numbers of students. This motivates the view that development of algorithms to support the application of a rubric is an important goal, as discussed in detail in the following section. The development of automated methods is facilitated by creation of training data for a specific rubric, consisting of a large number of examples where the rubric has been applied.

Two third-year undergraduates were trained to use the rubric over a period of seven weeks, by which point they had achieved measurably reliable performance in applying the rubric. Subsequently, each of them spent 10 h per week rescored half the essays written by students for the first assignment. Their training included understanding the structure of argument writing and completing both essay assignments (see Table II).

Pearson correlation [53] on the content and argument components (ArgC) of the rubric was used to assess rater agreement. After the raters applied the rubric to the first sample, their correlations with each other and with the assigned grades varied widely, from negative correlation to high correlation. This difference improved for the second round of three essays; the correlation between the raters was perfect on two, and poor on the third. After discussing these results, the raters were allowed to proceed independently with applying the rubric to the remaining essays.

To check reliability, each rater scored 28 essays per week for three weeks and 31 in the fourth week. Ten randomly selected essays were assigned to both raters for continued monitoring of their reliability. The correlations for the content and argument dimensions on the ten essays were generally high. They ranged from one low outlier of -0.52 to 1 , with an average of 0.75 ($n = 10$), or 0.89 after dropping the one outlier ($n = 9$). The high average correlation of 0.75 indicates raters are consistent with one another, and dropping the one outlier shows that apart from an exceptional case of relatively low correlation, the raters had very high average correlation. The reliable raters

TABLE III
EXAMPLE OF A SCU: FOUR OF FIVE REFERENCE SUMMARIES,
IDENTIFIED HERE AS A, B, C, AND D, EXPRESSED THE SAME
IDEA (PROPOSITION) ABOUT THE POTENTIAL NEGATIVE
IMPACT OF DRIVERLESS VEHICLES ON
PUBLIC TRANSPORTATION

Weight = 4	Label = Driverless vehicles will likely reduce reliance on public transport
Reference ID	Text
A	One of the main points is the displacement of public transport
B	the rise of autonomous vehicles will disrupt the current standing of public transport
C	even more people would switch from public transport
D	it could also have a negative impact upon the public transport systems

had lower correlations, however, with the assigned grades for the three-essay assignment, with averages ρ equal to 0.72, 0.63 and 0.59, respectively. This shows that rater training leads to greater consistency in application of the rubric, and therefore, room for improvement for in-class grading.

The reliability study shows that the rubric can be applied very reliably by specially trained raters and with moderate to low reliability in uncertain classroom contexts. It indicates the difficulty of using a fine-grained rubric in large classes, where teaching assistants do the grading, where students want to see their grades quickly, and where timely and specific feedback is beneficial.

B. Technology for Supporting Experiential Learning

1) *Content Annotation and Analysis (The Wise Crowd Method)*: Writing summaries of source texts has been found to be among the best instructional tools to develop students' reading and writing skills for conceptual knowledge [51]. Their use as a pedagogical tool requires a method to assess the conceptual quality of a summary, which, in turn, rests on the identification of the main ideas in the source texts being summarized. Many similar methods have been utilized in educational psychology, including expert consensus [54], ranking of propositional units in source texts [55], and successive elimination of less important propositional units in source texts [56]. All these methods elicit explicit judgments of propositions. The present study relies on exploiting the notion of a wise crowd of experts who summarize the same sources [57], [58]. Ideas expressed in more of the expert wise crowd summaries have higher weights. Content scores for a summary consist of summing the weights of the ideas expressed, and then normalizing, as explained below. The previous work on this method found that four to five expert wise crowd summaries are sufficient to ensure that the probability of misranking a pair of summaries is below 0.1 [58].

Table III illustrates a summary content unit (SCU) from five expert wise crowd summaries of the AV article. Four of the five summaries expressed the idea that the use of public transportation might decrease with increased reliance on autonomous vehicles. Although the idea is expressed in different ways, all four expert summaries express the same idea

clearly. The ideas in student summaries that match an SCU are credited with the corresponding SCU weight. Given five reference summaries, SCU weights can range from 1 to 5. The ideas in student summaries that do not match an SCU are assigned a weight of 0. The score assigned to a student summary normalizes the total sum of the weights of their ideas by the number of ideas in the student's summary, and by the average number of ideas in a reference summary. Thus, a summary gets a higher score if the ideas expressed by the student match more of the high-weighted SCUs, and if the student expressed a good proportion of them. The scores can then be explained to the student or instructor in terms of the overlap of ideas in the student's summary with the full repertoire of SCUs for a given text.

In the previous work, it was shown that the wise crowd method for identifying important ideas in source texts, ranking them, and using the resulting ranked list to assess student summaries, correlates very well with a main-ideas-rubric used in an educational intervention ($\rho = 0.88$) [54]. The method itself has been found to be highly reliable with respect to manual identification of content units, correlations of expert wise crowd scores from different annotators, and ranking of automated summarizers [59]. Originally, this method was applied through manual annotation procedure (see next paragraph). An automated approach to the assessment and feedback step was developed [57] and, more recently, a fully automated approach called PyrEval that identifies and ranks the SCUs from a set of reference summaries, then uses the weighted SCUs to assess new summaries [60]. PyrEval was tested on summaries from the AV and Cryptocurrency topics. Also, described here is an extension to this annotation linking the SCUs from the summaries to propositions in the argument portion of a student essay.

The analysis of the content of the students' essays asks how well the automated method of summary content analysis replicates the manual method, and how the automated method could support feedback on the rubric, either to help students revise the essay as a whole, or to give the instructor an overview of students' grasp of the content and their ability to draw on it to support their arguments. Recall that the assignments first asked students to summarize the source text or texts in 150–250 words, and then to construct an argument addressing one of the prompts. Fig. 1 illustrates the workflow of the manual annotation through the use of two annotation tools: 1) DUCView and 2) SEAView. The reference summaries are annotated first to identify the SCUs and to create a pyr(amid) file (a list of SCUs derived from reference summaries is referred to as a pyramid). The pyr file serves as a content model that is used to assess student summaries. After an annotator matches the propositions, the student expresses to the weighted SCUs, and DUCView can export a single pan file for each student summary. The new aspect of content annotation that has been added is for the argument part of a student's essay. In this step, which uses SEAView, elementary discourse units (EDUs) are annotated [61], [62]; these are effectively individual clauses or propositions stored in the sep file. In contrast to a summary of a source text, the quality of a student's argument is not expected to depend on how much of

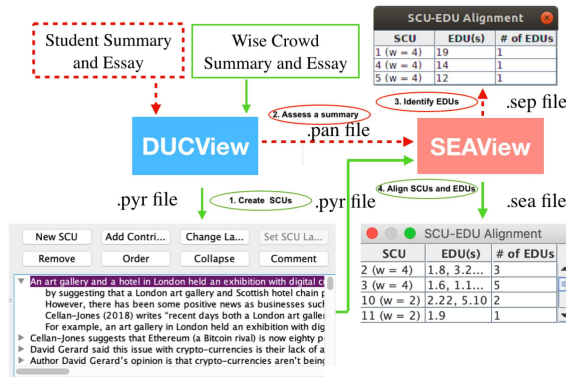


Fig. 1. Workflow diagram for content annotation, which use DUCView and SEAView. The green box and arrows indicate the flow of the expert wise crowd summaries to annotate the SCUs, and the box and arrows in dashed red lines show the flow of a student's essay, divided into summary and argument. The file extensions (pan, sep, etc.) indicate the underlying XML format, to differentiate the stages of annotation.

the same content is expressed as in a reference argument. On the other hand, of interest is how much of what the students' summarized from a source appears in their arguments, and what sort of content they use to frame their arguments. The last annotation step, therefore, involves matching the EDUs in a student's argument to the SCUs.

The automated wise crowd method performs fairly well on these summaries, as described in [60], with a Pearson correlation of 0.66 on the autonomous vehicle summaries when comparing the manual and automated summary content assessment, and a Pearson correlation of 0.72 for the cryptocurrency summaries. In a previous study [63], the instructor found the content scores and justifications to be very useful in understanding cases where the tool gave lower scores to written work compared to those given by the tutors. On inspection, this difference appeared to be due to some tutors assessing the writing fluency rather than the content of the essays. For the present study, neither the manual nor automated content scores on the summaries correlate well with the content dimensions of the rubric. This is because the rubric content dimensions of quality and coherence relate to the essay as a whole, not to the summaries alone, and the students consulted other sources of their choosing to find evidence to support their arguments. Clearly, the content assessment of the students' summaries reflects their reading skills, which suggests further investigation of whether summarization skills might provide insight into how well students use external sources in their arguments.

The ongoing work on the content analysis of the students' essays includes investigation of the essays on the third topic (Cybercrime) and analysis of the relationship between the content and argumentation, particularly with regard to the overall structure of the students' essays. The next section describes the analysis of students' arguments.

2) *Argumentation Annotation and Automated Analysis:* Effective argumentative writing presents a claim, considers evidence in support of and against the claim, and demonstrates how the pros outweigh the cons. The project aimed to test whether argumentation features derived from coarse-grained argumentative discourse structure correlate well with

TABLE IV
CORRELATION OF LR (5 FOLD CROSS-VALIDATION)
WITH ARGUMENT QUALITY SCORES

Features	Correlations
baseline	0.15
ArgC	0.27
ArgR	0.35
Str	0.17
baseline + ArgC	0.21
baseline + ArgR	0.26
baseline + Str	0.33
ArgC + ArgR + Str	0.41
baseline + ArgC + ArgR + Str	0.26

the 6-point scale rubric that rate the *quality* of the argument. To do this, the first step was to label the argumentative part of the 37 Cryptocurrency essays using an annotation scheme generally used in argument mining [64]: *main claim*, *claim*, and *premise/evidence* as ArgC, and *support* and *attack* as argument relations (ArgR). The advantage of a simple annotation scheme is twofold: 1) more reliable human annotation and 2) better performance of automatic methods to detect the argument structure. Two expert annotators with background in linguistics and argumentation performed the annotation that resulted in a gold-standard set of 36 main claims, 559 claims, 277 premises, 560 support relations, and 101 attack relations. A *proposition* was considered as the unit of annotation, given that premises are frequently propositions that conflate multiple clauses and sometimes even sentences [65].

The set of argumentative features introduced by Ghosh *et al.* [66] was used on the annotated essays to test whether they correlate with the argument dimension scores used in the reliability study. The features are grouped in three categories: 1) features related to ArgC, such as the proportion of argumentative sentences—those which contain a main claim, claims and/or premise—and the number of ArgC in an essay; 2) features related to ArgR, such as the number of supported and unsupported claims and the number of attack relations (counterarguments); and 3) features related to the *typology of argument structure* (Str)—the number of argument chains and argument trees (see [66] for more details).

While Ghosh *et al.* [66] showed that these features correlate with the holistic essay score (low, medium, and high) when applied to persuasive essays from standardized tests for teaching English as a second language (TOEFL), this study aims to test the effectiveness of these argumentation features in predicting the argument quality scores (scale of 0–5) obtained in the reliability study. Logistic regression (LR) learners were used and evaluated using quadratic-weighted kappa (QWK) against the human scores. QWK is a standard metric used for essay scoring [66], [67]. Table IV reports the results from a 5-fold cross-validation setting for the three argumentation feature groups and their combination. The baseline feature is the essay length in sentences, since it has been shown to be highly correlated with essay scores [68].

The best correlation is obtained when using all the argumentative features (ArgC + ArgR + Str), while ArgR is the best performing individual feature group. Moreover, all argumentation features outperform the baseline. Also, the *argument*

tree feature correlates with high-scoring essays, which is not surprising as these features capture the complexity of a well-written argument. In addition, top-scoring essays (with score 5) have a higher number of *attack* relations to the main claim, showing that these essays contain counterarguments, which is an aspect in the rubric. The number of claims supporting the main claim was negatively correlated with low-scoring essays since students who received a low score, although forming arguments, failed to link them to their main claim. Similar to the work of Ghosh *et al.* [66], the number of supported claims correlates negatively with lower scoring essays, which show that students who receive low scores do not provide evidence for their claims. Another interesting observation from this analysis is that in the best essays (score 5), the ratio of argumentative sentences to total number of sentences was higher than for essays with a score of 4, whereas essays with a score of 4 were generally longer. That could also explain why the baseline feature (essay length) performed so poorly, since length alone is not indicative of argument quality.

The correlations were lower than the ones reported by Ghosh *et al.* [66], a finding that could have several explanations: 1) the number of essays was smaller, 37 compared with 107; 2) a 6-point scale rather than a 3-point one was used; and 3) the scale used reflects the argument quality and not an overall score.

Looking at argument structure alone might not be enough; instead, both the structure and the semantics of arguments need to be examined in order to predict the argument quality more reliably [69]. Combining the work on content annotation and argument annotation achieves this goal, and is planned for future work.

IV. DISCUSSION

A. How and Why EL Can Be Used to Develop Writing

Foregoing discussions have highlighted the importance of providing timely, formative feedback to enhance students' understanding of what is expected in argumentative writing assignments. Rubrics provide an avenue for this and can be used to integrate commentary into students' grades to give them meaning beyond just a numerical value. Also, presented above are methods for using EL to facilitate the development of good writing skills in university STEM students. Reasons why EL advances student writing go beyond clarifying expectation. Prompt, consistent feedback also provides assessment standards and benchmarks, promotes independent learning and self-regulation, and raises student aspirations [70]. Thereby, it becomes possible to impact the broader range of students with differing abilities and capacities who are now entering HE.

B. How the Holistic Impact of EL Can Be Assessed

The experiential aspects of the learning cycle hinge on the premise that the writing activities students engage in include reflection. The expectancy-value framework [71] depicts the motivations of learners as being based on their expectancy of success and the value credited to an assigned task. Thus, it can be used to assess the holistic impact of EL. Students with low

self-efficacy typically find understanding and acting on formative feedback difficult. As a result, they tend not to engage in reflection. Motivation and engagement are both intrinsic to all learning. They are currently receiving much attention in HE because students' opinions now play a principal role in rating teaching and learning in tertiary education. Consequently, utilizing innovative teaching techniques to produce positive academic outcomes for students is no longer simply an option, rather it is now essential to HE and gives new impetus to EL.

C. Evaluating the EL Project

After receiving feedback on the first of two essay assignments, the students were asked to complete a questionnaire about the role of the rubric. First, they were asked the following questions requiring Yes/No responses.

1) Did you get the mark you were expecting on Argumentative Essay 1?

2) Did you use the rubric?

Those who had used the rubric were questioned further.

1) When was the rubric used? *before starting the assignment, while doing it, after completing it, or some combination of all of these?*

2) What was it used for? *to understand the requirements, as a guide, for checking, or some combination of all of these?*

3) Do you feel the rubric helped you achieve your desired score? If yes, explain how.

Out of the 84 respondents, almost two-thirds of them (63%) reported using the rubric in one or more of the ways suggested above, and of these, 34% believed that the passing score they received was due to having access to the rubric. Others said they probably would have achieved the same score without using the rubric, with only 11% suggesting that the rubric did not help them at all. What was even more striking is that more than 65% of students who attempted both essays scored the same or a higher mark on the second essay. There is limited evidence to suggest that the feedback from the first assignment aided their performance on the second essay. Instead, what seems more likely is that the second essay allowed students to master the EL approach they had been exposed to in the first assignment. The expert wise crowd were also questioned about their experiences with using the rubric. In addition to saying it helped them understand different aspects of the writing process, one said that it made "very clear what the expectations were." Another suggested that without the rubric, he "would not have known exactly what was expected." These responses confirm research showing that rubrics could help students to comprehend tacit knowledge embedded in assessment.

D. Developing Technology to Support EL

Automated analysis of student writing has great potential given the demands on instructors to provide feedback and the difficulty of reliable classroom grading for large classes. The ongoing work on assessing the content of student writing includes applying the wise crowd method for feedback to students and teachers during classroom instruction in middle school science courses. It is noted that designing the prompts

for student writing tasks needs to proceed hand-in-hand with refinements to the automated assessment so that the content of students' writing is more focused on the concepts associated with the learning objectives. Furthermore, designing automated feedback that is useful to students is an additional step beyond identifying which ideas the student needs to articulate more fully. This shows the need for deep collaboration between educators and technology developers.

While providing feedback on students' ability to construct an argument (claim, premises, support, and attack relations) is important, ongoing work by the authors points to the additional need to highlight whether or not the arguments have reasoning flaws. Three areas of need relating to source-based argumentative essays were identified: 1) detect whether or not the evidence provided contradicts the source texts; 2) detect the argument schemes used (e.g., analogy, from example, causal); and 3) generate the unstated premise (enthymeme) that links a claim to the evidence provided. While the first problem helps provide feedback whether students misunderstood the source-text material, the latter two help highlight better whether or not reasoning flaws (fallacies) occurred. To tackle this, joint modeling of content and argumentation will be paramount.

V. CONCLUSION

This article has discussed how and why argumentative writing instruction for STEM students should be aligned with the learn-by-doing approaches used in these fields. It argues that EL provides an alternative to simply telling these students what is expected of them in writing assignments. Furthering Dewey's conjecture—experience as reawakening—it suggests that conceptual understanding of writing arguments could be fostered by engaging students in transformative experiences that allow them to confirm and extend their ideas. A part of this process involves providing them with a rubric for instruction and assessment. The reliability study shows that rubrics can be applied reliably outside classroom contexts but classroom grading tends to be less reliable, which can be attributed to time pressures and lack of training. The study provides a benchmark for training and testing the algorithms being developed ultimately to support instructors or raters. Also shown is that the automated method for scoring summary content closely replicates hand scoring and supports feedback given via the rubric. This supports previous work demonstrating that automated methods for applying analytic rubrics can reduce the demands on instructors' time, and be used fruitfully to aid students in revising written work. The article goes further to suggest that the structure and semantics of arguments together could provide a more reliable prediction of the argument quality.

Source-based writing draws on reading comprehension as well as on writing skills, competencies which complement each other but require different kinds of instruction [72]. The results of the automated analysis of the student summaries show that the automated summary analysis performs well and could, therefore, be used by instructors to provide feedback on students' understanding of sources. The same features that have proved useful for automated analysis of argument in

previous work [73] are shown to be the most predictive of the feature sets used here, in the context of undergraduate writing courses designed for STEM students. The work on integrating automated assessment of argumentation and subject matter content is already in progress.

Teaching and assessment vary greatly across HEIs. As these institutions seek better ways of engaging students, the need for more personalized forms of assessment continues to gain relevance. The availability of 21st century educational technologies makes it possible to support new pedagogical approaches through automation; at least for transparency, uniformity, and competence. Although automated scoring is still subject to much debate, what is being advocated here is automation with human intervention; automation that is designed with student-centered learning in mind.

ACKNOWLEDGMENT

The authors would like to thank the two NSF-funded REUs, Connor Heaton and Annie Qin Sui, who participated in the reliability studies and other project activities. Thanks also to the reviewers for the careful review and helpful feedback.

REFERENCES

- [1] S. Cottrell, *Critical Thinking Skills: Effective Analysis, Argument and Reflection*. London, U.K.: Macmillan Int. High. Educ., 2017.
- [2] D. F. Radcliffe, "Innovation as a meta-attribute for graduate engineers," *Int. J. Eng. Educ.*, vol. 21, no. 2, pp. 194–199, 2005.
- [3] C. S. Nair, A. Patil, and P. Mertova, "Re-engineering graduate skills—A case study," *Eur. J. Eng. Educ.*, vol. 34, no. 2, pp. 131–139, 2009.
- [4] N. Gibb, *Reading: The Next Steps: Supporting Higher Standards in Schools*. London, U.K.: Dept. Educ., 2015.
- [5] D. Reilly, D. L. Neumann, and G. Andrews, "Gender differences in reading and writing achievement: Evidence from the national assessment of educational progress (NAEP)," *Amer. Psychol.*, vol. 74, no. 4, pp. 445–458, 2019.
- [6] L. A. Schindler, G. J. Burkholder, O. A. Morad, and C. Marsh, "Computer-based technology and student engagement: A critical review of the literature," *Int. J. Educ. Technol. High. Educ.*, vol. 14, no. 1, pp. 1–28, 2017.
- [7] J. T. E. Richardson, J. B. Slater, and J. Wilson, "The national student survey: Development, findings and implications," *Stud. High. Educ.*, vol. 32, no. 5, pp. 557–580, 2007.
- [8] J. Dewey, "Experience and education," *Educ. Forum*, vol. 50, no. 3, pp. 241–252, 1986.
- [9] L. S. Vygotsky, *Mind in Society: The Development of Higher Psychological Processes*. Cambridge, MA, USA: Harvard Univ. Press, 1980.
- [10] D. A. Kolb, *Experiential Learning: Experience as the Source of Learning and Development*. Englewood Cliffs, NJ, USA: Prentice-Hall, 1984.
- [11] G. Harfitt, "Community-based experiential learning in teacher education," in *Oxford Research Encyclopedia of Education*. Oxford, U.K.: Oxford Univ. Press, 2019.
- [12] Department for Business, Innovation and Skills. *National Strategy for Access and Student Success*. Accessed: Sep. 23, 2020. [Online]. Available: <https://www.gov.uk/government/publications/national-strategy-for-access-and-student-success>
- [13] S. Marginson, "Global trends in higher education financing: The United Kingdom," *Int. J. Educ. Develop.*, vol. 58, pp. 26–36, Jan. 2018.
- [14] J. Heywood, *Assessment in Higher Education: Student Learning, Teaching, Programmes and Institutions*, vol. 56. London, U.K.: Jessica Kingsley Publ., 2000.
- [15] J. C. Archer, "State of the science in health professional education: Effective feedback," *Med. Educ.*, vol. 44, no. 1, pp. 101–108, 2010.
- [16] D. Boud and E. Molloy, "Rethinking models of feedback for learning: The challenge of design," *Assess. Eval. High. Educ.*, vol. 38, no. 6, pp. 698–712, 2013.
- [17] I. I. Bejar, "Rater cognition: Implications for validity," *Educ. Meas. Issues Pract.*, vol. 31, no. 3, pp. 2–9, 2012.

- [18] X. Zhai, K. C. Haudek, M. A. M. Stuhlsatz, and C. Wilson, "Evaluation of construct-irrelevant variance yielded by machine and human scoring of a science teacher PCK constructed response assessment," *Stud. Educ. Eval.*, vol. 67, Dec. 2020, Art. no. 100916.
- [19] C. Williams and S. Beam, "Technology and writing: Review of research," *Comput. Educ.*, vol. 128, pp. 227–242, Jan. 2019.
- [20] C. Strobl *et al.*, "Digital support for academic writing: A review of technologies and pedagogies," *Comput. Educ.*, vol. 131, pp. 33–48, Apr. 2019.
- [21] D. Grimes and M. Warschauer, "Utility in a fallible tool: A multi-site case study of automated writing evaluation," *J. Technol. Learn. Assess.*, vol. 8, no. 6, pp. 4–44, 2010.
- [22] A. C. Graesser, D. S. McNamara, M. M. Louwerse, and Z. Cai, "Cohmetrix: Analysis of text on cohesion and language," *Behav. Res. Methods Instrum. Comput.*, vol. 36, pp. 193–202, May 2004.
- [23] M. Heilman and N. Madnani, "ETS: Domain adaptation and stacking for short answer scoring," in *Proc. 7th Int. Workshop Semantic Eval. (SemEval)*, Atlanta, GA, USA, Jun. 2013, pp. 275–279. [Online]. Available: <https://www.aclweb.org/anthology/S13-2046>
- [24] R. Olmos, G. Jorge-Botana, J. A. León, and I. Escudero, "Transforming selected concepts into dimensions in latent semantic analysis," *Discourse Process.*, vol. 51, nos. 5–6, pp. 494–510, 2014.
- [25] A. Lachner, C. Burkhart, and M. Nückles, "Mind the gap! automated concept map feedback supports students in writing cohesive explanations," *J. Exp. Psychol. Appl.*, vol. 23, no. 1, pp. 29–46, 2017.
- [26] E. B. Page, "Project essay grade: PEG," in *Automated Essay Scoring: A Cross-Disciplinary Perspective*, M. D. Shermis and J. C. Burstein, Eds. Mahwah, NJ, USA: Lawrence Erlbaum Assoc. Publ., 2003, pp. 43–54.
- [27] M. Liu, Y. Li, W. Xu, and L. Liu, "Automated essay feedback generation and its impact on revision," *IEEE Trans. Learn. Technol.*, vol. 10, no. 4, pp. 502–513, Oct./Dec. 2017.
- [28] J. Park and K. Cho, "Toward the integration of peer reviewing and computational linguistics approaches," *J. Educ. Comput. Res.*, vol. 55, no. 1, pp. 123–144, 2016.
- [29] J. Wilson and R. Roscoe, "Automated writing evaluation and feedback: Multiple metrics of efficacy," *J. Educ. Comput. Res.*, vol. 58, no. 1, pp. 87–125, 2019.
- [30] D. Perin and M. Lauterbach, "Assessing text-based writing of low-skilled college students," *Int. J. Artif. Intell. Educ.*, vol. 28, no. 1, pp. 56–78, 2018.
- [31] J. Wilson and A. Czik, "Automated essay evaluation software in English language arts classrooms: Effects on teacher feedback, student motivation, and writing quality," *Comput. Educ.*, vol. 100, pp. 94–109, Sep. 2016.
- [32] T. Sladoljev-Agejev and J. Šnajder, "Using analytic scoring rubrics in the automatic assessment of college-level summary writing tasks in L2," in *Proc. 8th Int. Joint Conf. Nat. Lang. Process. (IJCNLP)* vol. 2, Taipei, Taiwan, 2017, pp. 181–186. [Online]. Available: <https://www.aclweb.org/anthology/I17-2031>
- [33] Z. Rahimi, D. Litman, R. Correnti, E. Wang, and L. C. Matsumura, "Assessing students' use of evidence and organization in response-to-text writing: Using natural language processing for rubric-based automated scoring," *Int. J. Artif. Intell. Educ.*, vol. 27, no. 4, pp. 694–728, 2017.
- [34] R. Olmos, G. Jorge-Botana, J. M. Luzón, J. I. Martín-Cordero, and J. A. León, "Transforming LSA space dimensions into a rubric for an automatic assessment and feedback system," *Inf. Process. Manage.*, vol. 52, no. 3, pp. 359–373, 2016.
- [35] T. K. Landauer and S. T. Dumais, "A solution to Plato's problem: The latent semantic analysis theory of acquisition, induction, and representation of knowledge," *Psychol. Rev.*, vol. 104, no. 2, pp. 211–240, 1997.
- [36] M. S. Lanchos, M. Hernández, Á. S.-E. Paniagua, J. M. L. Encabo, and G. de Jorge-Botana, "Using semantic technologies for formative assessment and scoring in large courses and MOOCs," *J. Interact. Media Educ.*, vol. 2018, no. 1, p. 12, 2018, doi: [10.5334/jime.468](https://doi.org/10.5334/jime.468).
- [37] J. D. Novak and A. J. Cañas, "Theoretical origins of concept maps, how to construct them, and uses in education," *Reflecting Educ.*, vol. 3, no. 1, pp. 29–42, 2007.
- [38] Y.-T. Sung, C.-N. Liao, T.-H. Chang, C.-L. Chen, and K.-E. Chang, "The effect of online summary assessment and feedback system on the summary writing on 6th graders: The LSA-based technique," *Comput. Educ.*, vol. 95, pp. 1–18, Apr. 2016.
- [39] A. Lachner, C. Burkhart, and M. Nückles, "Formative computer-based feedback in the university classroom: Specific concept maps scaffold students' writing," *Comput. Human Behav.*, vol. 72, pp. 459–469, Jul. 2017.
- [40] L. Gerard, A. Kidron, and M. Linn, "Guiding collaborative revision of science explanations," *Int. J. Comput. Supported Collab. Learn.*, vol. 14, pp. 291–324, May 2019.
- [41] L. F. Gerard and M. C. Linn, "Using automated scores of student essays to support teacher guidance in classroom inquiry," *J. Sci. Teach. Educ.*, vol. 27, no. 1, pp. 111–129, 2016.
- [42] UNESCO. *Recommendation on Adult Learning and Education*. Accessed: Aug. 15, 2019. [Online]. Available: <https://unesdoc.unesco.org/ark:/48223/pf0000245179>
- [43] E. D. Turley and C. W. Gallagher, "On the 'uses' of rubrics: Reframing the great rubric debate," *English J.*, vol. 97, no. 4, pp. 87–92, 2008.
- [44] A. Jonsson and G. Svingby, "The use of scoring rubrics: Reliability, validity and educational consequences," *Educ. Res. Rev.*, vol. 2, no. 2, pp. 130–144, 2007.
- [45] J. Trace, V. Meier, and G. Janssen, "'I can see that': Developing shared rubric category interpretations through score negotiation," *Assessing Writ.*, vol. 30, pp. 32–43, Oct. 2016.
- [46] T. H. Sundeen, "Instructional rubrics: Effects of presentation options on writing quality," *Assessing Writ.*, vol. 21, pp. 74–88, Jul. 2014.
- [47] R. P. Ferretti, C. A. MacArthur, and N. S. Dowdy, "The effects of an elaborated goal on the persuasive writing of students with learning disabilities and their normally achieving peers," *J. Educ. Psychol.*, vol. 92, no. 4, pp. 694–702, 2000.
- [48] H. A. Gallagher, N. Arshan, and K. Woodworth, "Impact of the national writing project's college-ready writers program in high-need rural districts," *J. Res. Educ. Effect.*, vol. 10, no. 3, pp. 570–595, 2017.
- [49] M. J. Capp, "The effectiveness of universal design for learning: A meta-analysis of literature between 2013 and 2016," *Int. J. Inclusive Educ.*, vol. 21, no. 8, pp. 791–807, 2017.
- [50] S. Black, *Crack the Essay: Secrets of Argumentative Writing Revealed*, Gramercy House Publ., 2018.
- [51] S. Graham and D. Perin, "A meta-analysis of writing instruction for adolescent students," *J. Educ. Psychol.*, vol. 99, pp. 445–476, Aug. 2007.
- [52] T. Eckes, *Introduction to Many-Facet Rasch Measurement: Analyzing and Evaluating Rater-Mediated Assessment*. Frankfurt, Germany: Peter Lang, 2011.
- [53] D. Freedman, R. Pisani, and R. Purves, *Statistics*, 4th ed. New York, NY, USA: W. W. Norton, 2007.
- [54] D. Perin, R. H. Bork, S. T. Peverly, and L. H. Mason, "A contextualized curricular supplement for developmental reading and writing," *J. Coll. Read. Learn.*, vol. 43, no. 2, pp. 8–38, 2013.
- [55] A. L. Brown and J. D. Day, "Macrorules for summarizing texts: The development of expertise," *J. Verb. Learn. Verb. Behav.*, vol. 22, pp. 1–14, Feb. 1983.
- [56] R. E. Johnson, "Recall of prose as a function of the structural importance of the linguistic units," *J. Verb. Learn. Verb. Behav.*, vol. 9, no. 1, pp. 12–20, 1970.
- [57] R. J. Passonneau, A. Poddar, G. Gite, A. Krivokapic, Q. Yang, and D. Perin, "Wise crowd content assessment and educational rubrics," *Int. J. Artif. Intell. Educ.*, vol. 28, no. 1, pp. 29–55, 2018.
- [58] A. Nenkova, R. Passonneau, and K. McKeown, "The pyramid method: Incorporating human content selection variation in summarization evaluation," *ACM Trans. Speech Lang. Process.*, vol. 4, no. 2, p. 4, May 2007.
- [59] R. J. Passonneau, "Formal and functional assessment of the pyramid method for summary content evaluation," *Nat. Lang. Eng.*, vol. 16, no. 2, pp. 107–131, Apr. 2010.
- [60] Y. Gao, C. Sun, and R. J. Passonneau, "Automated pyramid summarization evaluation," in *Proc. 23rd Conf. Comput. Nat. Lang. Learn. (CoNLL)*, Hong Kong, Nov. 2019, pp. 404–418. [Online]. Available: <https://www.aclweb.org/anthology/K19-1038>
- [61] W. C. Mann and S. A. Thompson, "Rhetorical structure theory: Toward a functional theory of text organization," *Text Interdiscipl. J. Study Discourse*, vol. 8, no. 3, pp. 243–281, 1988.
- [62] D. Marcu, "The rhetorical parsing, summarization, and generation of natural language texts," Ph.D. dissertation, Dept. Comput. Sci., Univ. Toronto, Toronto, ON, Canada, 1997.
- [63] Y. Gao, P. M. Davies, and R. J. Passonneau, "Automated content analysis: A case study of computer science student summaries," in *Proc. 13th Workshop Innovat. Use NLP Build. Educ. Appl.*, New Orleans, LA, USA, Jun. 2018, pp. 264–272. [Online]. Available: <https://www.aclweb.org/anthology/W18-0531>
- [64] C. Stab and I. Gurevych, "Annotating argument components and relations in persuasive essays," in *Proc. 25th Int. Conf. Comput. Linguist. (COLING)*, 2014, pp. 1501–1510.

- [65] C. Hidey, E. Musi, A. Hwang, S. Muresan, and K. McKeown, "Analyzing the semantic types of claims and premises in an online persuasive forum," in *Proc. 4th Workshop Argument Min.*, Sep. 2017, pp. 11–21.
- [66] D. Ghosh, A. Khanam, Y. Han, and S. Muresan, "Coarse-grained argumentation features for scoring persuasive essays," in *Proc. 54th Annu. Meeting Assoc. Comput. Linguist.*, 2016, pp. 549–554.
- [67] N. Farra, S. Somasundaran, and J. Burstein, "Scoring persuasive essays using opinions and their targets," in *Proc. 10th Workshop Innovat. Use NLP Build. Educ. Appl.*, Denver, CO, USA, Jun. 2015, pp. 64–74. [Online]. Available: <http://www.aclweb.org/anthology/W15-0608>
- [68] M. Chodorow and J. Burstein, "Beyond essay length: Evaluating e-rater®'s performance on toefl® essays," ETS Res. Rep. Series, Princeton, NJ, USA, Rep. RR-04-04, 2004.
- [69] B. B. Klebanov, C. Stab, J. Burstein, Y. Song, B. Gyawali, and I. Gurevych, "Argumentation: Content, structure, and relationship with essay quality," in *Proc. 3rd Workshop Argument Min. (ArgMining)*, 2016, pp. 70–75.
- [70] C. Brooks, A. Carroll, R. M. Gillies, and J. Hattie, "A matrix of feedback for learning," *Aust. J. Teach. Educ.*, vol. 44, no. 4, p. 2, 2019.
- [71] A. Wigfield, "Expectancy-value theory of achievement motivation: A developmental perspective," *Educ. Psychol. Rev.*, vol. 6, no. 1, pp. 49–78, 1994.
- [72] J. Fitzgerald and T. Shanahan, "Reading and writing relations and their development," *Educ. Psychol.*, vol. 35, no. 1, pp. 39–50, 2000.
- [73] Y. Gao *et al.*, "Rubric reliability and annotation of content and argument in source-based argument essays," in *Proc. 14th Workshop Innovat. Use NLP Build. Educ. Appl.*, 2019, pp. 507–518. [Online]. Available: <https://www.aclweb.org/anthology/W19-4452>

Patricia Marybelle Davies (Member, IEEE) received the Dr.Edu. degree in educational technology from the University of Manchester, Manchester, U.K., in 2013.

She is an Associate Professor with the College of Science and Human Studies, Prince Mohammad bin Fahd University, Al Khobar, Saudi Arabia. Her research critically examines educational technology applications for advancing student learning.

Dr. Davies was awarded the Google Educator Grants from 2016 to 2018. She is a Fellow of the Higher Education Academy. She is Co-Convenor for the Educational Technology SIG of the British Educational Research Association and has membership of other professional bodies.

Rebecca Jane Passonneau received the Dr.Phil. degree from the Department of Linguistics, University of Chicago, Chicago, IL, USA, in 1985.

She joined the Department of Computer Science and Engineering with Penn State University, Pennsylvania, PA, USA, in 2016. Her work is reported in over 120 publications in journals and refereed conference proceedings, and has been funded through nearly 30 sponsored projects, from 14 sources, including government agencies, corporate sponsors, corporate gifts, and foundations. Her main area of research is natural language processing, which she has pursued at many academic and industry research labs.

Dr. Passonneau is a member of many professional organizations, and is currently on the editorial board of the *International Journal of Artificial Intelligence in Education*, and other academic journals.

Smaranda Muresan received the Dr.Phil. degree in computer science from Columbia University, Columbia University, New York, NY, USA, in 2006.

She is a Research Scientist with the Data Science Institute, Columbia University. From 2008 to 2013, she was a Faculty Member with the School of Communication and Information, Rutgers University, New Brunswick, NJ, USA, where she co-founded the Laboratory for the Study of Applied Language Technologies and Society and also she received the Distinguished Achievements in Research Award. Her research, funded by NSF, DARPA, and IARPA, has led to over 80 publications. Her research expertise is natural language processing (NLP), focusing on argument mining and persuasion, figurative language understanding and generation, and NLP applications in education and public health.

Dr. Muresan is a Board Member of the North American Association for Computational Linguistics from 2020 to 2021. She co-organized Workshops on Argument Mining and Figurative Language Processing at NAACL/ACL. She is the Co-Chair of the series of New York Academy of Sciences' Symposia on Natural Language, Dialog, and Speech.

Yanjun Gao received the Dr.Phil. degree from the Department of Computer Science and Engineering, Penn State University, Pennsylvania, PA, USA, in 2021.

She is currently a Postdoctoral Research Associate with the University of Wisconsin-Madison, Madison, WI, USA. From 2017 to 2021, she was a Research Assistant with NLP Lab, Penn State University. Her research focuses on developing AI applications for education using NLP techniques, including proposition identification, semantic representation for discourse, and summarization evaluation.

Dr. Gao recently served on the committee for the China National Conference on Chinese Computational Linguistics (CCL, 2020) and 1st Workshop in NLP Evaluation and Comparison (Eval4NLP, 2020).

EduPlanner: LLM-Based Multi-Agent Systems for Customized and Intelligent Instructional Design

Xueqiao Zhang[†], Chao Zhang[†], Jianwen Sun, Jun Xiao, Yi Yang, and Yawei Luo[‡]

Abstract—Large Language Models (LLMs) have significantly advanced smart education in the Artificial General Intelligence (AGI) era. A promising application lies in the automatic generalization of instructional design for curriculum and learning activities, focusing on two key aspects: (1) Customized Generation: generating niche-targeted teaching content based on students’ varying learning abilities and states, and (2) Intelligent Optimization: iteratively optimizing content based on feedback from learning effectiveness or test scores. Currently, a single large LLM cannot effectively manage the entire process, posing a challenge for designing intelligent teaching plans. To address these issues, we developed EduPlanner, an LLM-based multi-agent system comprising an evaluator agent, an optimizer agent, and a question analyst, working in adversarial collaboration to generate customized and intelligent instructional design for curriculum and learning activities. Taking mathematics lessons as our example, EduPlanner employs a novel Skill-Tree structure to accurately model the background mathematics knowledge of student groups, personalizing instructional design for curriculum and learning activities according to students’ knowledge levels and learning abilities. Additionally, we introduce the CDDP, an LLM-based five-dimensional evaluation module encompassing Clarity, Integrity, Depth, Practicality, and Pertinence, to comprehensively assess mathematics lesson plan quality and bootstrap intelligent optimization. Experiments conducted on the GSM8K and Algebra datasets demonstrate that EduPlanner excels in evaluating and optimizing instructional design for curriculum and learning activities. Ablation studies further validate the significance and effectiveness of each component within the framework. Our code is publicly available at https://github.com/Zc0812/Edu_Planner.

Index Terms—Large Language Models, Multiple Agents, Smart Education, Instructional Design, Intelligent Agent.

I. INTRODUCTION

IN the educational realm, formulating pedagogical tasks presents a challenge that involves numerous scenarios and test cases. The crafting of instructional design for curriculum and learning activities [1], [2] is an essential but difficult component of these tasks. It requires a considerable investment of time and energy from educators to understand the diverse

Key Points in Instructional Design for Curriculum and Learning Activities

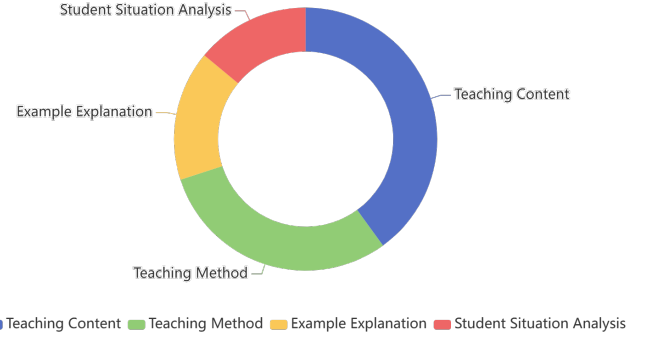


Fig. 1. The proportion of people who consider various factors critical in the task of generating lesson plans. Taking mathematics lessons as our example, in EduPlanner, we model the **student situation** using a newly proposed Skill-Tree ability module. We iteratively evaluate and optimize the **teaching method** and **teaching content** through the adversarial collaboration between the Evaluator Agent and the Optimizer Agent. Additionally, an Analyst Agent is employed to analyze **error-prone examples** and incorporate them into the subsequent lesson plan. EduPlanner effectively addresses the most pressing challenges in lesson planning.

learning profiles of their students before making an instructional design for curriculum and learning activities, ensuring a comprehensive structure, clarity of content, the inclusion of appropriate introductions and illustrative examples, as well as reflecting on and optimizing teaching behavior after class. To explore the pedagogical priorities that shape teachers’ instructional designs, Wu *et al.* [3] revealed that 40.2% educators consider teaching content the paramount aspect, 29.4% place heightened emphasis on the application of teaching methods, and 16.3% focus on example problem explanations. The remaining 14.1% focus on an in-depth analysis of the student situation, as depicted in Fig. 1. Traditionally, teachers have relied on their accumulated experience and innate intuition to devise instructional frameworks. This process often necessitates a trial-and-error approach, with iterative improvements made through classroom implementation. Moreover, the diversity of student dispositions and the unpredictability of the classroom environment compound the difficulty of crafting comprehensive instructional design for curriculum and learning activities prior to instruction. The absence of a standard metric to evaluate the quality of these plans further exacerbates the complexity of this task, rendering the entire procedure laborious and often inefficient.

The emerging trend in Generative AI, distinguished by its ability to generate a diverse range of content, including

[†] Equal contribution.

[‡] Corresponding author: Yawei Luo (yaweiluo@zju.edu.cn).

This work was supported by the National Natural Science Foundation of China (62293554, U2336212), National Key R&D Program of China (SQ2023AAA01005), Zhejiang Provincial Natural Science Foundation of China (LZ24F020002), Ningbo Innovation “Yongjiang 2035” Key Research and Development Programme (2024Z292), and Young Elite Scientists Sponsorship Program by CAST(2023QNRC001).

X. Zhang, C. Zhang and Y. Luo are with the School of Software Technology, Zhejiang University, Ningbo, 315048, China.

J. Sun is with the Faculty of Artificial Intelligence in Education, Central China Normal University, Wuhan, 430079, China

J. Xiao and Y. Yang are with the College of Computer Science and Technology, Zhejiang University, Hangzhou, 310007, China.

texts, audio, images, and videos, has gradually penetrated the realm of education [4]–[9], thus influencing various aspects of traditional pedagogy. Recent advances in GAI, such as LLM, have positioned them as promising solutions for instructional design [10], [11]. LLMs have the ability to develop sophisticated virtual environments and analyze complex social dynamics [12]–[14]. By observing agent responses and metric scores, researchers can gain insight into the effectiveness of various strategic approaches within instructional design. However, existing zero-shot instructional design generation methods [15], which depend entirely on the capabilities of LLM, often simply follow instructions to perform tasks without addressing the individual needs of students. In addition, the existing methods [16] only concentrate on the generation and evaluation of instructional designs, yet they fail to incorporate the application of LLM in the optimization process of instructional designs. Consequently, these instructional designs tend to be dull in knowledge transmission, deficient in case explanations, and devoid of a setting in course difficulty levels.

To address these challenges, this paper introduces “EduPlanner”, an automated multi-agent system designed for the generation, evaluation, and optimization of instructional design. Taking mathematics lessons as our example, “EduPlanner” enables teachers to employ a variety of mathematical instructional design drafts, simulate classroom scenarios, and make targeted adjustments to instructional strategies based on feedback. The system comprises an evaluator agent, an optimizer agent, and a question analyst, which collaborate adversarially to generate customized and intelligent instructional design for curriculum and learning activities.

“EduPlanner” begins by assessing the overall learning situation of the class using an innovative Skill-Tree structure, which informs the creation of teaching resources that align with individual learning trajectories, thus improving instructional efficacy and student engagement. Based on personal evaluation of the class, “EduPlanner” generates an initial instructional design for curriculum and learning activities and forwards it to the evaluator agent. The evaluator agent employs a five-dimensional evaluation system comprising **Clarity**, **Integrity**, **Depth**, **Practicality**, and **Pertinence** (CIDPP). Feedback from the evaluator is then used by the optimizer agent to improve the instructional design and by the analyst agent to record typical examples. This adversarial cycle mimics a real teacher’s iterative process of instructional design for curriculum and learning activities.

We evaluate the effectiveness of our framework using CIDPP-based experiments on the GSM8K [17] and Algebra [18] datasets, as illustrated in Fig. 2. Compared to conventional methods, EduPlanner significantly improves the efficiency of instructional design for curriculum and learning activities, reduces pedagogical burdens, and facilitates precise alignment with the diverse educational needs of students.

The main contributions of this study can be summarized as follows:

- We introduce “EduPlanner”, a novel framework for customized and intelligent instructional design including three agents: Evaluator Agent, Optimizer Agent, and Analyst Agent.

- We propose an innovative Skill-Tree structure to craft a personalized learning profile for each student which provides corresponding input for instructional generators, enabling a focus on tailored student needs when structuring educational content.
- We design a comprehensive five-dimensional instructional design evaluation system (CIDPP), and our framework achieves state-of-the-art results in experiments utilizing this system.

II. RELATED WORK

A. Smart Education

Smart education, a modern campus feature, employs AI and technology to create an interactive learning environment and improve academic achievement [19]. Gao *et al.* [20] employ AI for speech recognition and customized learning. Almogren *et al.* [21] use ChatGPT to provide instant assistance, encourage independent learning, and provide prompt feedback. Saurav Shrestha *et al.* [22] present the creation process of the Social Presence-Enabled Augmented Reality (SPEAR) tool, an innovative augmented reality (AR) learning application designed specifically for online engineering education. Nicolas Pope *et al.* [23] address the limit of educational tools to teach novice K-12 learners about data-driven systems by introducing the “GenAI Teachable Machine,” a visual, data-driven design platform. CV Suresh and Bhavesh *et al.* [24] explore the transformative potential of AI-driven instructional design to enhance personalized learning experiences and improve educational efficiency.

B. Generative AI for Educational Transformation

GAI has attracted great attention with the advancement of LLMs. Within the educational domain, GPTeach [25] employs GPT to simulate student behaviors. EduAgent [10] excels in anticipating the detailed learning actions of students and generating plausible learning behaviors for virtual students without ground truth. AgentVerse [12] enables the development, training, and interaction of agents across various domains, simulating complex human decision-making processes. Gradescope [26] improves the assessment process for educators by organizing assessment tasks, increasing student engagement, and providing valuable data on student performance.

C. LLMs-based Multi-agent Systems

Recent studies have explored the use of LLMs in agent systems, demonstrating excellent performance in accomplishing tasks [14], [27] upon prompts through collaborative interactions among multiple agents. The concept of generative agents [28] involves the creation of a simulated town with 25 agents. The approach relies on multiple language models [29], [30], often proves inefficient, particularly in open-ended settings. Managing the control aspects of LLM agents, Zheng *et al.* [31] presents a challenge, yet this is where reinforcement learning [32], [33] excels. MetaGPT [34] uses standard operating procedures to enhance the multi-agent systems and emulates a software team with agents

Question: A brownie recipe is asking for 350 grams of sugar, and a pound cake recipe requires 270 more grams of sugar than a brownie recipe. How many grams of sugar are needed for the pound cake?

Final Answer: 620

Section: Translation

Question: Sheila and Lenore were driving to their grandmother's house. Lenore left one hour after Sheila. Sheila drove at a rate of 45 mph, and Lenore drove at a rate of 60 mph. How many hours will it take for Lenore to catchup to Sheila?

Final Answer: 3

Section: Solve Applications with Systems of Equations

Question: The perimeter of a rectangle is 84. The length is 10 more than three times the width. Find the length.

Final Answer: 34

Section: Solving Systems of Equations by Substitution

(a)

Question: Janet 2019s ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder at the farmers' market daily for \$2 per fresh duck egg. How much in dollars does she make every day at the farmers' market?

Solution: Janet sells $16 - 3 - 4 = 9$ duck eggs a day. She makes $9 * 2 = 18$ every day at the farmers' market.

Answer: 18

Question: Cynthia eats one serving of icecream every night. She buys cartons of icecream with 15 servings of icecream per carton at a cost of \$4.00 per carton. After 60 days, how much will she spend on icecream?

Solution: Each container of icecream has 15 servings and she eats 1 serving a night so after 60 days she will need $60/15 = 4$ containers of icecream. If each carton costs \$4.00 and she needs 4 containers then it will cost her $4 * 4 = 16.00$.

Answer: 16

(b)

Fig. 2. Examples of datasets. On the left is GSM8K [17] and on the right is Algebra [18]. GSM8K is a dataset consisting of 8.5K high-quality, language-diversified elementary school mathematics word problems released by OpenAI. Algebra is a dataset comprised of 222 examples related to algebraic equations.

including product manager, architect, project manager, engineer, and quality engineer. Autogen [35] offers designable conversable agents and is capable of integrating with external tools to facilitate collaborative conversations within multi-agent systems. AutoGPT [36] works based on established goals without human intervention and employs a real-time loop evaluation strategy to assess the completion of objectives dynamically. CAMEL [37] develops a role-playing framework that generates dialogue datasets to foster communication and collaboration within multi-agent systems.

D. Evaluation using LLMs

Most of the evaluations in relative experiments are open-ended and lack definitive reference answers, designing a rule-based program to evaluate the output of LLM is challenging. Traditional evaluation metrics based on the similarity between the output and the reference answers (e.g., ROUGE [38], BLEU [39]) also fail to address these questions. Employing experts for manual evaluation is time-consuming and expensive. Therefore, in order to facilitate an efficient automated evaluation process, some works explore LLMs as a substitute for human evaluation. These models, often trained via Reinforcement Learning from Human Feedback (RLHF), have demonstrated great alignment with humans. The LMSys [40] explores the merits and limitations of utilizing diverse LLMs as judges for tasks ranging from writing and mathematics to general world knowledge. The self-rewarding LLM [41] trains itself through the data generated by its own and evaluated by the GPT-4. CharacterLLM [42] employs GPT as an evaluator to access the LLMs in role-playing scenarios, and Neeko [43] evaluates the performance on role switching of LLM in multi-role-playing contexts based on GPT. L-eval [44] takes the pair-wise battle format and reports the win rate vs. LLMs, augmenting the judgment prompt to mitigate the preference for more detailed and lengthy answers.

III. METHODOLOGY

A. Overview

We present EduPlanner, a multi-agent-based system designed for the evaluation and optimization of instructional designs. The system employs a modular architecture, depicted in Fig. 3. To tailor instructional design for specific student groups, we model students' knowledge backgrounds using a Skill-Tree structure. This structure facilitates the generation and optimization of instructional designs that meet the diverse learning needs of students. Our approach includes three collaborative agents: the Evaluator Agent, which assesses the quality of instructional designs; the Optimizer Agent, which improves lesson content; and the Analyst Agent, which identifies common error points in example questions within the instructional designs. These agents are integrated using an adversarial learning framework. The Evaluator Agent provides feedback on the advantages and disadvantages of the instructional designs, guiding the Optimizer Agent and Analyst Agent. The evaluation scores serve as targets for optimization, driving an iterative adversarial process that enhances the instructional design for curriculum and learning activities. When optimization is complete, the instructional design with the highest score in the queue is output and its content mainly includes two parts: knowledge point explanation and question explanation.

B. Skill-Tree Structure

To generate customized instructional design for students with different knowledge backgrounds, a key issue to be addressed is the modeling of students' abilities. In CGMI [11], a character tree structure was designed to solve the problem of inconsistency of roles in LLM after multiple rounds of dialogue. Following CGMI [11], we choose to employ a tree structure to portray the competencies¹ of students. In a novel

¹We use mathematics as an example to illustrate the role of the Skill-Tree structure, while other subjects can also be applied.

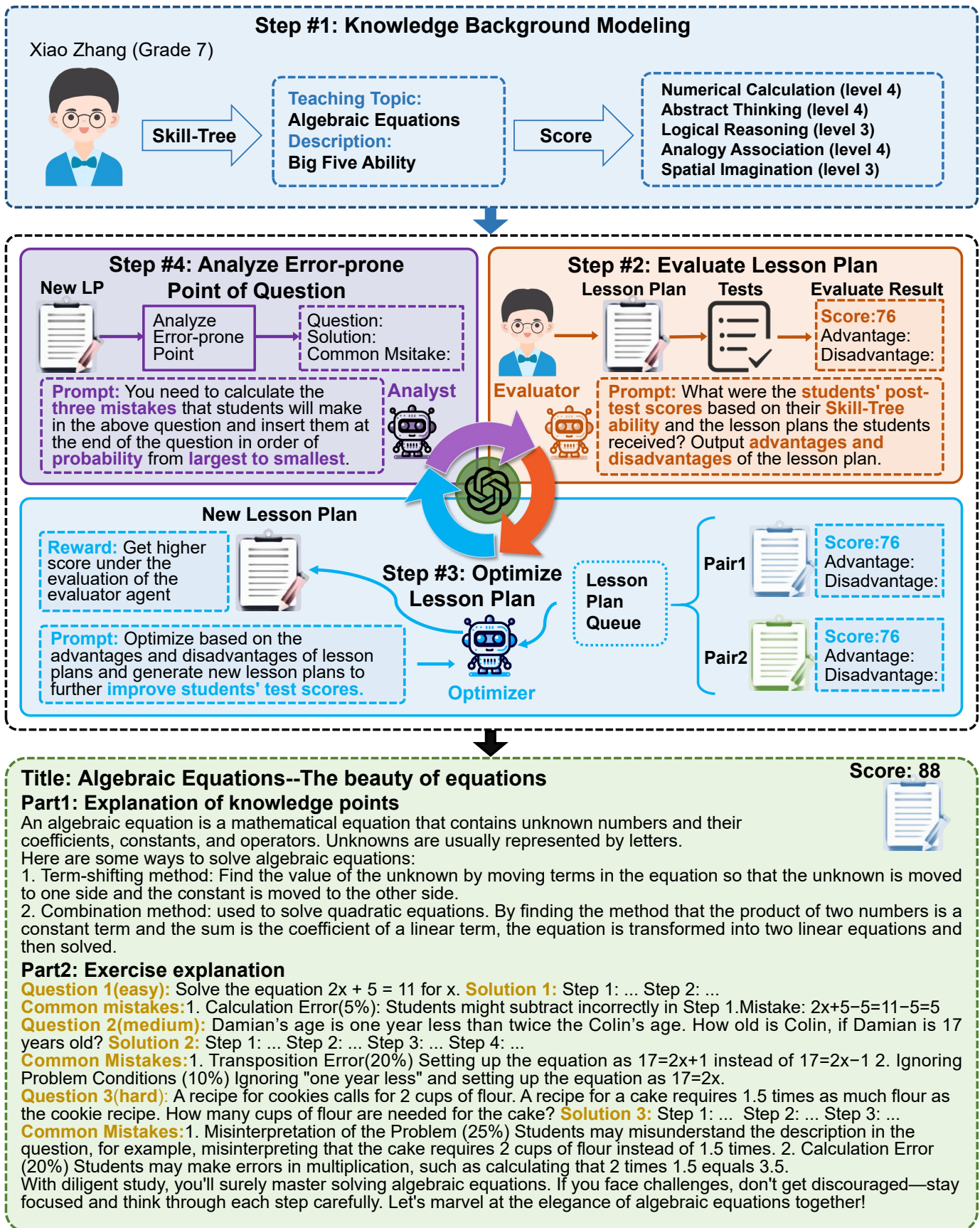


Fig. 3. EduPlanner: LLM-Multi-Agent-Based Instructional Design Generation Framework. When the optimization is completed, the instructional design for curriculum and learning activities with the highest score in the queue is output, and its content mainly includes two parts: knowledge point explanation and question explanation.

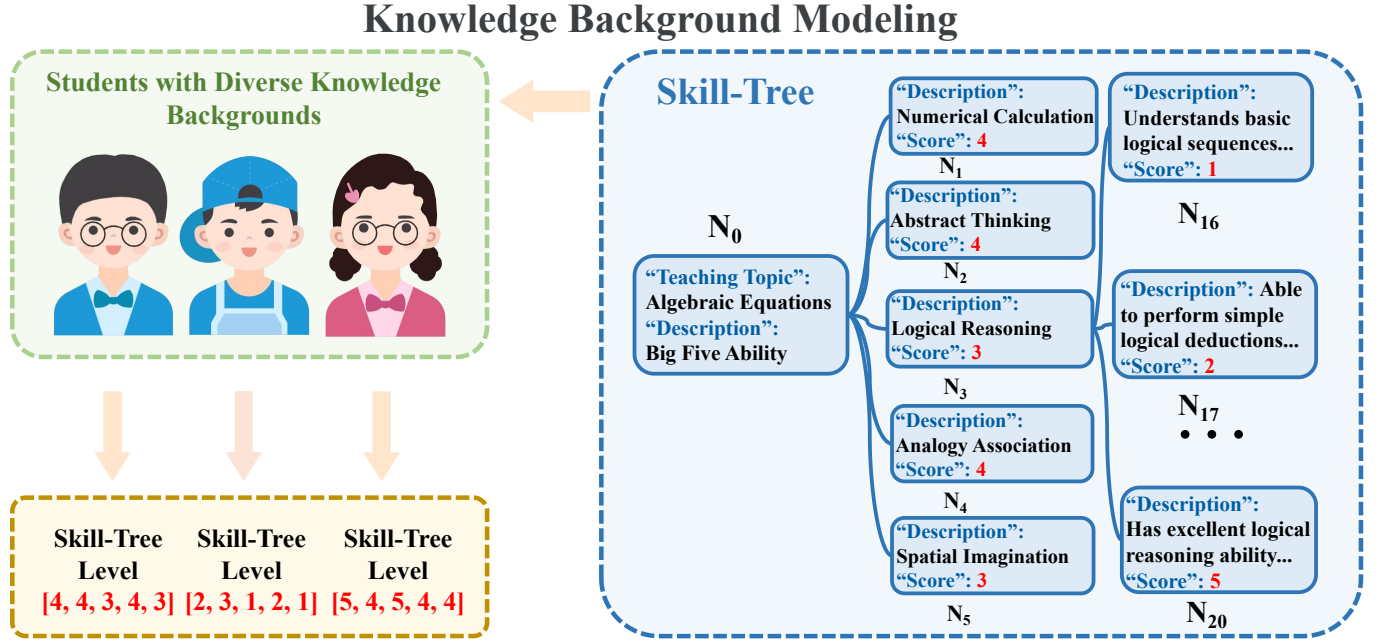


Fig. 4. Skill-Tree structure models the students' diverse knowledge background through five principal abilities, outputs the students' Skill-Tree level, and provides it to the agents so that the agents can capture the knowledge background of different students and ensure that targeted instructional design for curriculum and learning activities can be generated.

extension of this approach, we introduce five specialized sub-nodes under each primary node. This enhancement enables a detailed quantification of abilities across different dimensions, offering an in-depth description of the knowledge bases and learning abilities of individual students, as shown in Fig. 4. Compared to traditional binary evaluations (good or bad), this approach offers a more comprehensive analysis with multiple dimensions, enabling the evaluator agent to capture the differences in students' abilities.

Student Abilities (Mathematics as Example):

- Node N_0 : explains the content of the teaching topic and the description of nodes N_1 – N_5
- Node N_1 : represents Numerical Calculation ability, reflecting the accuracy and proficiency of students in addition, subtraction, multiplication, and division.
- Node N_2 : represents Abstract Thinking ability, one of the key factors in understanding and solving algebraic equations.
- Node N_3 : depicts Logical Reasoning ability, a core skill for solving mathematical problems.
- Node N_4 : represents Analogy Association ability, an important tool for students to expand knowledge boundaries.
- Node N_5 : refers to Spatial Imagination ability, which is particularly important for solving geometric problems in algebraic equations.

C. Instructional Design Evaluation

In educational practice, the effectiveness of instructional design is a challenging issue. Given the great performance on various tasks of LLM-Based-Agents, we are encouraged to utilize agents as experienced evaluation experts that provide

feedback on instructional designs, such as advantages, disadvantages, and suggestions for improvement.

1) **Evaluation Criteria:** To comprehensively evaluate instructional designs and catalyze intelligent optimization, we introduce the CDDP, an LLM-based five-dimensional evaluation module encompassing **Clarity**, **Integrity**, **Depth**, **Practicality**, and **Pertinence**. This evaluation system comprises the pedagogical philosophy in instructional design, with each dimension serving a distinct purpose:

Clarity is essential for making instructional designs clear and removing unnecessary information. It helps teachers establish clear teaching goals, ensuring that educational activities are aligned with pedagogical principles of purposeful instruction.

Integrity involves examining instructional designs to ensure they are detailed and complete, covering all necessary aspects like clear explanations of concepts and relevant examples. It aligns with the pedagogical philosophy that promotes a comprehensive and integrated approach to learning.

Depth measures how instructional designs engage students in deep thinking and present content that reveals the deep meanings and relationships within knowledge. It is in line with the pedagogical emphasis on fostering deep learning experiences.

Practicality assesses the real-world applicability of instructional designs and how well students can use their knowledge to solve practical problems. It corresponds with the pedagogical philosophy that focuses on applying learned knowledge.

Pertinence examines the efficacy of instructional designs in satisfying the diverse learning needs of students. It promotes the adaptation of different levels of knowledge and learning preferences, reflecting the personalized teaching approach cen-

Algorithm 1: Evaluator Agent Expert Evaluation (EAEE)

Input: Evaluator Agent $\mathcal{A}_e$, Initial Instruction Design $\mathcal{L}_{p_0}$, Knowledge Background, Evaluation Task, Test Questions, $\mathcal{X}_1, \mathcal{X}_2, \dots, \mathcal{X}_T$

Output: post score s , advantage a , disadvantage d

```

1 Set knowledge background || Instruction Design ||  $\mathcal{X}_1 \rightarrow \mathcal{P}_1$ 
2 Give  $\mathcal{P}_1$  as input to  $\mathcal{A}_e$  and receive output  $\mathcal{O}_1$ 
3 for  $i$  in  $T$  do
4   Set  $\mathcal{X}_i$  || evaluation task  $\rightarrow \mathcal{P}_i$ 
5   Give  $\mathcal{P}_i$  as input to  $\mathcal{A}_e$  and receive output  $\mathcal{O}_i$ 
6 Extract  $s_1, \dots, s_T$  from  $\mathcal{O}_1, \dots, \mathcal{O}_T$ 
7 Set  $\sum_i s_i / T$   $\rightarrow s$ 
8 Extract  $a_1, \dots, a_T, d_1, \dots, d_T$  from  $\mathcal{O}_1, \dots, \mathcal{O}_T$ 
9 Use  $\mathcal{A}_e$  to summarize  $a_1, \dots, a_T \rightarrow a, d_1, \dots, d_T \rightarrow d$ 
10 return  $(s, a, d)$ 

```

tral to modern pedagogical thought.

2) **Evaluator Agent:** The Evaluator Agent, built on the Meta-Llama-3-70B LLM [45], is designed to evaluate instructional designs and offer feedback. Trained on 100 carefully annotated instructional design for curriculum and learning activities, each assessed by education experts for effectiveness, scoring, advantages, and disadvantages, the Evaluator Agent demonstrates an evaluation capability comparable to that of human experts. The process involves analyzing students' Skill-Tree scores to measure their knowledge levels and accessing scores after instruction of instructional designs using randomly selected test questions. The agent provides detailed feedback that clearly points out the advantages and content that needs improvement, forming a crucial basis for the subsequent optimization of the instructional design.

3) **Instructional Design Evaluation Algorithm:** The Evaluator Agent assesses the quality of instructional design and provides feedback, denoted as $\mathbf{F}$ (e.g., advantages and disadvantages), to the Optimizer Agent. We refer to this algorithm as Evaluator Agent Expert Evaluation (EAEE) [15]. Specifically, we allow the Evaluator Agent understand students' background knowledge and learning abilities based on their Skill-Tree scores. Then, we provide the students with a specific instructional design for curriculum and learning activities and let the evaluator agent evaluate the students' average scores on exam questions. The specific mathematical Eq. 1 are as follows. The evaluator agent provides feedback on the instructional design for curriculum and learning activities. Alg. 1 details the EAEE process.

$$R(lp, S) = \frac{1}{|T|} \sum_{i=1}^{|T|} f(lp, S, x_i), \quad (1)$$

where x_i represents the i -th test question, lp is the instructional design to be evaluated, and S is the Skill-Tree score of the student, allowing the LLMs to capture differences in students' learning abilities. The function $f(\cdot)$ is the evaluation function of the Evaluator Agent. T is a hyperparameter we set, which represents the number of test questions used during an instructional design for evaluation. $R(\cdot)$ represents the average

test score evaluated by the Evaluator Agent for students with test questions.

The process of the EAEE algorithm is as follows. The student role description, instructional design, evaluation task description, and the first test question are combined to form the initial prompt input p_1 . The initial input p_1 is submitted to the Evaluator Agent, and its output o_1 is received, which includes predictions of the student's learning effectiveness. The process is repeated for the M test questions. Finally, the model extracts the predicted probabilities or scores for the student's learning success from each output and calculates the average. The advantages and disadvantages of the instructional design generated in each output are also extracted and summarized by $\mathcal{A}_e$. The summary is provided to the Optimizer Agent $\mathcal{A}_o$ as reference information for optimization of the instructional design.

D. Instructional Design Optimization

1) **Optimizer Agent:** The Optimizer Agent is designed to optimize personalized, high-quality instructional designs according to the students' knowledge backgrounds. The process involves using the feedback $\mathbf{F}$ provided by the Evaluator Agent (i.e., the advantages and disadvantages of the instructional design) for optimization. The goal is to maximize the instructional design's evaluation score assessed by the Evaluator Agent. Eq. 2 expresses this instructional design optimization strategy.

$$\begin{cases} F = f(lp, S) \\ lp_{new} = \arg \max_{R(lp_{new}, S)} \{g(D, F)\} \end{cases} \quad (2)$$

In this equation, F represents the feedback generated by the Evaluator Agent, the function $g(\cdot)$ is responsible for optimizing and generating a new instructional design using the feedback and the instructional designs in the queue D along with their scores, $R(\cdot)$ represents the evaluation score for the new instructional design. The final generated lp_{new} is optimized to maximize the instructional design's evaluation score.

Our Optimizer Agent employs a dynamically sorted instructional design queue to prioritize the most effective instructional design. After the Evaluator Agent evaluates the new instructional design, which involves generating scores and feedback, the Optimizer Agent inserts the instructional design for the curriculum and learning activities into the queue based on its score. The process of the instructional design queue is described by Eq. 3.

$$D_n = \text{Trim}(\text{Sort}(\text{Ins}(lp_n, D), R(lp_n, S)), P) \quad (3)$$

In Eq. 3, D denotes the instructional design queue, lp_n is the new instructional design, S represents the Skill-Tree scores of the students, P indicates the maximum capacity of the instructional design queue, $R(\cdot)$ represents the student's average test score, $\text{Ins}(\cdot)$ indicates inserting the new instructional design into the instructional design queue, $\text{Sort}(\cdot)$ means sorting the queue in ascending order, $\text{Trim}(\cdot)$ keeps only the top P instructional design.

Algorithm 2: Optimizer Agent Expert Optimization (OAEO)

Input: Evaluator Agent $\mathcal{A}_E$, Optimizer Agent $\mathcal{A}_O$, Question Analyst $\mathcal{A}_A$, Initial Instructional Design $\mathcal{L}_{p_0}$, Knowledge Background $\mathcal{K}$, Evaluation Task, Optimization Task, Test Questions $\mathcal{X}_1, \mathcal{X}_2, \dots, \mathcal{X}_T$, Optimization Rounds $\mathcal{N}$

Output: Instructional Design $\mathcal{L}_{p_0}, \mathcal{L}_{p_1}, \dots, \mathcal{L}_{p_{p-1}}$

- 1 Evaluate $\mathcal{L}_{p_0}$ using Algorithm 1 and receive score $\mathcal{R}_0$.
- 2 Set $\{(\mathcal{L}_{p_0}, \mathcal{R}_0)\} \rightarrow \mathcal{D}$.
- 3 Set knowledge background optimization task $\rightarrow \mathcal{P}_0$.
- 4 **for** i in $\mathcal{N}$ **do**
- 5 Set $[] \rightarrow \mathcal{D}'$.
- 6 **for** k in $\mathcal{K}$ **do**
- 7 Generate $\mathcal{C}_{n-1}^{\mathcal{K}+k}$ by feeding $\mathcal{P}_{n-1}$ into $\mathcal{A}_O$.
- 8 Insert q into $\mathcal{C}_{n-1}^{\mathcal{K}+k}$.
- 9 Evaluate $\mathcal{C}_{n-1}^{\mathcal{K}+k}$ using Algorithm 1 and receive score $\mathcal{R}_{n-1}^{\mathcal{K}+k}$.
- 10 Analyze error-prone points of questions in $\mathcal{C}_{n-1}^{\mathcal{K}+k}$ using Algorithm 3 and receive new $\mathcal{C}_{n-1}^{\mathcal{K}+k}$.
- 11 Set $\mathcal{D} \cup (\mathcal{C}_{n-1}^{\mathcal{K}+k}, \mathcal{R}_{n-1}^{\mathcal{K}+k}) \rightarrow \mathcal{D}$.
- 12 $\mathcal{D}.\text{append}(\mathcal{D}')$.
- 13 Sort $\mathcal{D}$ by scores in ascending order.
- 14 Set $\mathcal{D}[: \text{max length}] \rightarrow \mathcal{D}$.
- 15 Set knowledge background $\|\mathcal{D}\|$ optimization task $\rightarrow \mathcal{P}_n$.
- 16 **return** $\mathcal{D}$.

2) **Instructional Design Optimization Algorithm:** We refer to the instructional design optimization algorithm as Optimizer Agent Expert Optimization (OAEO) [15]. The agent optimizer ($\mathcal{A}_O$) and the agent example analyzer ($\mathcal{A}_A$) form the core of the optimization process. The student description, initial instructional design, evaluation task, optimization task, and the first test question are combined into the initial prompt (p_0). This prompt is submitted to the $\mathcal{A}_O$, which generates an optimized instructional design for curriculum and learning activities output (o_0), including specific suggestions for content improvement.

The process is repeated for each test question. The $\mathcal{A}_A$ extracts examples from each output, forming a set of examples. All suggestions and examples are combined into the final instructional design for curriculum and learning activities. Alg. 2 provides a detailed description of the optimization process.

E. Instructional Design Analysis

1) **Analyst Agent:** The Analyst Agent explains the common mistakes produced by questions in the instructional design to alert students. We compile 50 common mistakes in algebraic equations into five groups. For each example from Alg. 2, the Analyst Agent identifies three common error-prone points for students with corresponding knowledge backgrounds. These error points are inserted into the example explanations to help students avoid these errors.

2) **Error-prone Point Analysis Algorithm:** To identify common errors students may make, we develop the Analyst Agent Expert Analysis (AAEA) algorithm. The algorithm extracts examples from instructional designs. Utilizing annotated

Algorithm 3: Analyst Agent Expert Analysis (AAEA)

Input: Question Analyzer $\mathcal{A}_a$, Common Mistakes Database (CMD), Instructional Design (Lp), Example Questions (EQs) generated by Optimizer Agent, Knowledge Background (KB)

Output: new instructional designs with integrated error-prone points and probabilities(nlp)

- 1 $\mathcal{A}_a \leftarrow$ Initialize Analyst Agent
- 2 $EQs \leftarrow$ Retrieve example questions from Lp
- 3 Load CMD
- 4 **for** eq in EQs **do**
- 5 $KB \leftarrow$ Identify target student group knowledge background for eq
- 6 $CMs \leftarrow$ Retrieve top 3 common mistakes from CMD based on KB and mistake probability
- 7 **for** cm in CMs **do**
- 8 $mp \leftarrow$ Calculate mistake probability for eq
- 9 $me \leftarrow$ Generate explanation for cm
- 10 $CMs \leftarrow$ Sort CMs by mp in descending order
- 11 Insert CMs and me into eq section of Lp
- 12 **return** nlp

common errors as reference, the Analyst Agent generates potential errors that students might make for each example. These errors are inserted into the example explanations within the instructional design, serving as reminders to help students avoid these mistakes and correctly solve the problems. Alg. 3 provides a detailed description of this process.

IV. EXPERIMENTS

In this section, we verify the results of the generation framework. We propose an instructional design evaluation system (CIDPP) to analyze the quality of instructional designs. In addition, we conducted ablation experiments to demonstrate the roles of each module in the framework.

A. Hyperparameter Configuration

In this section, we tune the hyperparameters of each component to ensure optimal performance. For the evaluator agent's base model, *i.e.* Meta-Llama-3-70B-Instruct, we set the temperature parameter at 0.0 for stable output. The optimizer agent's base model, *i.e.* GPT-4, has a temperature parameter set at 1.0 for diverse instructional design content. The analyzer, also based on GPT-4, has a temperature parameter set at 0.7 for balance between stability and diversity.

B. Contrastive Results

To validate the efficacy of our framework, we compare our results with other methods by LLMs like GPT-3.5-turbo, Llama-3-70B-Instruct and GPT-4 as detailed in the Fig 5. For the direct generation of instructional designs through LLMs, we apply a prompt as below “Generate an instructional design with the teaching theme of algebraic equations. It contains two parts, the explanation of knowledge points and the explanation of examples. The explanation of knowledge points should be comprehensive, and the explanation of examples should be representative. The generated instructional designs must be concise and limited to 200 words.” Finally, we assess

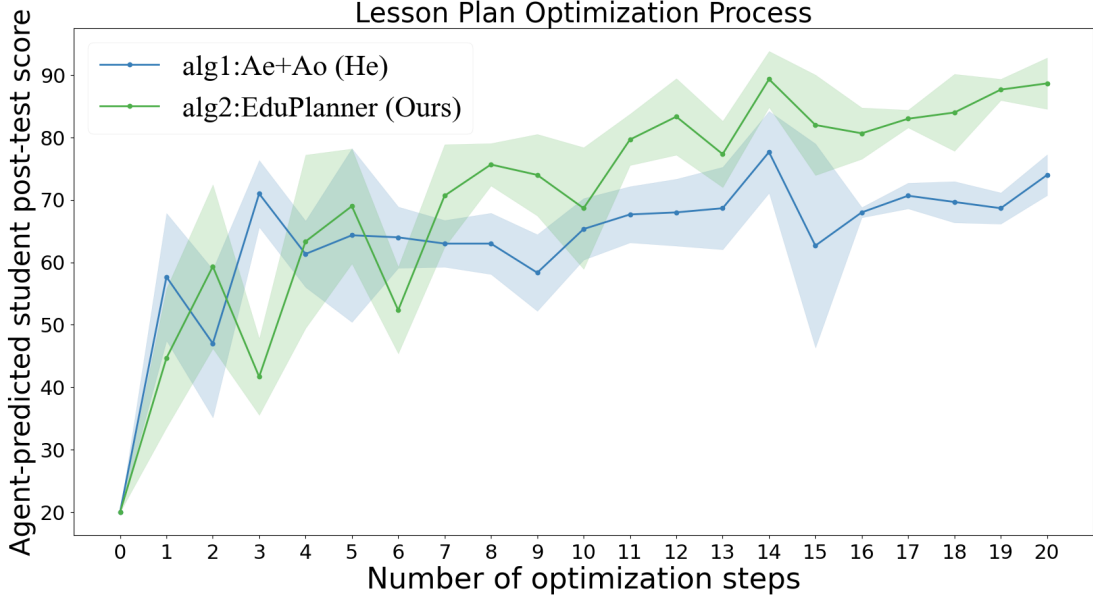


Fig. 5. In this figure, we obtained the optimization process curves of our framework and He’s method [15] under the premise that the hyperparameters are consistent with the initial instructional design for curriculum and learning activities. Our framework has a higher score in the instructional design than He’s [15] method, and the optimization process is smoother.

the instructional design based on the CIDPP as shown in Tab. I, where the results objectively demonstrate the effectiveness of our framework in improving the instructional designs. The prompt of evaluation is as follow:

“Role: You are an impartial evaluator, experienced in educational content analysis and instructional design evaluation.

Attention: You are responsible for assessing the quality of a given instructional design based on five specific evaluation criteria. Your evaluation should be objective and based solely on the Evaluation Standard provided below. **Lesson Plan:** {lessonplan}

Evaluation Standard: Clarity: The lesson plan’s directness and simplicity, ensuring it avoids unnecessary complexity and redundancy. Integrity: Whether the lesson plan is complete and systematic, covering both knowledge point explanations and exercise explanations in a complementary manner. Depth: The ability of the lesson plan to inspire deep thinking and facilitate understanding of the underlying connections between knowledge points. Practicality: The practical application value of the examples in the lesson plan, ensuring students can use the knowledge to solve real-life problems. Pertinence: The adaptability of the lesson plan to different students’ knowledge levels and learning needs to achieve optimal learning outcomes. **Constraints:** Avoid any bias in evaluation based on the content’s length or appearance. Be as objective as possible in assessing each aspect individually without favoring any specific structure or terminology. **Workflow:** Output your final verdict in the following format: “[A]: [points]; [short analyzes]”, “[B]: [points]; [short analyzes]”, “[C]: [points]; [short analyzes]”, “[D]: [points]; [short analyzes]”, “[E]: [points]; [short analyzes]”. Take a deep breath and think step by step!”

In our research, we introduce a novel framework designed to enhance the optimization of instructional design for curriculum and learning activities. Based on baseline [15], we preserve the

evaluator and optimizer agent, while integrating two crucial additions: a Skill-Tree structure and an analyst agent. As illustrated in Fig. 5, our method outperforms baseline [15] and exhibits a smoother process.

Furthermore, we conduct an ablation study on all components of our instructional design in an optimization framework. As shown in Tab. II and Fig. 6 and Fig. 7, we evaluate the individual impact of each component and measure its contribution to the optimization results using CIDPP.

These contrastive results emphasize that the integration of the evaluator agent, optimizer agent, analyst agent, Skill-Tree mechanisms, and CIDPP within our methodology improves the structural integrity of instructional design and enhances their quality and pedagogical resonance through personalized improvements. Our experiments have shown that utilizing a standalone evaluator agent or an optimizer agent does not produce the most favorable results. But integrating these agents establishes a great baseline, excelling in the dimensions of clarity, integrity, and depth. Moreover, the addition of an analyst agent has notably improved the practicality and pertinence of instructional design for curriculum and learning activities. Although the enhancement in clarity may not be pronounced, the Skill-Tree mechanism improves other performance metrics by creating a map of the student’s competencies and crafting educational content that is aligned with their individual knowledge background.

V. DISCUSSION

The advent of AGI has ushered a transformation across diverse sectors, presenting both opportunities and challenges for the field of education. It extends novel forms of support for teaching and learning, introducing innovative methodologies to educational research with a focus on multidisciplinary

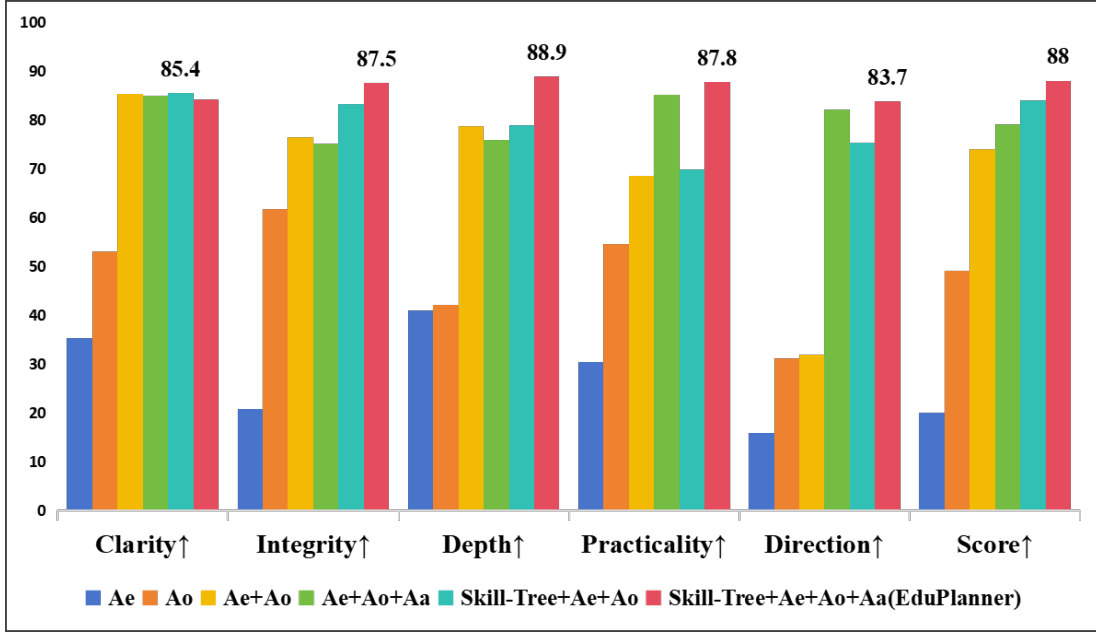


Fig. 6. The figure illustrates the influence of each framework component on each evaluation metric in CIDDP via a bar chart.

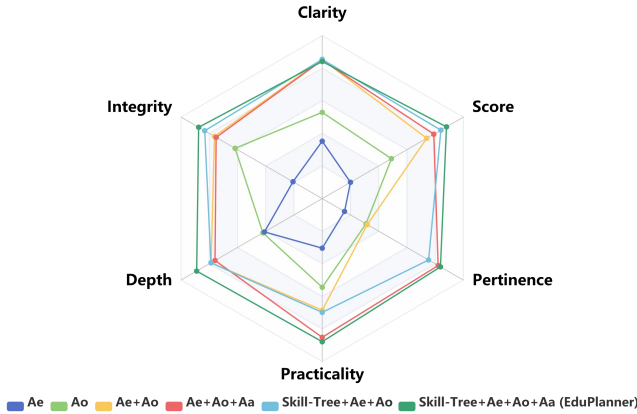


Fig. 7. The figure employs a radar chart to portray the incremental effects of each component.

integrated approaches like pedagogical [46] and AI. Smart education [47]–[50], identified as a pivotal trend in the future educational landscape, is obtaining increasing attention and importance. Central to this trend is personalized education, which disrupts traditional teacher-student roles by dynamically adjusting educational resources and teaching strategies to meet individual students' needs.

Developing instructional design for curriculum and learning activities is a crucial aspect of educational practice that must adapt to changing requirements. The traditional method of creating instructional designs is time-consuming for teachers, which involves understanding each student's individual characteristics and knowledge level to effectively align subject matter with their needs. As educational concepts evolve, previous teaching methods may no longer meet the needs of current students. Developing effective instructional design requires the

integration of clear assessment methods and feedback mechanisms for quick adjustments and improvements in teaching strategies, which still presents a challenge. Modern instructional design often involves using educational technologies like multimedia tools and online platforms. Educators face the added challenge of mastering these tools and integrating them into their teaching practices.

Considering the existing issues, our framework introduces an evaluative system for instructional designs, a Skill-Tree model for student ability, and a structure with three collaborative agents. These advancements facilitate a comprehensive evaluation and improvement of instructional design quality, customized to address individual student needs.

In our studies, we propose a novel evaluation system for instructional designs that considers clarity, integrity, depth, practicality, and pertinence. This evaluation system reflects the pedagogical principles in instructional design. The instructional design's clarity helps teachers establish clear teaching objectives, ensuring that teaching activities are purposeful and in line with goal-oriented teaching principles in pedagogical philosophy. Ensuring the integrity of instructional designs emphasizes essential elements such as detailed explanations of key concepts and relevant examples, aligning with the comprehensive learning principles promoted in educational philosophy. The depth of instructional designs measures how well the instructional design engages students in deep thinking and provides detailed explanations of the deep interrelations of knowledge, aligning with the pedagogical emphasis on deep-level learning. The practicality of instructional designs evaluates how effectively students can apply their knowledge to solve real-world problems. It corresponds with the pedagogical philosophy that emphasizes the application of knowledge to address practical issues. The pertinence focuses on tailoring instructional designs to meet students' varying

TABLE I
QUALITY INDICATORS

Strategy	Clarity↑	Integrity↑	Depth↑	Practicality↑	Pertinence↑	Score↑
GPT-3.5-turbo	28.2	28.1	33.6	33.8	10.0	17
Llama-3-70B-Instruct	35.2	20.7	40.9	30.4	15.8	20
GPT-4	52.9	61.7	42.0	54.4	31.0	49
EduPlanner	84.1	87.5	88.9	87.8	83.7	88

TABLE II
ABLATION STUDY ON THE GSM8K AND ALGEBRA DATASETS.

Strategy	Clarity↑	Integrity↑	Depth↑	Practicality↑	Pertinence↑	Score↑
Ae	35.2	20.7	40.9	30.4	15.8	20
Ao	52.9	61.7	42.0	54.4	31.0	49
Ae+Ao (He [15])	85.3	76.3	78.7	68.5	31.9	74
Ae+Ao+Aa	84.9	75.1	75.9	85.1	82.1	79
Skill-Tree+Ae+Ao	85.4	83.2	78.8	69.8	75.3	84
Skill-Tree+Ae+Ao+Aa (EduPlanner)	84.1	87.5	88.9	87.8	83.7	88

knowledge levels and learning requirements, aligning with the personalized teaching philosophy promoted by pedagogical philosophy.

VI. CONCLUSION

In this paper, we develop EduPlanner, a multi-agent system comprising three key intelligent agents: the Evaluator, the Optimizer, and the Example Analyzer, for the automated evaluation and optimization of instructional designs. To meet students with varying knowledge levels, we design a Skill-Tree structure to measure their knowledge bases, which allows the system to capture the differences in each student’s learning abilities and generate corresponding instructional designs. Through a series of experiments, we validate our framework’s superior performance over other methods. We also conducted ablation experiments to examine the roles of the Skill-Tree structure and the Analyst Agent. The results confirm that these novel components are crucial in generating remarkable instructional designs that show the transformative potential of AGI to address complex problems and complex challenges in the domain of education.

VII. LIMITATIONS AND FUTURE WORK

A. Limitations

While EduPlanner has achieved notable results in the generation and improvement of instructional design, it faces various limitations due to the constantly changing educational technology environment and the increasing diversity of teaching needs.

1) The skill tree is specifically tailored to generate instructional designs within certain subjects, which may limit its applicability to other disciplines.

2) EduPlanner’s reliance on LLM resources is marked by substantial consumption, which leads to a huge cost in terms of time and money.

3) There is an absence of adaptive integration with existing tools within practical teaching environments, which indicates a demand for ensuring alignment with current educational practices.

B. Future Work

GAI is revolutionizing instructional designs evaluation and optimization, aiming to better address the evolving needs of educational systems [51]–[55]. Despite current challenges, continuous improvements in these systems are expected to boost the educational content. Future research should focus on creating models tailored to specific subjects and improving LLMs to better replicate real-world educational settings. Another key strategy involves the combination of various disciplines, improving students’ skills in integrating knowledge across different subjects. The knowledge developed through optimizing instructional designs influences other smart education domains. Incorporating LLMs into instructional designs can enhance the adaptability and effectiveness of educational practices in response to the dynamic educational environment.

REFERENCES

- [1] M. Cevikbas, J. Koenig, and M. Rothland, “Empirical research on teacher competence in mathematics lesson planning: Recent developments,” *ZDM—Mathematics Education*, vol. 56, no. 1, pp. 101–113, 2024. 1
- [2] S. Liu, Q. Li, X. Shen, J. Sun, and Z. Yang, “Automated discovery of symbolic laws governing skill acquisition from naturally occurring data,” *Nature Computational Science*, vol. 4, no. 5, pp. 334–345, 2024. 1
- [3] Y. Wu, “A research on teaching management of teachers in tianzhi normal university affiliated technical school,” Master’s thesis, Tianjin University of Technology and Education, 2018. 1
- [4] C. C. Tossell, N. L. Tenhundfeld, A. Momen, K. Cooley, and E. J. de Visser, “Student perceptions of chatgpt use in a college essay assignment: Implications for learning, grading, and trust in artificial intelligence,” *IEEE Transactions on Learning Technologies*, 2024. 2
- [5] S. Liu, J. Feng, Z. Yang, Y. Luo, Q. Wan, X. Shen, and J. Sun, “Comet: ‘cone of experience’ enhanced large multimodal model for mathematical problem generation,” *Science China Information Sciences*, vol. 67, no. 12, pp. 1–2, 2024. 2
- [6] Z. Liu, S. X. Yin, C. Lee, and N. F. Chen, “Scaffolding language learning via multi-modal tutoring systems with pedagogical instructions,” *arXiv preprint arXiv:2404.03429*, 2024. 2
- [7] S. Ma, Y. Luo, and Y. Yang, “Personas-based student grouping using reinforcement learning and linear programming,” *Knowledge-Based Systems*, vol. 281, p. 111071, 2023. 2
- [8] J. Sun, F. Yu, S. Liu, Y. Luo, R. Liang, and X. Shen, “Adversarial bootstrapped question representation learning for knowledge tracing,” in *Proceedings of the 31st ACM International Conference on Multimedia*, pp. 8016–8025, 2023. 2

- [9] R. Zou, S. Liu, Y. Luo, Y. Liu, J. Feng, M. Wei, and J. Sun, "Balancing humans and machines: A study on integration scale and its impact on collaborative performance," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, pp. 17628–17636, 2024. 2
- [10] S. Xu, X. Zhang, and L. Qin, "Eduagent: Generative student agents in learning," *arXiv preprint arXiv:2404.07963*, 2024. 2
- [11] S. Jinxin, Z. Jiabao, W. Yilei, W. Xingjiao, L. Jiawen, and H. Liang, "Cgmi: Configurable general multi-agent interaction framework," *arXiv preprint arXiv:2308.12503*, 2023. 2, 3
- [12] W. Chen, Y. Su, J. Zuo, C. Yang, C. Yuan, C. Qian, C.-M. Chan, Y. Qin, Y. Lu, R. Xie, *et al.*, "Agentverse: Facilitating multi-agent collaboration and exploring emergent behaviors in agents," *arXiv preprint arXiv:2308.10848*, 2023. 2
- [13] J. S. Park, J. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, pp. 1–22, 2023. 2
- [14] C. Qian, X. Cong, C. Yang, W. Chen, Y. Su, J. Xu, Z. Liu, and M. Sun, "Communicative agents for software development," *arXiv preprint arXiv:2307.07924*, 2023. 2
- [15] J. He-Yueya, N. D. Goodman, and E. Brunskill, "Evaluating and optimizing educational content with large language model judgments," *arXiv preprint arXiv:2403.02795*, 2024. 2, 6, 7, 8, 10
- [16] B. Hu, L. Zheng, J. Zhu, L. Ding, Y. Wang, and X. Gu, "Teaching plan generation and evaluation with gpt-4: Unleashing the potential of llm in instructional design," *IEEE Transactions on Learning Technologies*, 2024. 2
- [17] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano, *et al.*, "Training verifiers to solve math word problems," *arXiv preprint arXiv:2110.14168*, 2021. 2, 3
- [18] J. He-Yueya, G. Poesia, R. E. Wang, and N. D. Goodman, "Solving math word problems by combining language models with symbolic solvers," *arXiv preprint arXiv:2304.09102*, 2023. 2, 3
- [19] H. U. Khan, M. Abbas, O. Alruwaili, S. Nazir, M. H. Siddiqi, and S. Alanazi, "Selection of a smart and secure education school system based on the internet of things using entropy and topsis approaches," *Computers in Human Behavior*, p. 108346, 2024. 2
- [20] B. Gao, "Understanding smart education continuance intention in a delayed benefit context: An integration of sensory stimuli, utaut, and flow theory," *Acta Psychologica*, vol. 234, p. 103856, 2023. 2
- [21] A. S. Almogren, W. M. Al-Rahmi, and N. A. Dahri, "Exploring factors influencing the acceptance of chatgpt in higher education: A smart education perspective," *Heliyon*, 2024. 2
- [22] S. Shrestha, Y. Shan, R. Emerson, and Z. Hosseini, "Developing and usability testing of an augmented reality tool for online engineering education," *IEEE Transactions on Learning Technologies*, 2024. 2
- [23] N. Pope, J. Kahila, H. Vartiainen, and M. Tedre, "Children's ai design platform for making and deploying ml-driven apps: Design, testing, and development," *IEEE Transactions on Learning Technologies*, 2025. 2
- [24] C. S. Babu, M. Bhavesh, T. Venkatesh, *et al.*, "Ai-driven instructional design: Enhancing learning experiences through personalization and efficiency," in *Effective Instructional Design Informed by AI*, pp. 87–114, IGI Global Scientific Publishing, 2025. 2
- [25] J. M. Markel, S. G. Opferman, J. A. Landay, and C. Piech, "Gpteach: Interactive training with gpt-based students," in *Proceedings of the tenth acm conference on learning@ scale*, pp. 226–236, 2023. 2
- [26] A. Singh, S. Karayev, K. Gutowski, and P. Abbeel, "Gradescope: a fast, flexible, and fair system for scalable assessment of handwritten work," in *Proceedings of the fourth (2017) acm conference on learning@ scale*, pp. 81–88, 2017. 2
- [27] J. Li, S. Wang, M. Zhang, W. Li, Y. Lai, X. Kang, W. Ma, and Y. Liu, "Agent hospital: A simulacrum of hospital with evolvable medical agents," *arXiv preprint arXiv:2405.02957*, 2024. 2
- [28] J. S. Park, J. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, pp. 1–22, 2023. 2
- [29] G. Wang, Y. Xie, Y. Jiang, A. Mandlekar, C. Xiao, Y. Zhu, L. Fan, and A. Anandkumar, "Voyager: An open-ended embodied agent with large language models," *arXiv preprint arXiv:2305.16291*, 2023. 2
- [30] Y. Luo and Y. Yang, "Large language model and domain-specific model collaboration for smart education," *Frontiers of Information Technology & Electronic Engineering*, vol. 25, no. 3, pp. 333–341, 2024. 2
- [31] S. Zheng, Y. Feng, Z. Lu, *et al.*, "Steve-eye: Equipping llm-based embodied agents with visual perception in open worlds," in *The Twelfth International Conference on Learning Representations*, 2023. 2
- [32] S. Lifshitz, K. Paster, H. Chan, J. Ba, and S. McIlraith, "Steve-1: A generative model for text-to-behavior in minecraft," *Advances in Neural Information Processing Systems*, vol. 36, 2024. 2
- [33] R. Krishna, D. Lee, L. Fei-Fei, and M. S. Bernstein, "Socially situated artificial intelligence enables learning from human interaction," *Proceedings of the National Academy of Sciences*, vol. 119, no. 39, p. e2115730119, 2022. 2
- [34] S. Hong, X. Zheng, J. Chen, Y. Cheng, J. Wang, C. Zhang, Z. Wang, S. K. S. Yau, Z. Lin, L. Zhou, *et al.*, "Metagpt: Meta programming for multi-agent collaborative framework," *arXiv preprint arXiv:2308.00352*, 2023. 2
- [35] Q. Wu, G. Bansal, J. Zhang, Y. Wu, S. Zhang, E. Zhu, B. Li, L. Jiang, X. Zhang, and C. Wang, "Autogen: Enabling next-gen llm applications via multi-agent conversation framework," *arXiv preprint arXiv:2308.08155*, 2023. 3
- [36] H. Yang, S. Yue, and Y. He, "Auto-gpt for online decision making: Benchmarks and additional opinions," *arXiv preprint arXiv:2306.02224*, 2023. 3
- [37] G. Li, H. A. A. K. Hammoud, H. Itani, D. Khizbullin, and B. Ghanem, "Camel: Communicative agents for" mind" exploration of large scale language model society," 2023. 3
- [38] C.-Y. Lin, "Rouge: A package for automatic evaluation of summaries," in *Text summarization branches out*, pp. 74–81, 2004. 3
- [39] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "Bleu: a method for automatic evaluation of machine translation," in *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, pp. 311–318, 2002. 3
- [40] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. Xing, *et al.*, "Judging llm-as-a-judge with mt-bench and chatbot arena," *Advances in Neural Information Processing Systems*, vol. 36, 2024. 3
- [41] W. Yuan, R. Y. Pang, K. Cho, S. Sukhbaatar, J. Xu, and J. Weston, "Self-rewarding language models," *arXiv preprint arXiv:2401.10020*, 2024. 3
- [42] Y. Shao, L. Li, J. Dai, and X. Qiu, "Character-llm: A trainable agent for role-playing," *arXiv preprint arXiv:2310.10158*, 2023. 3
- [43] X. Yu, T. Luo, Y. Wei, F. Lei, Y. Huang, P. Hao, and L. Zhu, "Neeko: Leveraging dynamic lora for efficient multi-character role-playing agent," *arXiv preprint arXiv:2402.13717*, 2024. 3
- [44] C. An, S. Gong, M. Zhong, M. Li, J. Zhang, L. Kong, and X. Qiu, "L-eval: Instituting standardized evaluation for long context language models," *arXiv preprint arXiv:2307.11088*, 2023. 3
- [45] Meta, "Introducing meta llama 3: The most capable openly available llm to date," <https://ai.meta.com/blog/meta-llama-3/>, 2024. 6
- [46] M. I. Ciolacu and P. Svasta, "Education 4.0: Ai empowers smart blended learning process with biofeedback," in *2021 IEEE Global Engineering Education Conference (EDUCON)*, pp. 1443–1448, IEEE, 2021. 9
- [47] Z. Yang, J. Yang, K. Rice, J.-L. Hung, and X. Du, "Using convolutional neural network to recognize learning images for early warning of at-risk students," *IEEE Transactions on Learning Technologies*, vol. 13, no. 3, pp. 617–630, 2020. 9
- [48] J. C. Tham and G. Verhulsdonck, "Smart education in smart cities: Layered implications for networked and ubiquitous learning," *IEEE Transactions on Technology and Society*, vol. 4, no. 1, pp. 87–95, 2023. 9
- [49] H. Wan, Z. Zhong, L. Tang, and X. Gao, "Pedagogical interventions in spocs: Learning behavior dashboards and knowledge tracing support exercise recommendation," *IEEE Transactions on Learning Technologies*, 2023. 9
- [50] H. Singh and S. J. Miah, "Smart education literature: A theoretical analysis," *Education and Information Technologies*, vol. 25, no. 4, pp. 3299–3328, 2020. 9
- [51] J.-L. Hung, B. E. Shelton, J. Yang, and X. Du, "Improving predictive modeling for at-risk student identification: A multistage approach," *IEEE Transactions on Learning Technologies*, vol. 12, no. 2, pp. 148–157, 2019. 10
- [52] J. Zhu, X. Ma, and C. Huang, "Stable knowledge tracing using causal inference," *IEEE Transactions on Learning Technologies*, 2023. 10
- [53] A.-B. M. Salem and A. Y. Nikitaeva, "Knowledge engineering paradigms for smart education and learning systems," in *2019 42nd international convention on information and communication technology, electronics and microelectronics (MIPRO)*, pp. 1571–1574, IEEE, 2019. 10
- [54] D. P. Wangoo and S. N. Reddy, "Smart learning environments framework for educational applications in iot enabled educational ecosystems: a review on ai based gui tools for iot wearables," in *2020 IEEE 17th India Council International Conference (INDICON)*, pp. 1–8, IEEE, 2020. 10

- [55] T. Sutjarittham, H. H. Gharakheili, S. S. Kanhere, and V. Sivaraman, "Experiences with iot and ai in a smart campus for optimizing classroom usage," *IEEE Internet of Things Journal*, vol. 6, no. 5, pp. 7595–7607, 2019. 10

Generative AI-Driven Knowledge Base Question Answering with Differentiable Knowledge Graphs for Sustainable Engineering Education

1st Sami Ahmed Haider

School of Engineering and Physical Sciences
Heriot-Watt University
Edinburgh, UK
S.Haider@hw.ac.uk

2nd Mukesh Soni

Division of Research and Development
Lovely Professional University
Phagwara, India
Mukesh.soni@gmail.com

3rd Jehan Akbar

School of Engineering
University of the West of England
Bristol, United Kingdom
Jehan.akbar@uwe.ac.uk

4th Samar Abbas

School of Information and Control Engineering
Southwest University of Science and Technology
Mianyang, P.R. China
sammarhashmi@gmail.com

5th Khwaja Muthair Ahmad

School of Engineering and Physical Sciences
Heriot-Watt University
Edinburgh, UK
muthair@swjtu.edu.cn

6th Faisal Tariq

James Watt School of Engineering
University of Glasgow
Glasgow, United Kingdom
Faisal.Tariq@glasgow.ac.uk

7th Masood-ur-Rehman

James Watt School of Engineering
University of Glasgow
Glasgow, United Kingdom
masood.urrehman@glasgow.ac.uk

8th Adnan Zahid

School of Engineering and Physical Sciences
Heriot-Watt University
Edinburgh, UK
a.zahid@hw.ac.uk

Abstract—Knowledge Base Question Answering (KBQA) is a challenging and widely studied research direction. Approaches that rely on embedding use implicit reasoning to obtain answers, but they are not well suited to producing complete, transparent reasoning paths. Models built upon differentiable knowledge graphs, by contrast, can deliver explainable outputs with minimal supervision signals, requiring only question–answer pairs. Building on differentiable knowledge graphs, we propose an encoder–decoder model called EDKBQA. The encoder fine-grainedly and sequentially models questions using a multi-head attention mechanism and an LSTM to generate query vectors that better capture the semantic properties of each reasoning hop. The decoder incorporates a feedforward neural network with an attention mechanism for multi-hop reasoning, effectively representing the relative importance of each hop when determining the final answer. The proposed model directly addresses information loss arising from coarse-grained, non-sequential modelling. Experiments on the MetaQA-1hop, MetaQA-2hop, MetaQA-3hop, WebQSP, and CWQ datasets achieve accuracies of 97.5%, 100%, 100%, 77.8%, and 51.4%, respectively. Ablation studies confirm that every module contributes to overall performance. Furthermore, the paper demonstrates the model’s practical utility through a case study in sustainable engineering education, showcasing its ability to provide transparent, multi-hop explanations for complex engineering concepts.

Index Terms—Generative AI, Multi-Hop Knowledge, Question Answering, Differentiable Knowledge Graph, LSTM, Encoder–Decoder.

I. INTRODUCTION

Knowledge Base Question Answering (KBQA) [1] has become an important technology widely used in generative AI-based applications such as search engines and voice assistants. Improving the accuracy and interpretability of KBQA has therefore become an active research area. A knowledge base (KB) is a data structure that stores facts as triples, using edges to represent relationships between entity nodes; each triple consists of a head entity, a tail entity, and the relationship connecting them. By searching the KB, a KBQA system retrieves the answer to a natural-language query. The two principal categories are single-hop KBQA, in which the subject and answer are connected by a single head–relation–tail triple, and multi-hop KBQA, in which the subject may be several hops away from the answer.

In single-hop KBQA, the straightforward triple structure yields near-human performance. Multi-hop KBQA, however, requires navigating multiple linked triples, substantially increasing semantic complexity and demanding deeper reasoning. As a result, multi-hop KBQA has attracted growing research interest, with current models seeking to improve both accuracy, by identifying the correct answer entity, and interpretability, by explicitly modelling the reasoning path between the subject and the final answer.

Answering complex multi-hop queries requires specialised methods [2]. Existing approaches fall broadly into two cat-

egories: semantic parsing and information retrieval. Semantic parsing methods [3] translate questions into intermediate structured forms—such as query graphs or logical statements—which are then executed over the KB. However, they require costly question-logic annotation and domain expertise in query languages; moreover, the intermediate structured queries do not always faithfully capture the question’s semantics, leading to answer failures. Information retrieval-based methods [4] directly retrieve answers by learning semantic representations of both the question and the KB. These are further divided into embedding-based methods [5] and path-based methods [6].

Embedding-based methods score the embeddings of the subject and candidate entities, but their reasoning is implicit and lacks interpretability. Path-based methods [7] trace explicit reasoning paths from the subject entity to the answer, offering strong interpretability, though they struggle to scale when the KB or path space is large. Multi-hop questions exhibit distinct semantics at each hop and a sequential reasoning path that is both transparent and visualisable [8], with different hops carrying different importance weights when determining the final answer.

This paper presents an end-to-end EDKBQA model built on a differentiable knowledge graph [9], [10]. The encoder employs fine-grained sequential modelling, combining multi-head attention [11] to capture detailed semantics with an LSTM [12] to learn the logical ordering of hops. The decoder employs a multi-hop attention module with two feedforward networks to assign hop-importance weights, thereby enabling accurate semantic retention and effective reasoning. Experiments demonstrate that EDKBQA significantly improves accuracy and interpretability over baseline models, with its fully differentiable architecture ensuring high computational efficiency, strong scalability, and straightforward portability.

The contributions of this paper are summarised as follows:

- We propose an encoder–decoder model based on a differentiable knowledge graph that achieves excellent question-answering performance across three datasets, highly competitive with current state-of-the-art models.
- We develop a fine-grained sequential module that uses an LSTM to propagate information across hops and multi-head attention to capture precise hop-level semantics, producing more coherent and expressive query vectors.
- We design a multi-hop reasoning attention module in the decoder that uses a GRU-enhanced relation scorer and dual feedforward networks to determine hop-importance weights and generate final entity scores.
- We strengthen the link to sustainable engineering education with a concrete case study, demonstrating how the model’s explainable reasoning paths can serve as an AI tutor for complex engineering systems.

II. PROPOSED EDKBQA MODEL

This section first introduces relevant background and task definitions, including knowledge bases, KBQA, and differentiable knowledge graphs, before presenting the encoder–decoder architecture.

A. Background and Definitions

A KB represents factual information as a set of triples:

$$\mathcal{KB} = \{\langle H, R, T \rangle \mid H, T \in \mathcal{E}, r \in \mathcal{R}\},$$

where $\mathcal{E}$ is the entity set and $\mathcal{R}$ is the relation set. A triple $\langle H, R, T \rangle$ indicates that relation R holds between head entity H and tail entity T . The neighbourhood of an entity in the KB is defined as $\mathcal{N} = \{\langle H, R, T \rangle \in \mathcal{KB}\}$, combining all triples—including both incoming and outgoing edges—around each entity. For brevity, an inverse triple $\langle T, R, H \rangle$ is written as $\langle H, R^{-1}, T \rangle$.

A KBQA task [26] requires, given a natural-language question $q = w_1, w_2, \dots, w_l$ and a KB, finding the answer entity set $\mathcal{A}_q$ from the candidate entity set, starting from the subject entities mentioned in q . Multi-hop KBQA handles complex questions where the subject and answer are separated by multiple hops, requiring multi-step reasoning across several entities and relations.

A differentiable knowledge graph [28] represents a KB as a sparse matrix structure, allowing question–KB interaction through differentiable mathematical operations. Formally, $\mathcal{KB} = \{\langle H, P, T \rangle \mid H \in \mathcal{E}, T \in \mathcal{E}, P \in \mathcal{R}\}$, where triples are encoded by three binary matrices: the head-entity matrix $\mathbf{H}_M \in \{0, 1\}^{N_T \times N_E}$, the relation matrix $\mathbf{P}_M \in \{0, 1\}^{N_T \times N_R}$, and the tail-entity matrix $\mathbf{T}_M \in \{0, 1\}^{N_T \times N_E}$, defined as:

$$\mathbf{H}_M(i, j) = \mathbb{I}[e_j = s_i], \quad \mathbf{P}_M(i, j) = \mathbb{I}[p_j = p_i], \quad \mathbf{T}_M(i, j) = \mathbb{I}[e_i = o_i]$$

Given an entity score vector $\mathbf{a}_{t-1} \in \mathbb{R}^{N_E}$ and a relation score vector $\mathbf{r}_t \in \mathbb{R}^{N_R}$ at hop $t - 1$, the entity score vector at hop t is computed as:

$$\mathbf{a}_t = f(\mathbf{a}_{t-1}, \mathbf{r}_t). \quad (1)$$

An alternative representation [26] uses a fact matrix $\mathbf{Z} \in \mathbb{R}^{N_E \times N_E}$, where element $Z_{i,j}$ encodes the relationship between head entity E_i and tail entity E_j . Specifically:

$$Z_{i,j}^{(t)} = \sum_b \text{pr}_{i,j,b}^{(t)}, \quad (2)$$

where $\text{pr}_{i,j,b}^{(t)}$ is the b -th probability value of the relation between E_i and E_j at hop t . The entity score vector at hop t is then:

$$\mathbf{a}_t = \mathbf{Z}^{(t)\top} \mathbf{a}_{t-1}. \quad (3)$$

B. Model Architecture

The EDKBQA model leverages a differentiable knowledge graph to enable fully end-to-end reasoning, executing multi-hop question–KB interactions through differentiable, generative AI operations. This path-based design is both computationally efficient and capable of explicitly modelling intermediate reasoning paths. The model can also be parallelised across multiple GPUs, providing strong interpretability and excellent scalability.

Fig. 1 illustrates the EDKBQA architecture, which comprises an encoder for question representation and a decoder for

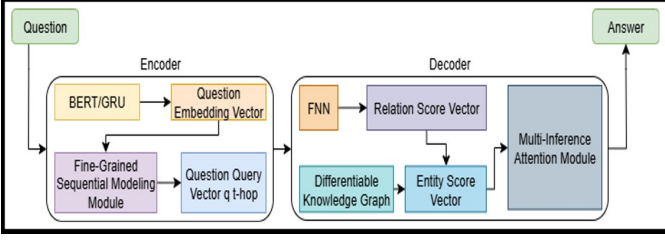


Fig. 1. Encoder-Decoder Model Architecture.

multi-hop reasoning. The encoder produces a per-hop query vector through a fine-grained sequential module that combines multi-head attention for detailed semantic representation and an LSTM for capturing hop-level logical ordering. The decoder then reasons over these query vectors to derive the final answer.

The model first derives a per-hop relation score vector from the query representation, then computes a per-hop entity score vector through differentiable knowledge graph operations, and finally aggregates these to produce the final entity score used to select the answer. The multi-hop attention module employs two feedforward networks to weight hop importance.

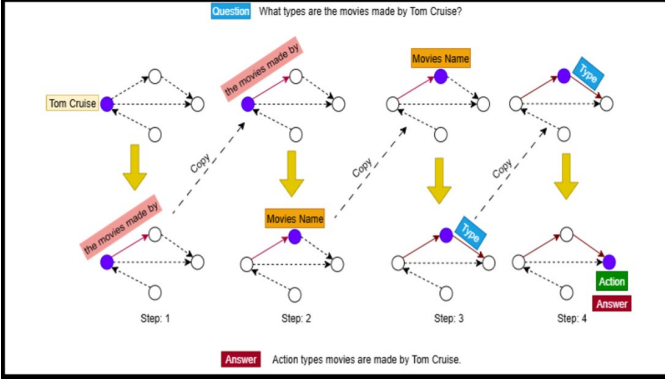


Fig. 2. Multi-Hop Reasoning Process.

Fig. 2 illustrates the multi-hop reasoning process using the example question “What types of movies are made by Tom Cruise?” The model uses named entity recognition (NER) to identify the subject entity “Tom Cruise” and links it to the corresponding KB entity. It then initialises relation and entity score vectors, and iteratively refines these over multiple reasoning hops. In the final step, the decoder’s multi-hop attention module computes importance weights for each hop and takes a weighted sum of the per-hop entity score vectors to identify the highest-scoring answer entity, “Action”.

For questions requiring more than two hops, the single-hop reasoning step is simply repeated until the answer is reached, giving the model clear interpretability through explicit intermediate paths. Its fully differentiable knowledge graph further ensures strong scalability and efficiency.

C. Encoder

The encoder’s task is to generate a per-hop query vector that better represents the semantic information at each reasoning step. Existing models tend to produce coarse-grained, sentence-level representations that lose key hop-level information and ignore the sequential logical dependencies between hops. To address this, we introduce an encoder that produces fine-grained, hop-aware query vectors by integrating past and current reasoning states via an LSTM, while multi-head attention captures precise semantic details at the token level.

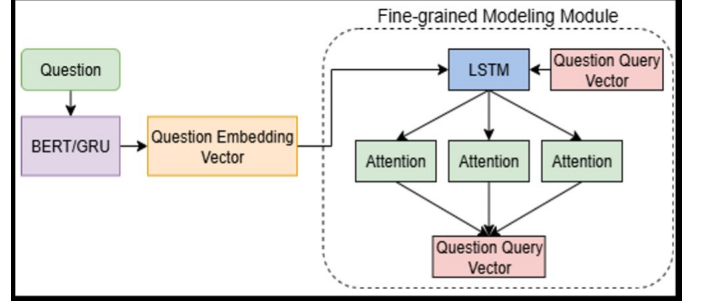


Fig. 3. Encoder Architecture.

Fig. 3 shows the encoder architecture. First, a pre-trained language model (BERT or GRU) generates the initial question embedding $\mathbf{k}$ and token-level embeddings $\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_n$:

$$\mathbf{k}, \mathbf{t}_1, \dots, \mathbf{t}_n = \text{Encoder}(q; \theta_e). \quad (4)$$

The LSTM then takes the question embedding $\mathbf{k}$ and the query vector from hop $t-1$ to generate the query vector for hop t :

$$\mathbf{k}_t = \text{LSTM}(\mathbf{k} \mid \mathbf{k}_{t-1}; \theta_w). \quad (5)$$

The query vector $\mathbf{k}_t$ serves as the query Q matrix, while the token embeddings serve as both the key K and value V matrices. Multi-head attention computes:

$$b_j^{(t)} = \text{Softmax}(\mathbf{k}_t \cdot [\mathbf{t}_1, \dots, \mathbf{t}_n]^\top), \quad (6)$$

$$\hat{\mathbf{k}}_t^{(i)} = \sum_{j=1}^n b_j^{(t)} \mathbf{t}_j. \quad (7)$$

Finally, the outputs of m attention heads are averaged to obtain the hop-level query vector:

$$\mathbf{k}_t = \frac{1}{m} \sum_{i=1}^m \hat{\mathbf{k}}_t^{(i)}. \quad (8)$$

D. Decoder

The decoder transforms the encoder’s query vectors into candidate entity scores through differentiable operations on the knowledge graph. For each hop, it generates a relation vector from the query vector, then combines it with the previous hop’s entity vector via differentiable knowledge graph reasoning to produce the next hop’s entity vector. A weighted sum of all hop-level entity vectors forms the final score vector.

The multi-hop reasoning attention module uses two feed-forward neural networks to learn the relative importance of different hops. One network is trained on the hop-specific query vectors; the other is trained on the global question embedding. The final hop-importance weight is obtained by averaging the two independently computed weights, combining local hop semantics with global question semantics to yield more reliable entity score estimates.

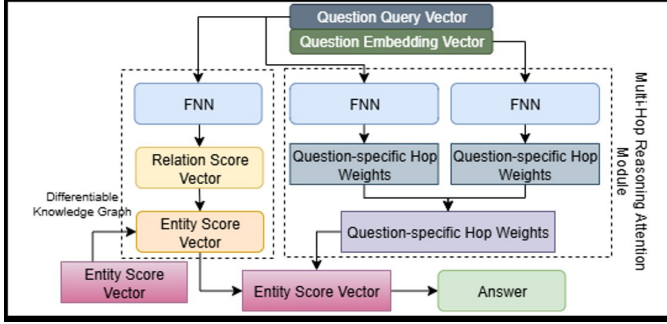


Fig. 4. Decoder Architecture.

Fig. 4 illustrates the decoder. The operations are summarised as follows. For each hop t , a feedforward network transforms the query vector into a relation score vector:

$$\mathbf{r}_t = \text{FNN}(\mathbf{k}_t). \quad (9)$$

Differentiable knowledge graph operations then produce the hop-level entity score vector from the previous entity vector $\mathbf{w}_{t-1}$ and $\mathbf{r}_t$ (only the topic entity is assigned a score of 1 in $\mathbf{w}_0$):

$$\mathbf{w}_t = f(\mathbf{w}_{t-1}, \mathbf{r}_t). \quad (10)$$

Two feedforward networks compute hop-importance weights from the global embedding $\mathbf{k}$ and the per-hop query $\mathbf{k}_t$, respectively:

$$\tilde{w}_t^{(2)} = \text{FNN}(\mathbf{k}), \quad (11)$$

$$\tilde{w}_t^{(1)} = \text{Softmax}(\text{FNN}(\mathbf{k}_t)). \quad (12)$$

These are averaged to obtain the final hop weight:

$$\tilde{w}_t = \frac{\tilde{w}_t^{(1)} + \tilde{w}_t^{(2)}}{2}. \quad (13)$$

The final entity score vector is then the weighted sum across all hops:

$$\mathbf{ev} = \sum_{t=1}^T \mathbf{w}_t \tilde{w}_t. \quad (14)$$

The entity with the highest score in $\mathbf{ev}$ is selected as the answer.

III. EXPERIMENTS

A. Datasets

We evaluate our approach on MetaQA, WebQuestionsSP (WebQSP), and ComplexWebQuestions (CWQ), with statistics shown in Table I. MetaQA is a large multi-hop KGQA

benchmark derived from WikiMovies, containing nearly 400k questions (up to 3 hops) built over a movie-domain knowledge graph with 43k entities, 9 predicates, and 135k triples. WebQSP is a refined subset of WebQuestions providing 4,737 questions linked to Freebase; unanswered questions are removed during training and counted as errors during evaluation. All WebQSP questions can be answered via one- or two-hop reasoning. Following prior work, we prune Freebase to a subgraph containing all facts within two hops of the topic entity, yielding a processed KB with 1.8M entities, 572 predicates, and 5.7M triples.

TABLE I
SUMMARY OF BENCHMARK DATASETS

Dataset	Train	Validation	Test
MetaQA-1hop	97,127	9,996	9,965
MetaQA-2hop	119,335	14,976	14,872
MetaQA-3hop	114,245	14,307	14,274
WebQSP	3,125	100	1,705
CWQ	27,525	3,518	3,631

CWQ extends WebQSP with deeper hops and richer constraints. Following the approach in [31], we pre-process CWQ using PageRank to retrieve a question-specific subgraph for each query [32], yielding subgraphs averaging 1,948 entities with a 64% answer recall rate. Experiments on CWQ are conducted on the validation set.

B. Baselines

Our model is compared against the following baseline methods: KGT5 [16], GraftNet [17], PullNet [18], Relation-Learning [19], EMQL [20], NSM [23], SR+NSM [24], Rigel-Baseline [25], TransferNet [26], CFG [28], and Struct-GPT [30].

C. Evaluation Metrics

Hits@1 ranks all candidate entities by score and assigns 1 if the correct answer is ranked first, and 0 otherwise. As the strongest indicator of a model's ability to identify the correct entity, Hits@1 is averaged over all test questions, with unanswered cases counted as errors. This paper adopts Hits@1 as the primary evaluation metric.

F1 treats all entities scoring above a chosen threshold as predicted answers and computes the harmonic mean of precision and recall against the ground truth. The threshold is tuned on the validation set. Because F1 is highly sensitive to the threshold and many baselines do not report it, F1 comparisons are provided only for WebQSP.

D. Experimental Results

The MetaQA dataset uses a bidirectional GRU as the question encoder, whilst WebQSP and CWQ use bert-base-uncased owing to their greater complexity. Hop counts are set to 3, 2, and 3 for MetaQA, WebQSP, and CWQ, respectively. All experiments were run on an NVIDIA RTX 3080 GPU. The model was trained using RAdam (learning rate = 1×10^{-3}); BERT parameters were fine-tuned with

a smaller learning rate of 3×10^{-5} . Multi-head attention used 3 heads.

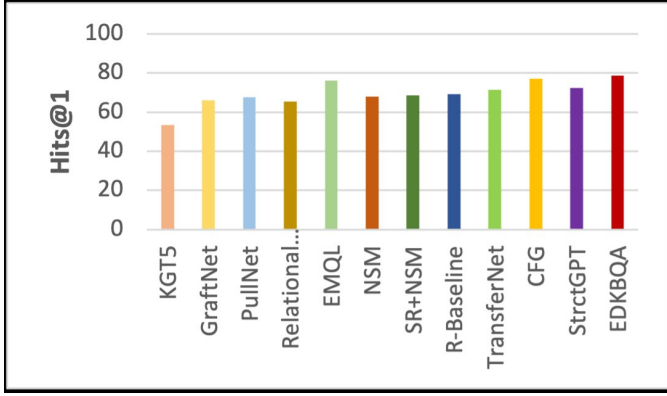


Fig. 5. Hits@1 Results on WebQSP.

Fig. 5 shows the Hits@1 results on WebQSP. On MetaQA, EDKBQA correctly answers all 2-hop and 3-hop questions and achieves 97.5% accuracy on 1-hop queries; minor errors arise from entities sharing identical names. Multi-hop queries benefit from stronger relational constraints. Our model attains 77.8% accuracy on WebQSP and 51.4% on CWQ, both substantially exceeding baseline performance.

The baseline models fall into two categories: embedding-based and path-based methods. Results show that path-based EDKBQA outperforms all embedding-based approaches, demonstrating superior interpretability, scalability, and overall performance. In the encoder, multi-head attention captures fine-grained token semantics whilst the LSTM models hop-to-hop dependencies, producing query vectors that more accurately reflect the question’s meaning. In the decoder, two feedforward networks compute hop-importance weights, allowing the model to capture the contribution of each hop to the final answer.

The EDKBQA model is built on a fully differentiable knowledge graph, making it computationally efficient, fast to train, scalable across multiple GPUs, and easily extensible without retraining when new entities are added. Analysis of failure cases reveals two main issues: (i) the encoder occasionally misidentifies the subject entity, preventing correct relation-path discovery—an issue mitigable by adding missing entities to BERT’s vocabulary; and (ii) semantically similar relations with limited training examples cause the model to confuse relation types and fail to derive the correct answer path.

Recent work has explored how large language models (LLMs) perform on multi-hop KBQA; our experiments show that EDKBQA substantially outperforms StrctGPT. LLMs that rely solely on their internal knowledge perform poorly on specialised KB reasoning and, even with fine-tuning, remain less stable and less reliable than dedicated KBQA models.

E. Ablation Study

Fig. 6 shows the ablation results. Removing any component—the encoder’s LSTM, the multi-head attention

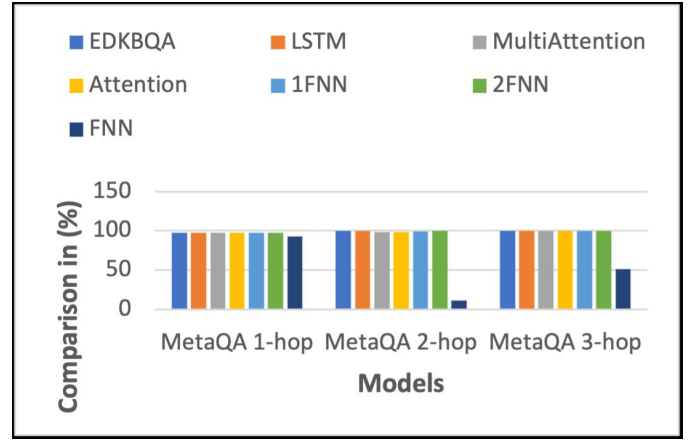


Fig. 6. Ablation Study Results on MetaQA.

mechanism, or either of the decoder’s two feedforward networks—causes substantial performance drops across all datasets. Replacing multi-head attention with a single-head variant also noticeably weakens results, underscoring the value of fine-grained sequential and semantic modelling. The encoder’s LSTM and multi-head attention, together with the decoder’s dual feedforward networks, deliver the largest performance gains on the most challenging multi-hop tasks.

IV. CONCLUSION

This paper proposes EDKBQA, an end-to-end differentiable knowledge graph-based encoder–decoder framework for question answering. The encoder fuses fine-grained sequential modelling with an LSTM to capture hop-wise dependencies, whilst multi-head attention highlights important semantic cues when constructing per-hop query vectors. The decoder improves accuracy with a multi-hop attention module that uses two feedforward networks to estimate hop-importance weights, yielding more reliable entity scores and final answers. Extensive experiments confirm the effectiveness of EDKBQA, demonstrating strong improvements in answer accuracy. Future work will focus on improving relation-vector generation, currently performed by a simple linear classifier in the decoder, to further enhance multi-hop reasoning performance.

REFERENCES

- [1] J. Qadir, “Engineering education in the era of ChatGPT: Promise and pitfalls of generative AI for education,” in *Proc. IEEE Global Eng. Educ. Conf. (EDUCON)*, 2025.
- [2] S. Wang and B. Qin, “A novel joint training model for knowledge base question answering,” *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 32, pp. 666–679, 2024, doi: 10.1109/TASLP.2023.3336526.
- [3] C. Wang, K. Xue, and C. Shi, “An optimized model with encoder-decoder ConvLSTM for global ionospheric forecasting,” *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025, Art no. 1001605, doi: 10.1109/LGRS.2025.3565645.
- [4] Y. Lin and H. Shen, “SmartQ: A question and answer system for supplying high-quality and trustworthy answers,” *IEEE Trans. Big Data*, vol. 4, no. 4, pp. 600–613, Dec. 2018, doi: 10.1109/TBDATA.2017.2735442.
- [5] A. Serrano-Laguna, I. Martinez-Ortiz, P. Moreno-Ger, and B. Fernandez-Manjon, “AI as yet another tool in undergraduate student projects: Preliminary results,” in *Proc. IEEE Global Eng. Educ. Conf. (EDUCON)*, 2024, pp. 1337–1342, doi: 10.1109/EDUCON60312.2024.10511606.

- [6] U. Naseem, M. Khushi, A. G. Dunn, and J. Kim, "K-PathVQA: Knowledge-aware multimodal representation for pathology visual question answering," *IEEE J. Biomed. Health Inform.*, vol. 28, no. 4, pp. 1886–1895, Apr. 2024, doi: 10.1109/JBHI.2023.3294249.
- [7] Y. Zheng, L. Lu, Y. Hu, Y. Chen, and A. Wang, "Thoughtful and cautious reasoning: A fine-tuned knowledge graph-based multi-hop question answering framework," *Eng. Appl. Artif. Intell.*, vol. 150, p. 110479, 2025, doi: 10.1016/j.engappai.2025.110479.
- [8] Z. Wang, X. Su, and Z. Ding, "Long-term traffic prediction based on LSTM encoder-decoder architecture," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 10, pp. 6561–6571, Oct. 2021, doi: 10.1109/TITS.2020.2995546.
- [9] H. Zhou, T. He, Y.-S. Ong, G. Cong, and Q. Chen, "Differentiable clustering for graph attention," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 8, pp. 3751–3764, Aug. 2024, doi: 10.1109/TKDE.2024.3363703.
- [10] W. Wu, Z. Zhu, J. Qi, W. Wang, G. Zhang, and P. Liu, "A dynamic graph expansion network for multi-hop knowledge base question answering," *Neurocomputing*, vol. 515, pp. 37–47, 2023, doi: 10.1016/j.neucom.2022.09.127.
- [11] J. Ma, J. Liu, Q. Lin, B. Wu, Y. Wang, and Y. You, "Multi-task learning for visual question answering," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 3, pp. 1380–1394, Mar. 2023, doi: 10.1109/TNNLS.2021.3105284.
- [12] L. Yang, H. Chen, Z. Li, X. Ding, and X. Wu, "Give us the facts: Enhancing large language models with knowledge graphs for fact-aware language modeling," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 7, pp. 3091–3110, Jul. 2024, doi: 10.1109/TKDE.2024.3360454.
- [13] H. Ren, Z. Chen, X. Liang, C. Yang, and W. Gui, "Association hierarchical representation learning for plant-wide process monitoring using multilevel knowledge graph," *IEEE Trans. Artif. Intell.*, vol. 4, no. 4, pp. 636–649, Aug. 2023, doi: 10.1109/TAI.2022.3161860.
- [14] Y. Wen, X. Zhu, and L. Zhang, "CQACD: A concept question-answering system for intelligent tutoring using a domain ontology with rich semantics," *IEEE Access*, vol. 10, pp. 67247–67261, 2022, doi: 10.1109/ACCESS.2022.3185400.
- [15] J. Xu, H. Zhang, H. Zhang, J. Lu, and G. Xiao, "ChatTf: A knowledge graph-enhanced intelligent Q&A system for mitigating factuality hallucinations in traditional folklore," *IEEE Access*, vol. 12, pp. 162638–162650, 2024, doi: 10.1109/ACCESS.2024.3485877.
- [16] A. Ait-Mlouk and L. Jiang, "KBot: A knowledge graph based chatbot for natural language understanding over linked data," *IEEE Access*, vol. 8, pp. 149220–149230, 2020, doi: 10.1109/ACCESS.2020.3016142.
- [17] S. Zhang, Z. Hu, A. Subramonian, and Y. Sun, "Motif-driven contrastive learning of graph representations," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 8, pp. 4063–4075, Aug. 2024, doi: 10.1109/TKDE.2024.3364059.
- [18] L. Nie, X. Wei, D. Zhang, X. Wang, Z. Gao, and Y. Yang, "Data-driven answer selection in community QA systems," *IEEE Trans. Knowl. Data Eng.*, vol. 29, no. 6, pp. 1186–1198, Jun. 2017, doi: 10.1109/TKDE.2017.2669982.
- [19] Z. Li, H. Wang, and W. Zhang, "Translational relation embeddings for multi-hop knowledge base question answering," *J. Web Semant.*, vol. 74, p. 100723, 2022, doi: 10.1016/j.websem.2022.100723.
- [20] H. Xiang, Y. Wang, X. Zhang, and Z. Chen, "Two-level battery health diagnosis using encoder-decoder framework and Gaussian mixture ensemble learning based on relaxation voltage," *IEEE Trans. Transp. Electr.*, vol. 10, no. 2, pp. 3966–3975, Jun. 2024, doi: 10.1109/TTE.2023.3317449.
- [21] S. Hu, L. Zou, J. X. Yu, H. Wang, and D. Zhao, "Answering natural language questions by subgraph matching over knowledge graphs," *IEEE Trans. Knowl. Data Eng.*, vol. 30, no. 5, pp. 824–837, May 2018, doi: 10.1109/TKDE.2017.2766634.
- [22] H. Liu, T. Wang, C. Lang, Y. Li, and H. Ling, "Deep probabilistic graph matching," *IEEE Trans. Knowl. Data Eng.*, vol. 37, no. 9, pp. 5127–5140, Sept. 2025, doi: 10.1109/TKDE.2025.3578989.
- [23] J. Castro et al., "A big data semantic driven context aware recommendation method for question-answer items," *IEEE Access*, vol. 7, pp. 182664–182678, 2019, doi: 10.1109/ACCESS.2019.2957881.
- [24] H. Li, F. Qi, and G. Shi, "A novel spatio-temporal 3D convolutional encoder-decoder network for dynamic saliency prediction," *IEEE Access*, vol. 9, pp. 36328–36341, 2021, doi: 10.1109/ACCESS.2021.3063372.
- [25] J. Chen, X. Xin, X. Liang, X. He, and J. Liu, "GDSRec: Graph-based decentralized collaborative filtering for social recommendation," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 5, pp. 4813–4824, May 2023, doi: 10.1109/TKDE.2022.3153284.
- [26] J. Yu, Q. Su, X. Quan, and J. Yin, "Multi-hop reasoning question generation and its application," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 1, pp. 725–740, Jan. 2023, doi: 10.1109/TKDE.2021.3073227.
- [27] W. Zheng, L. Yan, W. Zhang, L. Ouyang, and D. Wen, "D→K→I: Data-knowledge-driven group intelligence framework for smart service in education metaverse," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 53, no. 4, pp. 2056–2061, Apr. 2023, doi: 10.1109/TSMC.2022.3228849.
- [28] Y. Feng, Y. Sun, G. G. Yen, and K. C. Tan, "REP: An interpretable robustness enhanced plugin for differentiable neural architecture search," *IEEE Trans. Knowl. Data Eng.*, vol. 37, no. 5, pp. 2888–2902, May 2025, doi: 10.1109/TKDE.2025.3543503.
- [29] R.-Z. Wang, Z.-H. Ling, and Y. Hu, "Knowledge base question answering with attentive pooling for question representation," *IEEE Access*, vol. 7, pp. 46773–46784, 2019, doi: 10.1109/ACCESS.2019.2909826.
- [30] J. Van der Putten, F. Van der Sommen, and P. H. N. De With, "Efficient decoder reduction for a variety of encoder-decoder problems," *IEEE Access*, vol. 8, pp. 169444–169455, 2020, doi: 10.1109/ACCESS.2020.3020360.
- [31] H. M. Yao, L. Jiang, and M. Ng, "Implementing the fast full-wave electromagnetic forward solver using the deep convolutional encoder-decoder architecture," *IEEE Trans. Antennas Propag.*, vol. 71, no. 1, pp. 1152–1157, Jan. 2023, doi: 10.1109/TAP.2022.3216920.
- [32] R. Li, D. Li, H. Yang, X. Liu, H. Jin, R. Pu, and Q. Miao, "RECoT: Relation-enhanced chains-of-thoughts for knowledge-intensive multi-hop questions answering," *Neurocomputing*, vol. 637, p. 129903, 2025, doi: 10.1016/j.neucom.2025.129903.



Haider, Sami Ahmed, Soni, Mukesh, Akbar, Jehan, Abbas, Samar, Ahmad, Khwaja Muthair, Tariq, Faisal, Ur Rehman, Masood and Zahid, Adnan (2026) Generative AI-Driven Knowledge Base Question Answering with Differentiable Knowledge Graphs for Sustainable Engineering Education. In: 2026 IEEE Global Engineering Education Conference (EDUCON), Cairo, Egypt, 27-30 April 2026, ISBN 9798331576714 (doi: [10.1109/educon67543.2026.11574212](https://doi.org/10.1109/educon67543.2026.11574212))



Copyright © 2026 IEEE

This is the author accepted manuscript reproduced under a [Creative Commons Attribution 4.0 International License](https://creativecommons.org/licenses/by/4.0/).

<https://eprints.gla.ac.uk/390225/>

Deposited on: 23 July 2026

Balancing Generative AI and Critical Thinking to Develop Written Communication Skills in Cybersecurity

Apostolos Charalambous
Faculty of Pure and
Applied Sciences
Open University of Cyprus
Cyprus
apostolos.charalambous@s
t.ouc.ac.cy

Andriani Piki
School of Sciences
University of Central
Lancashire Cyprus
Cyprus
apiki@uclan.ac.uk

Joakim Kävrestad
School of Engineering
Jönköping University
Sveeden
joakim.kavrestad@ju.se

Eliana Stavrou
Faculty of Pure and
Applied Sciences
Open University of Cyprus
Cyprus
eliana.stavrou@ouc.ac.cy

Abstract—As cybersecurity education continues to evolve, the need for curricula that effectively balance the capabilities of generative Artificial Intelligence (AI) tools with the development of critical thinking and active learning skills has become increasingly urgent. This study addresses this challenge by proposing a curriculum for postgraduate cybersecurity education that focuses on developing transferable skills, particularly critical thinking and written communication. These skills are essential for cybersecurity professionals to excel in both their technical and communication-oriented responsibilities, meeting the growing demand of the cybersecurity industry in the age of AI. The proposed curriculum emphasizes the integration of constructivist learning principles and Bloom's taxonomy, two widely applied pedagogical models, to enhance learners' critical thinking and written communication skills. Designed for a penetration testing module, the curriculum follows a structured, step-by-step approach to build the necessary competences and empower aspiring cybersecurity professionals to meet the expectations of the cybersecurity industry. Through targeted activities, learners develop foundational knowledge while refining advanced written communication skills, equipping them to produce professional-level documentation, such as penetration testing reports. Generative AI is incorporated in the curriculum, providing opportunities for learners to experiment with AI-generated content while fostering the cognitive skills needed to critically assess its accuracy, relevance, and alignment with professional standards. This study contributes to cybersecurity education by presenting a replicable curriculum model that equips learners with vital skills, preparing them to navigate the complexities of written communication responsibilities in cybersecurity roles and adapt to the evolving demands of the AI era.

Keywords—*Cybersecurity, penetration testing report, generative AI, transferable skills, written communication skills, postgraduate education, curriculum design*

I. INTRODUCTION

In the fast-evolving field of cybersecurity, technical proficiency alone is no longer sufficient [1][2] for navigating the complexities of technological advances such as generative Artificial Intelligence (genAI). This presents challenges for educational programmes which must remain resilient in the face of this evolution, by embracing innovative pedagogical approaches [3] and going beyond traditional teaching methods [4][5][6]. Furthermore, the interdisciplinary nature of this domain means that, alongside technical expertise, cybersecurity professionals must possess effective communication skills, particularly in written communication [7][8][9][10].

Transferable skills, such as written communication, are fundamental across various cybersecurity roles, whether

documenting forensic evidence, reporting penetration testing results, or delivering comprehensive risk assessments. Written communication is vital for effectively presenting complex ideas across different contexts and audiences. The European Cybersecurity Skills Framework (ECSF) [11] highlights the importance of this skillset in ensuring that cybersecurity experts across several cybersecurity career roles can convey critical information effectively to a diverse range of stakeholders. However, mastering writing skills is not straightforward. It requires the combination of diverse knowledge areas and skills including domain expertise, topic comprehension, critical thinking, attention to detail, a clear understanding of the target audience, logical structuring, analysing and evaluating existing documentation and policies, and synthesising multiple resources to create new reports [12]. These abilities are essential for demonstrating writing proficiency and constitute a fundamental part of the cybersecurity profession [13].

Cultivating critical thinking and written communication skills is crucial, yet challenging, particularly in an era marked by rapid digital transformation and technological disruption. Specifically, in the field of education, the rise of genAI has intensified the efforts in cultivating these skills, presenting both opportunities and challenges [14] for the future of education. On one hand, genAI has the potential to enhance learning by enabling learners to quickly grasp complex ideas and refine their writing [15]. On the other hand, it introduces risks, such as over-dependence on AI-generated content and diminished creativity, undermining writing and critical thinking skills. Biases in AI algorithms [16] and reduced attention to detail further complicate its use in educational contexts. These challenges, if not addressed, can directly hinder the development of a skilful workforce, as graduates may struggle to demonstrate the proficiency needed to meet professional standards. In light of these challenges, the need for effective interventions to develop and sustain written communication skills has never been more pressing. Cybersecurity education, especially at the postgraduate level, must equip learners with strong critical thinking abilities while also presenting opportunities for embracing the genAI era professionally and ethically, ensuring they are prepared to balance the use of genAI tools with their continuous personal skill development [6][17].

This paper focuses on the importance of written communication skills within cybersecurity and proposes a new curriculum tailored to postgraduate cybersecurity education. The curriculum aims to enhance students' writing skills by leveraging the capabilities of genAI [17] while also

fostering their self-efficacy [18]. The curriculum further seeks to help students become self-aware of their strengths and weaknesses, promote continuous improvement of their communication and critical thinking skills, and maintain an ethical approach to using genAI tools. By blending these key components into cybersecurity education, we aim to bridge the current gap in writing proficiency and ensure that students are better prepared to excel in both their technical and communication-based responsibilities, meeting the expectations of the cybersecurity industry in the age of AI.

The paper is organized as follows: Section II discusses background and related research focusing on the importance of written communication skills; Section III presents the methodology implemented in this work; Section IV presents the proposed curriculum, followed by the discussion of empirical findings and evaluation of curriculum design in Section V. The last two sections critically discuss the key observations and provide study conclusions, respectively.

II. RELATED WORK

A. Importance of Written Communication Skills in the genAI Era

Effective verbal and written communication skills are fundamental in social and professional contexts alike. Constructing and communicating new content entails demonstrating an array of skills such as critical evaluation and interpretation of existing knowledge, analytical thinking, the ability to apply prior knowledge in new situations, being able to explain and describe ideas to different audiences and demonstrate a holistic understanding (these are broadly captured by Bloom's Taxonomy of cognitive thinking skills).

Recently, equipping learners with critical thinking and effective written communication skills has attained greater impetus in view of the digitalisation and rising complexity of the information landscape [19]. Trends such as information overload, mis- and dis-information [14], the increasing distribution of fast, unfiltered and often biased content, alongside the growing promotion of unstructured, unformatted, or purpose-less social interaction styles, impede the cultivation of critical thinking and effective, professional written communication. Another worrying trend is the increasing misuse of genAI tools and the lack of critical evaluation of AI-generated content. The ease with which genAI tools can generate high-quality and believable content has led to over-reliance on these tools, reducing users' creative thinking skills [20] and their ability to methodically review and critique their own writing [14]. Many users, including students and professionals, use the generated content 'as is' without reflecting on, or questioning, its accuracy, logic, practical usefulness, completeness or ethical soundness [3][14]. Unquestionably, in the field of education, genAI tools such as ChatGPT have sparked both admiration and controversy [15].

In response to these trends, academics, instructional designers and professional trainers have recognized the importance of leveraging the capabilities of genAI tools in education [15][17]. As AI-enabled applications increasingly permeate various aspects of our social and professional lives, the need for critical thinking skills and the ability to produce thoughtful, inquiry-based, well-structured, and purposive content has never been more crucial [3]. However, developing written communication skills requires leveraging genAI effectively and purposefully [19], while at the same time

balancing the use of these tools with the cultivation of critical thinking skills. When embedded in purposefully designed curricula [19] which emphasize the balance between self-efficacy [18], informed engagement [15], robust critical thinking skills [3], and appropriate prompt engineering techniques [17], genAI can help streamline various aspects of the writing process and help enhance learners' competence in written communication. The curriculum proposed in this study is a step towards this direction.

B. Written Communication Skills in Cybersecurity

Transferable skills, also known as soft skills, cognitive skills, or complementary skills, encompass a set of personality traits, competencies, or behaviors that professionals display in various situations [21] and which extend beyond technical or domain expertise. Transferable skills can be developed through experience over time [21]. Cultivating these skills enables cybersecurity professionals to effectively address and mitigate cyber threats, protect assets, and strengthen organizational resilience against cyberattacks. In cybersecurity, essential transferable skills include communication, problem-solving, teamwork, analytical thinking, and writing skills [22] – skills which are considered critical for professionals to perform their roles successfully.

Among the essential transferable skills, effective communication, particularly written communication, is a key prerequisite for success in many areas of life and is essential for career advancement [23]. The ability to interpret complex situations and convey insights clearly and concisely to support informed decision-making is highly valued by employers. The ISC2 Cybersecurity Workforce report [22] highlights the importance of effective communication among cybersecurity professionals, senior management, and board members, emphasizing the need to present facts in accessible language when interacting with diverse audiences [24]. Beyond verbal communication, it is equally essential to emphasize the development of strong writing skills.

Written communication is a key function in business environments, taking various forms ranging from emails and technical reports to memos and press releases. Hence, it plays a crucial role in ensuring clarity and professionalism during internal and external communications. In complex fields such as cybersecurity where clarity, accuracy, accountability, and professionalism are imperative, written communication skills are vital for promoting effective information sharing, risk mitigation, and operational consistency [25]. Cybersecurity professionals must articulate and describe complex technical findings to both technical and non-technical audiences, ensuring that vulnerabilities, risks, and mitigation strategies are accurately understood and appropriately acted upon. Clear and concise cybersecurity technical documentation standardizes protocols across the organization, enabling all employees to follow procedures that protect sensitive information and systems. Additionally, well-structured reports – such as vulnerability assessments and penetration testing reports – serve as formal records, helping organizations assess their security posture over time and comply with regulatory requirements. Well-structured security policies and plans play a crucial role in incident response, providing a clear reference during security events and reducing response time by guiding teams through established protocols. Accessible documentation also supports collaboration between technical teams and other stakeholders, promoting a culture of security awareness and facilitating faster, more coordinated responses

to security incidents, ultimately enhancing operational efficiency. In a field where miscommunication can lead to significant security lapses, strong writing skills are invaluable for articulating detailed insights, fostering trust with stakeholders, and enhancing organizational resilience against cyber threats.

C. ENISA ECSF Communication Skill Needs

The ECSF, developed by the European agency for cybersecurity (ENISA), aims to define the skills relevant for various cybersecurity roles. ECSF defines 12 cybersecurity roles and lists communication as a key competence for all of them (Table I), showcasing its importance for cybersecurity experts [11]. The communication skills needed differ between roles but typically include both the ability to write technical reports and communicating technical concepts to non-technical audiences such as stakeholder boards. While technical skills are important for the cyber workforce, they are often overemphasized and the ability to communicate with different stakeholders is often overlooked [2]. When participants at Black Hat 2016 and DEF CON 24 were asked about the most important skills for their job, communication was rated as a core skill, but also one that was rarely included in education programs [26]. Although the need for communication skills in the cybersecurity field has been advocated for years, it is still largely considered a service-oriented field, suggesting that more emphasis should be placed on discipline diversity [24]. Cybersecurity professionals must be able to advocate for security at different levels of the organization, from the board of managers down to each employee. Considering that the human aspects in cybersecurity are increasingly emphasized, the focus on cultivating transferable skills in a systematic way should be elevated.

TABLE I. COMMUNICATION SKILLS IN ECSF

ECSF Profile	Communication Skills	Deliverables
#1 CISO	Communicate, coordinate and cooperate with internal and external stakeholder, Develop, champion and lead the execution of a cybersecurity strategy	Cybersecurity strategy, cybersecurity policy
#2 Cyber incident responder	Communicate, present and report to relevant stakeholders	Incident response plan, cyber incident report
#3 Cyber legal, policy & compliance officer	Explain and communicate data protection and privacy topics to stakeholders and users	Compliance manual, compliance report
#4 Cyber threat intelligence specialist	Conduct technical analysis and reporting, Communicate, coordinate and cooperate with internal and external stakeholders, Communicate, present and report to relevant stakeholders	Cyber threat intelligence manual, cyber threat report
#5 Cybersecurity architect	Communicate, present and report to relevant stakeholders, Design systems and architectures based on security and privacy by design and by defaults cybersecurity principles	Cybersecurity architecture diagram, cybersecurity requirements report
#6 Cybersecurity auditor	Communicate, explain and adapt legal and regulatory requirements and business needs	Cybersecurity audit plan/report

#7 Cybersecurity educator	Develop evaluation programs for the awareness, training and education activities, Communicate, present and report to relevant stakeholders	Cybersecurity awareness program, cybersecurity training material
#8 Cybersecurity implementer	Communicate, present and report to relevant stakeholders	Cybersecurity solutions
#9 Cybersecurity researcher	Communicate, present and report to relevant stakeholders	Publication in cybersecurity
#10 Cybersecurity risk manager	Communicate, present and report to relevant stakeholders	Cybersecurity risk assessment report / risk remediation action plan
#11 Digital forensics investigator	Explain and present digital evidence in a simple, straightforward and easy to understand way, Develop and communicate, detailed and reasoned investigation reports	Digital forensics analysis results, electronic evidence
#12 Penetration tester	Communicate, present and report to relevant stakeholders, Conduct technical analysis and reporting	Vulnerability assessment results report, penetration testing Report

D. AI in Cybersecurity

AI has become deeply integrated into the field of cybersecurity [6], with AI applications typically centered on enhancing technical capabilities such as predicting future cyberattacks [27], constructing intelligent models for malware analysis and classification, real-time intrusion detection [6], and threat intelligence sensing [28], and supporting smart cybersecurity services and management [29]. While these applications strengthen defenses against cyber threats, they also highlight a growing need for upskilling, as the efficiency of AI-driven tools emphasizes the importance of advancing human capabilities to effectively manage and mitigate evolving risks arising from AI itself [27]. Thus, the focus on human empowerment and upskilling has become crucial in AI-cybersecurity research, emphasizing the importance of “bringing humans into the loop” [30]. Despite AI’s extensive use in technical applications, and studies highlighting the importance of incorporating AI into network security curricula at the undergraduate level [6], genAI’s potential for enhancing cybersecurity education and upskilling in the context of postgraduate education remains largely unexplored, highlighting an area well-suited for further research. This study is motivated by these gaps, aiming to investigate the application of genAI to support skills development in cybersecurity.

III. METHODOLOGY

This research addresses the challenges of an evolving cybersecurity landscape and the emerging role of genAI in education, by proposing a postgraduate cybersecurity curriculum that balances genAI and critical thinking to develop learners’ written communication skills and foster a mindset focused on reflection and critical analysis. The goal is to develop a more informed and critically engaged cybersecurity workforce. Specifically, the proposed curriculum focuses on developing written communication competencies in the context of penetration testing.

To address the research objectives, background research was initially conducted, to explore alternative pedagogical

frameworks and written communication needs in cybersecurity and guide the design of the curriculum. Background research informed curriculum development, guiding the selection of key learning topics to address critical aspects of written communication essential for penetration testers. The curriculum was designed following constructivist learning principles and Bloom's taxonomy, gradually introducing learners to both theoretical and practical components to foster skill development. The curriculum was implemented with a cohort of 60 learners enrolled in a distance learning program as part of a postgraduate cybersecurity course focusing on penetration testing. Following the implementation of the curriculum, an anonymous evaluation questionnaire was administered to assess the curriculum's effectiveness and usefulness in developing written communication skills relevant to penetration testing. In addition to quantitative data highlighting the students' assessment of different aspects of their learning, qualitative insights were captured through an open-ended question. In the analysis below verbatim quotes are used to enrich the discussion of the findings. The combination of different types of data provided a comprehensive understanding of students' learning experiences. Overall, the gathered feedback provided valuable insights for refining the curriculum and motivating future research in this area.

IV. CURRICULUM DESIGN

This section discusses the scope of the curriculum and the target audience, the pedagogical philosophy and the learning objectives that informed the curriculum design. Finally, the curriculum thematic areas are presented.

A. Audience and Scope

The proposed curriculum focuses on postgraduate cybersecurity education delivered through distance learning. Postgraduate programmes in cybersecurity often include individuals who have completed a computing-related Bachelor's degree, professionals that pursue a career change and originate from different backgrounds or cybersecurity professionals interested in advancing their career. This means that programmes include cohorts with varying degrees of knowledge and skills, and this should be considered when designing curricula for postgraduate cybersecurity education, so all learners stay engaged and motivated independent of their expertise [31]. Specifically, the learning content should be structured and developed in a way that will empower all learners to master their written communication competences to a professional level.

B. Penetration tester career role

Penetration testers, also known as ethical hackers, play a key role in cybersecurity. Their mission is to simulate cyberattacks to identify potential vulnerabilities in an organization's systems, networks, and applications, utilizing a range of tools and techniques [32]. Their primary goal is to uncover security weaknesses before malicious actors can exploit them, while also assessing the potential severity of any vulnerabilities that could be exploited [11]. Upon completing a penetration test, penetration testers provide a vulnerability assessment report and a penetration testing report.

To be effective in their roles, penetration testers require a range of transferable skills, including analytical thinking, attention to detail, and the ability to collaborate effectively with others [11]. Among the most essential skills are

communication abilities, particularly in written communication. Clear and effective communication enables penetration testers to present their findings in a way that ensures vulnerabilities are fully understood and addressed appropriately by organizations [33]. Written communication is especially crucial for preparing and presenting detailed reports of the penetration test findings. These reports must be clear, precise, and actionable so that the organization can take the necessary steps to remediate identified vulnerabilities and strengthen its security posture.

C. Curriculum pedagogical philosophy

Cultivating written communication skills in cybersecurity, especially in the context of penetration testing, can be challenging as an individual needs to have a clear view of different aspects that need to be synthesized to produce a professional-level report. In the genAI era, this challenge can become even greater if the learners are not mindful of inherent biases, ethical considerations [16] and the risks that emerge when over-relying on the capabilities of genAI, without exercising their critical thinking in the process. New curricula should focus on empowering learners to navigate these risks by becoming knowledgeable of the written communication aspects that can lead to a professional-level report and by developing a high-degree of cognitive skills when utilizing genAI.

Given the potential of genAI tools as well as the importance of critical thinking, the authors suggest that purposefully balancing these elements can contribute to achieving the required learning outcomes. The proposed curriculum draws threads from constructivist learning and Bloom's Taxonomy of cognitive thinking skills with the aim to create engaging cybersecurity learning experiences. On one hand, constructivism is based on the tenet that learners construct knowledge based on their experiences. According to Piaget [34], constructivism theoretically captures how individuals build, acquire, or construct their knowledge through their real-world experiences and interactions. Hence the learners, rather than the educators, are the centre of constructivist pedagogies. Learning environments based on constructivist philosophy present genuine opportunities for learners to generate their own meaning and encourage them to establish connections between new knowledge and prior experiences. This emphasises the role of active student engagement, critical thinking, reflective reasoning, and self-directed learning, rather than passively consuming information as traditionally done in many classrooms [35]. We believe this fits well specifically in postgraduate education. On the other hand, Bloom's Taxonomy defines six cognitive levels of learning, starting with remembering basic knowledge to combining new and prior knowledge to form new concepts or create new artefacts [36]. The emphasis of both models on creation and construction is well-aligned with the learning objectives of a curriculum on penetration testing which emphasises written communication skills, as explored in this study. By combining constructivist learning and Bloom's taxonomy, the proposed curriculum aims to create a dynamic and effective cybersecurity education experience that equips learners not only with the knowledge but also with the written communication skills they need for performing their versatile roles.

D. Learning Objectives

The proposed curriculum (Table II) follows the guidelines provided in [37] to formulate its learning objectives. Emphasis

was placed on ensuring that learners develop a holistic understanding of the penetration testing reporting process. Each objective is designed to help learners grasp core aspects of the reporting process, from understanding the scope and reporting requirements to applying best practices in written communication. Additionally, with the growing role of genAI in education and industry, specific objectives target the responsible use of AI tools, encouraging learners to critically evaluate AI-generated content and integrate it effectively into their work.

The learning objectives are structured using constructivism principles and Bloom's taxonomy to progressively develop cognitive skills. To achieve this, the curriculum begins by focusing on the lower cognitive layers of Bloom's taxonomy, to build a strong foundation regarding written communication aspects in penetration testing. As learners advance, the curriculum shifts to higher-order thinking skills, including analysis, evaluation, and creation, enabling them to apply their knowledge critically, synthesize information, and produce professional-level penetration test reports (often called pentest reports). This scaffolded approach ensures a gradual, yet comprehensive mastery of the diverse skills required for effective written communication, ultimately preparing learners to meet the specific communication demands of the penetration tester career role.

TABLE II. LEARNING OBJECTIVES

ID	Learning Objective	Bloom's taxonomy
LO1	Recognize the scope and professional obligations involved in conducting and reporting a pentest	Remember
LO2	Explain how the scope of the pentest informs the reporting requirements, ensuring alignment between the test's objectives and the content of the final report	Understand
LO3	Distinguish between the expectations of senior management and the technical team to ensure the report content is tailored to meet the specific needs of each audience	Understand
LO4	Summarize the core qualities of effective written communication in a pentest report	Understand
LO5	Explain the risks associated with using genAI in written communication and the importance of developing cognitive skills to critically evaluate AI-generated content	Understand
LO6	Apply the appropriate structure and content to a pentest report, based on audience expectations	Apply
LO7	Use a report template to create a well-structured and professional-level pentest report	Apply
LO8	Experiment with genAI to enhance the executive summary of a pentest report	Apply
LO9	Analyze written communication practices in existing pentest reports to identify good and bad practices that influence the overall quality of a report	Analyze
LO10	Analyze the expectations of different audiences to determine and align content appropriately for the executive summary and technical sections of a pentest report	Analyze
LO11	Critically evaluate the quality of a pentest report and recommend improvements	Evaluate
LO12	Judge the effectiveness of genAI-enhanced content and determine where human revision is necessary	Evaluate
LO13	Create a comprehensive and professional pentest report by synthesizing information from an OSINT investigation and applying effective written communication practices	Create

LO14	Formulate effective prompt engineering techniques to enhance the quality of a pentest report	Create
------	--	--------

E. Curriculum Content Structure

The curriculum was structured considering four thematic areas as presented below.

1) Scope and professional obligations

Initially, the curriculum placed emphasis on the importance of the penetration testing scope and how it should be considered when documenting the test findings. This direction was considered critical for the learners to understand that they need to demonstrate a professional stand in terms of how they handle the overall testing and then how they report the findings. The scope of the penetration test is an important component that drives the activities to be performed, and which needs to be given the appropriate attention, both during the initial stages when the scope needs to be defined and subsequently when the testing team needs to document the findings.

The outcome of the scope definition phase is the Rules of Engagement (RoE), which outlines essential guidelines that direct the activities of penetration testers. These include the objective of the penetration test, the systems to be tested, those explicitly excluded, involved parties, the testing timeframe, and the permissible extent of exploitation. The curriculum covers these elements, along with penetration testing types that vary based on the information provided to the test team prior to testing. Accordingly, black, white, and gray box approaches were introduced, emphasizing the importance of clearly indicating in the report which information was provided in advance and what was gathered during the test. Overall, the content covered was carefully selected to help learners understand how the planning phase is closely linked to the report-writing process in penetration testing.

Additional professional and ethical considerations were emphasized in the curriculum. Learners were guided to understand that the report contains confidential information and must be handled with diligence, implementing all necessary measures to protect its integrity and confidentiality. Understanding professional obligations is crucial, as it ensures that learners recognize their responsibility to uphold ethical standards and protect sensitive information. This awareness not only safeguards client trust but also reinforces their credibility and integrity within the cybersecurity field.

2) Audience expectations and knowledge areas

The key objective of this thematic area was to enable the learners to structure their report based on the audience's expectations. The curriculum delivered content to highlight that the report is read by the senior management and by the technical team of the organization. The expectations of each audience are different regarding the information documented in the report. Considering the audience's expectations, the learners should demonstrate different competences to effectively document their findings and meet expectations.

Specifically, the senior management is interested in the overall security posture of the organization, and the impact that the organization will face due to exploitation that might happen of vulnerabilities that have been identified during the test. Typically, this audience does not have technical knowledge, thus technical jargon should be avoided. Moreover, it is expected that high-level recommendations will

be provided, considering the severity of the identified vulnerabilities. The senior management expectations call for specific competences that the learners should possess. It should not be taken for granted that the learners have a clear view of the knowledge areas and skills they should cultivate. These competences were identified from background research and discussed in the curriculum. Specifically, learners should be knowledgeable regarding the organization's business context. Even though they might have technical expertise, at the same time they need to identify the valuable assets of the organization and the impact that might occur if an asset is affected. This knowledge is essential so they can discuss this aspect in their report and convey the situation to the senior management. Another knowledge area that is essential for the purpose of report writing is risk assessment, as the learners need to be able to prioritize the risk and provide appropriate recommendations. This means that they need to be able to evaluate the severity of the vulnerabilities and to do so, they need to specify the relevant severity scale that should be utilized. This is a core aspect in risk assessment.

The expectations of the organization's technical team are very different compared to those of the senior management. The technical team is interested in all the technical details that led to the identification and exploitation of the listed vulnerabilities. They need to be able to reproduce the steps taken during the penetration testing. This means that an appropriate level of detail should be provided, with clear and accurate information about the methodology that was followed and the tools that have been utilized. The analysis of the findings should be accompanied by appropriate screenshots taken during the testing process as proof of evidence of the vulnerabilities. Learners should realize that the screenshots serve as secondary sources of information, and that the primary source is the analysis they perform and document appropriately in the report. All screenshots should be accompanied by an appropriate caption and referenced directly in the report. Directions towards resolving the vulnerabilities should also be provided. Given the information that the technical team expects the report to contain, the penetration testers must demonstrate proficiency across several knowledge areas. These include identifying various types of vulnerabilities, evaluating them based on severity and impact, and prioritizing them effectively. Additionally, the penetration testers must be able to provide recommendations to address specific vulnerabilities. To maintain proficiency in a constantly shifting cyber threat landscape, it is also essential to locate and utilize diverse information sources, ensuring they remain current with emerging threats and best practices.

In this thematic area, activities were designed to help learners identify poor writing practices specific to the executive summary and technical analysis sections. By recognizing these ineffective practices in contrast to professional standards, the target was for learners to gain a clearer understanding of how a poorly written report that fails to meet expectations can leave potential clients dissatisfied and damage their reputation.

3) *Report structure and content*

A penetration testing report should be structured into two parts, the executive summary and the technical analysis. Considering the information that needs to be reported under each part, structuring the report in a logical way is essential to effectively present the findings to the intended audience. In terms of the report's structure, the target should be to achieve

a logical flow of information, providing a clear walk-through of the steps that have been taken. This can promote reproducibility of tasks. Organizing the presentation of findings in a consistent approach is also necessary, producing a professionally looking report. Results that can be grouped together, for example port scanning results for several hosts, should be presented in a similar way. This means that relevant sections and subsections should all have the same structure. Sectioning should be visible, thus numbering sections is advisable. Furthermore, various elements, such as tables and graphs, can be used to effectively present the findings, enhancing the report's readability and supporting decision-making at both business and technical levels. This approach helps convey critical information clearly, enabling the organization to address findings efficiently and improve its security posture. Additionally, the table of contents, list of figures, and list of tables should accurately reflect the report's structure and content, free from indexing errors.

Organizing penetration testing notes is crucial to be able to handle report writing and structure it in an effective and efficient manner. Different tools have been presented, discussing how they can assist in the development of structured notes. The value of creating a template for note taking and for the overall report has also been highlighted. Understanding that a proactive, repeatable note-taking process can help standardize sections of the report and save time during the writing phase is key to producing consistent, high-quality documentation. This approach not only streamlines the report-writing process but also ensures accuracy and clarity, enabling the final report to meet professional standards effectively.

Overall, the curriculum gave emphasis on discussing the core written communication qualities needed to report the findings effectively, conveying the key details of the test that are of interest to the senior management and the technical teams, demonstrating a professional approach. Specifically, core qualities of written communication that have been covered in the curriculum include: 1) Appropriate – making sure that it has the right tone and the necessary level of formality, 2) Comprehensive – it includes the expected details, 3) Presentation – it is consistent in terms of formatting aspects and has correct spelling and grammar, 4) Accurate and Clear – it includes correct information, and it is understandable.

The curriculum covered all aspects that might be ignored during report writing, and which can affect the report's readability and value, even though the actual testing was performed with accuracy. The quote "If you do not document it, it did not happen", reflects how important it is to handle all aspects of the reporting phase with attention and professionalism.

A practical activity was conducted in which learners performed an OSINT investigation and documented their findings in a penetration testing report. They applied a report template to structure the content, addressing the needs of two distinct audiences: senior management and the organization's technical team. After creating the report, learners engaged in a quality control process, evaluating the report's clarity, accuracy, and alignment with professional standards.

4) *GenAI competences*

The final thematic area of the curriculum focused on genAI to familiarize learners with its capabilities and applications in professional report writing. Learners were

introduced to prompt engineering within the context of report writing in cybersecurity, exploring how genAI can enhance key elements such as structure, language, syntax, and summary quality to produce a professional-level report. Additionally, the curriculum addressed the risks associated with genAI, including over-reliance on AI-generated content and potential impacts on writing and critical thinking skills. Emphasis was placed on the importance of cultivating cognitive skills—such as critical thinking, attention to detail, and analytical abilities—which genAI can support but should not replace. Learners were cautioned that lack of foundational skills may lead to an overdependence on genAI, ultimately undermining proficiency and professionalism in the long run.

An activity was conducted to explore how generative AI can be leveraged to enhance the executive summary of a report. Learners were asked to reflect on the changes made, identifying specific aspects of the original report that were improved, such as clarity, conciseness, and overall readability. A key point of their reflection focused on determining where human revision is necessary to ensure accuracy, maintain the intended tone, and address specific areas that AI may overlook. This exercise aimed to assist learners critically assess the balance between AI-generated content and human oversight in producing a professional-level report.

TABLE III. LEARNING CONTENT AND PEDAGOGY

#	Learning Content (LC)	Pedagogy
1	Penetration testing scope and professional obligations	L/CD
2	Audience expectations	L/SGA
3	Report structure and content	L/CSA/CD
4	GenAI competences	L/D/PA

L= Lecture, CD=Critical Discussion, CSA= Case Study Analysis, D=Demonstration, SGA=Small Group Activity, PA= Practical Activity

V. EVALUATION

The proposed curriculum was evaluated using a questionnaire that included a set of multiple-choice questions and an open-ended question. The questionnaire was completed by 60 learners. The purpose of the questionnaire was to evaluate the delivered content and its effectiveness to enhance the learners' written communication skills. Responses were submitted anonymously, and no sensitive information was recorded as part of the responders' feedback.

Initially, it was important to acquire a clear view of whether the participants had prior experience with writing a penetration testing report and the context in which they developed the report (Fig.1). The results indicate that the majority of participants (74%) had not previously created a penetration testing report before engaging in the course unit. Among those with prior experience, most had developed penetration testing reports in an academic setting, with 4% doing so at the undergraduate level and 10% at the postgraduate level. Interestingly, only a small percentage (3%) had created a report independently or for certification purposes.

A core aspect of the evaluation was to assess the usefulness of the curriculum content to assist learners realizing good and bad documentation practices and writing a penetration testing report that meet professional standards. Five topical areas have been included in the evaluation scope (Fig.2) which are mapped to key topics covered by the curriculum:

- Presentation elements (cover page, graphics, text consistency, tables, etc.)
- Report information (title, version, author, reviewed/approved by, classification, version control, etc.)
- Executive summary structure (information to include, practices to avoid)
- Methodology (reference to widely recognized standards/guidelines, stages, vulnerability severity scale, tools, etc.)
- Technical analysis structure (documentation of vulnerabilities, recommendations, screenshots, appendices, etc.)

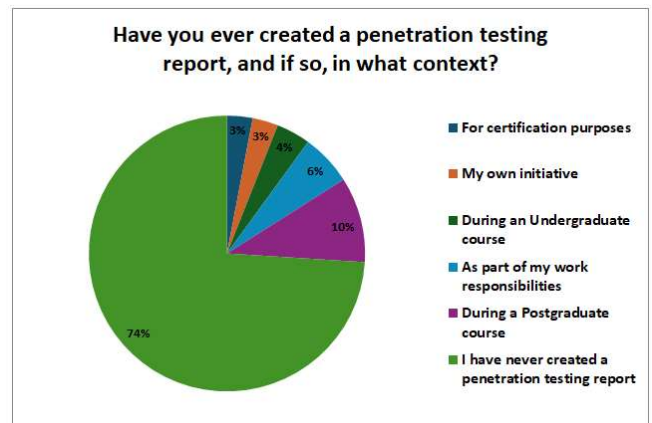


Fig. 1. Creation of a penetration testing report.

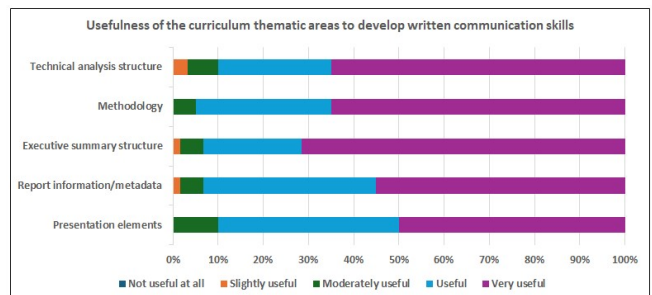


Fig. 2. Usefulness of the curriculum thematic areas.

Results reveal that, across all categories, a significant majority of respondents found the provided material and discussion to be either "Useful" or "Very Useful" in developing their understanding of effective documentation and presentation practices in penetration testing (Fig.2). The highest rating for "Very Useful" responses were found in Executive Summary Structure (71.7%), indicating that this is an area that is particularly impactful for learners aiming to master professional reporting standards. The inclusion of Methodology and Technical Analysis Structure, also scored highly, with 65% rating these areas as "Very Useful." Although the Presentation Elements category was rated "Very Useful" by 50% of the respondents, this percentage was lower compared to previous categories, with 40% rating it as "Useful" and 10% as "Moderately Useful." Report Information had a similar percentage of "Very Useful" ratings (55%). Interestingly, a very small percentage that ranged between 1.7-10%, rated some of the categories as "Slightly Useful" and/or "Moderately Useful," with Report Information

and Executive Summary Structure areas receiving minimal “Slightly Useful” feedback (1.7%).

Learners also responded to how useful the learning material and activities were to develop their skills and knowledge in written communication and to effectively present results from a penetration test (Fig. 3). The feedback indicates a strong positive response to the material and activities provided, with a substantial majority of participants (91%) agreeing or strongly agreeing that these resources were useful in enhancing their skills and knowledge in documentation and professional report writing for OSINT-based findings. With 53% “Strongly Agreeing” and another 38% “Agreeing,” it is a good indication that the curriculum’s approach effectively supported learners in achieving a high standard of professional documentation skills in the context of penetration testing.

This positive outcome suggests that the curriculum’s content, focused activities, and hands-on approach were well-aligned with the learners’ needs for professional development. The small percentage of “Slightly Disagree” (2%) and “Neutral” (7%) responses may indicate minor gaps or individual variations in learning preferences or prior knowledge levels. Such feedback can inform potential adjustments, such as incorporating even more diverse instructional methods or tailored support, to ensure all learners fully benefit from the curriculum.

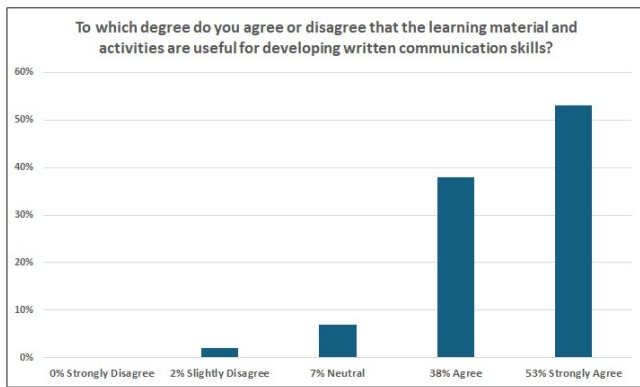


Fig. 3. Curriculum effectiveness.

The questionnaire also included an open-ended question inviting learners to share additional feedback. Some representative comments extracted from the responses include: “sufficient material to bring us as close as possible to a professional level”, “the material provided and taught in the course serves as a very good guide for creating a penetration testing report”, “...it was very interesting... we gained considerable knowledge, especially on how to structure a proper and well-documented investigation”, “helped us understand poor practices that may exist”, “the coverage of topics aligns perfectly with the focus on creating a detailed and effective report”. Overall, the results validate the effectiveness of the curriculum in meeting professional documentation and reporting expectations within the context of penetration testing, demonstrating that most learners felt equipped to produce reports that meet industry standards.

VI. DISCUSSION

A. Balancing genAI and Critical Thinking for Developing Written Communication Skills

Going beyond traditional teaching methods contributes to ensuring educational programmes remain resilient in the face of evolving challenges [4] by emphasising upskilling and reskilling [5][18]. In the evolving field of cybersecurity education, the integration of genAI alongside critical thinking is emerging as a powerful approach to developing key written communication skills among other competencies. The writing assistance features of genAI tools, when purposefully embedded in the learning process, can enhance learners’ understanding of their capabilities, limitations and autoregressive nature [15], hence becoming more critical of their own writing skills. As genAI continues to influence various domains, its potential to enhance educational experiences, particularly in postgraduate settings, is under active exploration. In the context of teaching how to produce effective cybersecurity documentation, genAI can provide substantial support in structuring and refining complex content, while critical thinking remains essential for learners to evaluate and ensure accuracy, clarity, and appropriateness. Combining genAI with a focus on critical analysis and theoretical exploration [6] equips learners with the ability to leverage technology responsibly and effectively, promoting both skill development and a deeper understanding of professional standards in cybersecurity communication. To be effective, this approach should be implemented in a systematic and structured manner.

Given the diversity of experience levels common in postgraduate education, cybersecurity cohorts often include recent bachelor’s graduates continuing with postgraduate studies, professionals pursuing career changes, and those already working in the field who wish to advance their competencies. This range of experience must be carefully considered to create a curriculum that caters to the diverse needs of learners. Evaluation results indicated that most learners entered the module with limited practical experience in professional documentation specific to penetration testing. This finding highlights the importance of incorporating foundational elements of report writing into a curriculum focused on developing written communication skills, especially for learners unfamiliar with industry-standard cybersecurity reporting practices.

Additionally, findings suggest that while some learners acquired relevant experience through informal education, this is less common, with academic and workplace settings remaining the primary avenues for developing transferable skills. These results highlight the need to design curricula that emphasize professional writing standards and systematically build relevant written communication skills within higher education. For learners with limited experience in penetration testing report creation, a course should prioritize documentation, structure, and content requirements to build foundational competencies. For those with prior experience, the curriculum can serve to refine and elevate existing skills, ensuring that all learners meet a consistent professional standard in reporting practices.

This structured approach can effectively balance the benefits of genAI with the essential skill of critical thinking to develop effective written communication skills that meet professional standards. By integrating genAI, learners can

access tools that help them structure reports and enhance their language precision. The emphasis on critical thinking encourages learners to evaluate AI-generated content, identifying areas where human insight is crucial to meet industry-specific requirements. This dual focus prepares learners not only to utilize AI responsibly but also to refine their skills independently, resulting in a curriculum that addresses the diverse competencies within the cohort and aligns with cybersecurity professional standards.

B. Learning Topics

Findings indicate that high value is placed on Executive Summary Structure, which may reflect its importance in aligning report content with the expectations of different audiences, particularly in distinguishing between content suitable for senior management versus technical stakeholders. Similarly, the emphasis on Technical Analysis Structure suggests that learners recognize the value of documenting vulnerabilities, recommendations, and supplemental details like screenshots, which are crucial for a comprehensive and professional penetration testing report. Based on learners' perception, Methodology scores second in terms of usefulness, alongside Technical Analysis Structure. This highlights the importance of emphasizing widely accepted standards and guidelines in the curriculum, to ensure consistency and credibility in reporting. Less emphasis was given to Presentation Aspects and Report Information. This may indicate that while presentation is essential, learners may view it as secondary to content-specific components, such as the executive summary and technical analysis. Similarly, given that Report Information received a lower percentage of being useful compared to all the other categories, this might suggest that while it is essential for professionalism, report metadata is considered less impactful on the overall quality of the report's substantive content. A few learners perceived some of the categories as slightly useful or moderately useful. This could reflect diverse learning needs or existing knowledge among learners. Authors will investigate this aspect further as part of their future work.

Overall, the results suggest that the curriculum effectively targets the critical areas learners perceive as most valuable for professional-level reporting, particularly in structuring and presenting both technical findings and executive summaries. Although all categories scored high, it is important to recognize that secondary elements, such as presentation aspects, play a significant role in establishing effective written communication. These elements contribute to the clarity, professionalism, and overall readability of the report, enhancing the impact of the technical content and ensuring that the report meets professional standards.

C. Limitations

One limitation of this work is the small cohort size, which affects the generalizability of the findings. With a limited sample it is challenging to fully understand how different experience levels might shape learners' understanding and expectations of professional written communication standards. Further research with a larger and more diverse cohort would provide valuable insights into how prior experience influences learners' perceptions and competencies, allowing for a more tailored approach in curriculum design to meet the needs of learners with varying levels of experience.

D. Future directions

The approach utilized in this study provides a solid foundation for identifying essential focus areas to effectively support the development of written communication skills in cybersecurity. The proposed approach can inform the creation of targeted learning activities that address both foundational and advanced competencies, ensuring a comprehensive skill-building experience. Future interventions should focus on creating adaptive and resilient curricula that accommodate the diverse experience levels of learners. Integrating real-world scenarios, such as case studies and simulated reporting tasks, can help bridge the gap between academic learning and industry expectations. Additionally, expanding the use of genAI tools in a controlled environment can support learners in refining their writing skills while fostering critical thinking about AI-generated content. Innovative cybersecurity programmes that emphasize personalized cybersecurity learning [17], design AI-augmented hands-on learning activities [6], foster intensive training and reskilling of cybersecurity professionals [5], and seamlessly integrate academic pursuits with real-world industry skill needs [4], present learners with unique opportunities to learn about cybersecurity while benefiting companies in terms of future talent recruitment [4]. Thus, they contribute to efforts towards addressing the skills gaps in the field of cybersecurity [22] and the broader cybersecurity talent shortage [5][6][37][38].

VII. CONCLUSIONS

Written communication skills are crucial for current and future generations of cybersecurity professionals. Even if there are diverse roles in the cybersecurity industry, a consistent task is to deliver written information to different stakeholders. This task encompasses a range of skills that cybersecurity professionals are expected to cultivate through formal or informal education, and through working experience. Current cybersecurity education programmes often overlook the importance of written communications skills and, as a result, they may fail to equip learners with this competence. Acquiring written communication skills in a consistent manner through working experience is also challenging if proper mentoring is not offered, especially for junior employees. In this paper, we are addressing the challenges of developing written communication skills for cybersecurity professionals and the challenges of utilizing generative AI. We do so by developing a curriculum where learners enhance their critical thinking skills while exploring the use of generative AI as a tool to help improve written communication in the context of penetration testing reports.

Our evaluation shows that by combining constructivist learning principles and Bloom's levels of cognitive learning, we can create a dynamic and effective cybersecurity education experience that equips learners with the knowledge and skillset they need to excel in their versatile roles. The evaluation also shows that balancing genAI-infused curricula with activities that trigger critical thinking can help postgraduate students enrich the written communication competencies needed to thrive in the everchanging cybersecurity jobs landscape. We envision that the proposed curriculum can inspire educators across the world and provide guidance on how to integrate communication skills into existing education plans in a novel way. As part of our future work, we plan to conduct studies with a larger sample size and employ mixed methods to assess the long-term impact of the

curriculum on learners' written communication skills and professional competency in cybersecurity.

REFERENCES

- [1] J. L. Hall and A. Rao, "Non-Technical skills needed by cyber security graduates," in 2020 IEEE Global Engineering Education Conference (EDUCON), 2020.
- [2] J. Dawson and R. Thomson, "The future cybersecurity workforce: Going beyond technical skills for successful cyber performance," *Front. Psychol.*, vol. 9, p. 744, 2018.
- [3] C. Bhuman, C. Khawar, "Fostering Critical Thinking in the AI Era: Innovative Educational Approaches for a Data-Driven Society," 2024.
- [4] J. Rajamäki, P. Rathod and P. Kämpfi, "Integrating International Research-Innovation Projects and Working Life Partners into Cybersecurity Degree Programme," 2024 IEEE Global Engineering Education Conference (EDUCON), Kos Island, Greece, 2024, pp. 1-9, doi: 10.1109/EDUCON60312.2024.10578728.
- [5] G. Karlsson, "From Campus to Boot Camp — Lessons from Extramural Teaching in Cybersecurity," 2024 IEEE Global Engineering Education Conference (EDUCON), Kos Island, Greece, 2024, pp. 1-8, doi: 10.1109/EDUCON60312.2024.10578866.
- [6] B. Alomar, Z. Trabelsi, T. Qayyum and M. M. Ambali Parambil, "AI and Network Security Curricula: Minding the Gap," 2024 IEEE Global Engineering Education Conference (EDUCON), Kos Island, Greece, 2024, pp. 1-7, doi: 10.1109/EDUCON60312.2024.10578888.
- [7] C. Maurer, M. Sumner, D. Mazzola, K. Pearson, and T. Jacks, "The Cybersecurity Skills Survey: Response to the 2020 SIM IT Trends Study", In *Proceedings of the 2021 on Computers and People Research Conference*, pp. 35-37, 2021.
- [8] O. Ozyurt, A. Ayaz, "Identifying cyber security competencies and skills from online job advertisements through topic modelling", *Security Journal*, pp.1-21, 2024.
- [9] C.M. Graham, Y. Lu, "Skills expectations in cybersecurity: semantic network analysis of job advertisements", *Journal of Computer Information Systems*, 63(4), pp.937-949, 2023.
- [10] C. Somers, E. Byrne, "Cyber Security Skills Report 2021 National Survey", *Cyber Ireland*, 2021. [Online]. Available: <https://www.cyberireland.ie/wp-content/uploads/2021/02/Cyber-Ireland-Skills-Report-2021.pdf>
- [11] ENISA, "European Cybersecurity Skills Framework Role Profiles", 2022. [Online]. Available: <https://www.enisa.europa.eu/publications/european-cybersecurity-skills-framework-role-profiles>
- [12] N. McNulty, Bloom's Taxonomy Reimagined: Digital Strategies for Today's Teachers. 2020.
- [13] G. White, "Higher education model for security literacy using bloom's revised taxonomy," *Cybersecurity Pedagogy and Practice Journal*, pp. 27-36, 2024.
- [14] M. M. Thiga, "Generative AI and the Development of Critical Thinking Skills," *Iconic Research and Engineering (IRE) Journals*, vol. 7, no. 9, pp. 83-90, 2024.
- [15] S. Hammer, S. Ottinger, B. Zönnchen, M. Hohendanner, M. Hobelsberger and V. Thurner, "ChatGPT in Higher Education: Perceptions of Computer Science-Related Students," 2024 IEEE Global Engineering Education Conference (EDUCON), Kos Island, Greece, 2024, pp. 01-08.
- [16] J. Otterbacher and Y. Manolopoulos, "Machine Ethics Research: Promises and Potential Pitfalls," *IEEE Intelligent Systems*, vol. 38, no. 4, pp. 62-68, 2023.
- [17] C. Kallonas, A. Piki, E. Stavrou, "Empowering Professionals: A Generative AI Approach to Personalized Cybersecurity Learning", *IEEE Global Engineering Education Conference (EDUCON)*, pp. 1-10, Kos Island, Greece, 2024.
- [18] E. Stavrou and A. Piki, "Cultivating self-efficacy to empower professionals' re-up skilling in cybersecurity," *Inf. Comput. Secur.*, vol. 32, no. 4, pp. 523-541, 2024.
- [19] S. S. Lim, T. Makany, "Deploying chatbots to build students' critical thinking skills: Leveraging generative AI effectively and purposefully in higher education," in *Encyclopedia of Educational Innovation*, M. A. Peters and R. Heraud, Eds. Springer, 2023.
- [20] H. B. Essel, D. Vlachopoulos, A. B. Essuman, and J. O. Amankwa, "ChatGPT effects on cognitive skills of undergraduate students: Receiving instant responses from AI-based conversational large language models (LLMs)," *Computers and Education: Artificial Intelligence*, vol. 6, 2024.
- [21] B. Tulgan, "Bridging the Soft Skills Gap: How to Teach the Missing Basics to Today's Young Talent," John Wiley & Sons, 2015.
- [22] ISC2, *Cybersecurity Workforce Study*, Report, www.isc2.org/research, 2023.
- [23] J.R. Sparks, Y. Song, W. Brantley, O. L. Liu, "Assessing Written Communication in Higher Education: Review and Recommendations for Next-Generation Assessment," *ETS Research Report Series*, no.2, pp.1-52, 2014.
- [24] J. M. Haney, W.G. Lutters, "Skills and Characteristics of Successful Cybersecurity Advocates," *In SOUPS*, 2017.
- [25] R. Augusto, *The Role of Technical Writing in Cybersecurity*. Doakio. <https://doakio.com/blog/the-role-of-technical-writing-in-cybersecurity/>, 2023
- [26] K. S. Jones., A. S. Namin, M. E. Armstrong, "The core cyber-defense knowledge, skills, and abilities that cybersecurity students should learn in school: Results from interviews with cybersecurity professionals," *ACM Transactions on Computing Education (TOCE)* 18, no.3, pp.1-12, 2018.
- [27] M. F. Ansari, B. Dash, P. Sharma, and N. Yathiraju, "The Impact and Limitations of Artificial Intelligence in Cybersecurity: A Literature Review," *International Journal of Advanced Research in Computer and Communication Engineering*, 2022.
- [28] J.-H. Li, "Cyber security meets artificial intelligence: a survey," *Front. Inf. Technol. Electron. Eng.*, vol. 19, no. 12, pp. 1462-1474, 2018.
- [29] I. H. Sarker, M. H. Furhad, and R. Nowrozy, "AI-driven cybersecurity: An overview, security intelligence modeling and research directions," *SN Comput. Sci.*, vol. 2, no. 3, 2021.
- [30] "Artificial Intelligence and Cybersecurity Research", ENISA, 2023. [Online]. Available: <https://www.enisa.europa.eu/publications/artificial-intelligence-and-cybersecurity-research>.
- [31] T. Crick, J.H. Davenport, P. Hanna, A.Irons, T. Prickett, "Overcoming the Challenges of Teaching Cybersecurity in UK Computer Science Degree Programmes", 2020 IEEE Frontiers in Education Conference (FIE), pp. 1-9, 2020.
- [32] M. Alhamed, and M.M.H. Rahman. "A Systematic Literature Review on Penetration Testing in Networks: Future Research Directions" *Applied Sciences*, 13, no. 12: 6986, 2023.
- [33] D. J. Beukes, "Evaluating the Cyber Security Skills Gap relating to Penetration Testing", Master Thesis, Rhodes University, 2021.
- [34] J. Piaget, "The Theory of Stages in Cognitive Development," in *Measurement and Piaget*, D. Green, M. P. Ford, and G. B. Flamer, Eds. New York, NY: McGraw-Hill, 1971, pp. 1-11.
- [35] H. V. Le and L. Q. Nguyen, "Promoting L2 Learners' Critical Thinking Skills: The Role of Social Constructivism in Reading Class," *Frontiers in Education*, vol. 9, 2024.
- [36] L. W. Anderson and D. R. Krathwohl, *A Taxonomy for Learning, Teaching, and Assessing: A Revision of Bloom's Taxonomy of Educational Objectives*. New York: Addison Wesley Longman, Inc, 2001.
- [37] E. Stavrou, "Planning for Professional Development in Cybersecurity: A New Curriculum Design," in *Human Aspects of Information Security and Assurance. HAISA 2023. IFIP Advances in Information and Communication Technology*, vol. 674, S. Furnell and N. Clarke, Eds. Cham: Springer, 2023.
- [38] H.-J. Kam, P. Menard, D. Ormond, and R. E. Crossler, "Cultivating cybersecurity learning: An integration of self-determination and flow," *Computers & Security*, vol. 96, no. 101875, p. 101875, 2020.

Adoption, Usage, and Perceptions of LLM-Generated Feedback in Computer Science Education: A Cross-Cultural Study of Students and Academics

1st Eyman Alyahyan

University of Glasgow

United Kingdom

e.alyahyan.1@research.gla.ac.uk

Imam Abdulrahman Bin Faisal University

Saudi Arabia

eaalyahyan@iau.edu.sa

2nd Mireilla Bikanga Ada

University of Glasgow

United Kingdom

mireilla.bikangaada@glasgow.ac.uk

3rd Jake Lever

University of Glasgow

United Kingdom

jake.lever@glasgow.ac.uk

Abstract—This full paper explores the use of Large Language Models (LLMs) for generating assessment feedback in Computer Science (CS) education, comparing adoption patterns and perceptions of students and academics in the UK and Saudi Arabia (KSA). The study investigates (1) differences in LLM use and adoption between students and academics in both countries, and (2) participants’ perceptions of LLM-generated feedback, focusing on benefits, challenges, and areas for improvement. A mixed-methods approach was employed, involving 251 participants. The collected data were analyzed through descriptive statistics and thematic analysis. Findings indicate that students in KSA demonstrate higher usage rates of LLMs, with ChatGPT identified as the most frequently used tool. Although KSA academics show limited use of LLMs, UK academics did not use them for feedback generation. Thematic analysis revealed three key themes: (a) *“It saves effort”*, (b) *“It is not as smart and genuine as a human”*, and (c) *“Making it more like humans and less like a machine”*. The findings emphasize the need for institutional policies to guide the responsible integration of LLM-generated feedback, ensuring alignment with teaching goals and improving feedback quality.

Index Terms—Computing education, Mixed methods research, Feedback, Higher education, perceptions, Large Language Model, Generative AI.

I. INTRODUCTION

Large Language Models (LLMs) are AI systems designed to interpret and generate human-like language using vast amounts of text data [1], [2]. Based on transformer architectures [3], these models are able to capture complex language structures and perform a variety of tasks. Techniques such as in-context learning [2] and reinforcement learning from human feedback (RLHF) [4] improve the quality, clarity, and relevance of their outputs. ChatGPT, in particular, has emerged as a widely recognized example, demonstrating strong abilities in natural language generation and flexibility in handling a wide range of

tasks without task-specific training [5]. Although LLMs have reshaped natural language processing, their adaptable features are now gaining attention in the field of education. A recent survey [6] grouped their educational applications into three key areas: helping students (e.g., answering questions, correcting grammar), supporting instruction (e.g., grading, content generation) and enabling personalized learning (e.g., knowledge tracing and adaptive pathways). These uses highlight the growing potential of LLMs in enhancing teaching and learning in higher education, with increasing interest in their use for formative feedback as an emerging area of research.

Feedback is widely recognized as a fundamental element of effective teaching and learning in higher education [7]. It plays a vital role in helping students assess their performance, identify areas of growth, and make significant progress in their learning journey [8]. This is particularly important in computer science (CS) education, where effective feedback plays a central role in helping students develop coding skills and solve problems [9], [10]. However, providing timely and personalized feedback remains a significant challenge, particularly due to logistical constraints such as large class sizes [11], [12]. To address this gap, recent studies have explored the use of generative AI tools, particularly LLMs, to automate and improve feedback provision in computing education [13]–[22].

Although these efforts have made substantial contributions by investigating LLM-generated feedback in real classroom settings and evaluating its accuracy and usefulness, most of them focus mainly on the technical aspects and student experience during the actual use of the tools. Moreover, there is still a limited understanding of how institutional and cultural factors may shape key perceptions of LLM-generated feedback in computer science education, particularly regarding its usage,

adoption, perceived usefulness, and satisfaction.

This study addresses this gap by investigating the adoption, use, and perceptions of LLM-generated feedback in computer science education, employing a cross-cultural design that includes participants from two distinct higher education systems. The study aims to contribute to a deeper understanding of how LLM-generated feedback is integrated and experienced in practice, shedding light on both its perceived benefits and the challenges that may hinder its effective use.

This paper is structured as follows: Section II reviews the literature, Section IV describes the methodology, Section V presents the results, Section VI discusses the key findings, and Section VII concludes the paper.

II. LITERATURE REVIEW

The growing use of generative AI tools in higher education has prompted researchers to investigate how students and educators perceive, adopt, and utilise these technologies, particularly in teaching, assessment, and feedback. The literature reflects a variety of perspectives shaped by institutional, cultural, and disciplinary factors.

In Saudi Arabia, several studies have demonstrated that students actively utilise tools like ChatGPT in their academic work. An extensive national study [23] reported that most students use generative AI to clarify concepts, summarize texts, translate content, and generate ideas. They found it helpful due to its speed and accessibility, but concerns were raised about subscription costs, unreliable output, and reduced learning independence. Similarly, a study of healthcare students [24] found high awareness of ChatGPT and generally positive attitudes. Many students believed that it improved their productivity and helped them learn more effectively. Among students of English as a Foreign Language (EFL), [25] highlighted that ChatGPT was considered moderately effective in improving language understanding and communication, especially due to its ease of use. Students were more willing to continue using it when they found it convenient and straightforward. [26] explored its use in writing courses, noting perceived benefits such as increased creativity, better learning experiences, helpful academic support, and increased productivity. However, the students also pointed out risks such as a lack of human tone, potential plagiarism, and anxiety about relying too much on AI.

In the UK, [27] found that computer science students frequently used ChatGPT, often on a weekly basis, to support their learning. They used it to access information, understand problems, and work more efficiently. Although students believed it was fair to use AI tools in academia, many were concerned about plagiarism and reported moderate trust in the accuracy of the tool. The study emphasized the need for students to have clear guidelines on responsible use. [28] found that postgraduate researchers used generative AI as a time-saving tool, a writing assistant, or a collaborative partner. Most users accepted its practical value, but there was uncertainty about its limitations and ethical boundaries. In another study [29], interviews with students and instructors

in computing education revealed shared views on the benefits of generative AI to improve learning and reduce workload. However, both groups raised concerns about overreliance and the trustworthiness of AI tools. The findings highlighted the need for curriculum adjustments that support the effective integration of AI tools while addressing potential risks such as plagiarism and enhancing equitable access to educational support.

Internationally, several studies offered comparative or multi-country insights. [30] compared students and educators in Australia, Cyprus, and the United States to evaluate the quality of online assessment. Both groups viewed essay and coding tasks as the most impacted by generative AI. Educators were more likely to suggest adapting assessments to encourage higher-order thinking, while students had mixed reactions, including concerns about reduced creativity. In another study, educators highlighted the value of using AI to support learning processes, but emphasized the importance of ethics, human interaction, and focusing on how students learn [31]. In the Middle East, [32] investigated how university instructors approached generative AI for assessment. Instructors reported positive views on generative AI because it helps reduce marking workload and increase student engagement. Many also described redesigning assignments to more effectively integrate AI tools. However, concerns were raised about the potential harm to student writing abilities and critical thinking. In the United States, [33] surveyed 328 college students and showed that ChatGPT's efficiency and compatibility positively influenced its ease of use, usefulness, and students' intention to keep using it, based on the Technology Acceptance Model (TAM).

Some studies also explored LLMs as feedback providers. [20] showed that students generally viewed ChatGPT-generated feedback on programming tasks as helpful and aligned with formative feedback principles. The students preferred feedback that included their code as it was more precise. The study also addressed practical concerns, including and more relevant. However, preferences varied with respect to the tone of feedback. Similarly, [16] presented a large-scale study in a global CS1 course where real-time style feedback generated by LLMs improved student engagement and prompted significant improvements in code quality. Students who received instant feedback were significantly more likely to revise their code, and the study also addressed practical concerns such as consistency, fairness, and demographic bias. In a more focused context, [19] examined how GPT-4 generated hints for compiler errors in an introductory programming course. Students reported reduced frustration and increased focus when receiving GPT-generated hints, and feedback was generally perceived as useful. Although performance gains were mixed depending on the error type, students still outperformed the control group in later stages without hints, suggesting a learning effect from earlier feedback.

Recent studies in Computing Education Research (CER) have highlighted several benefits of using LLM-generated feedback in higher education. These include the ability to

deliver personalized code correction [18], tiered or multilevel feedback to support students at different stages of learning [14], and real-time feedback on coding style [16]. Other studies have shown the usefulness of LLMs in generating automated next-step hints [15], responding to student help requests [13], offering consistent and reliable feedback [17], and improving students' ability to resolve syntax-related errors [19]. Despite these advantages, concerns persist regarding the reliability of AI-generated feedback. Several studies [13], [15], [18] have reported issues with inaccurate suggestions or hallucinated errors. Furthermore, ensuring that the feedback is adapted to the level and context of each student, often referred to as context-sensitive feedback, remains a challenge [16].

In summary, research on LLM-generated feedback in CS education remains limited, particularly across diverse cultural and academic contexts. This study addresses these gaps by exploring how such feedback is adopted, used, and perceived in two distinct higher education settings.

III. RESEARCH QUESTIONS

In light of the focus of the study, the following research questions were formulated.

RQ 1: How do students and instructors in computer science education in the UK and KSA use, perceive, and intend to adopt LLM-generated feedback?

RQ 2: What are the perceived benefits, challenges, and suggestions to improve the use of LLM-generated assessment feedback in CS education in the UK and KSA?

IV. METHODOLOGY

This study used a mixed-methods survey design [34] to examine the use of LLMs to generate formative feedback on computer science assessment tasks. Closed-ended items provided measurable data, while open-ended questions allowed participants to elaborate on their views, offering richer insights into perceived benefits, challenges, and future intentions associated with LLM-based feedback practices [35].

A. Participants

A total of 251 participants were involved in the study, comprising 174 students and 77 instructors from public universities in the UK and KSA. All participants came from disciplines related to computing, including computer science, software engineering, and information technology, at both the undergraduate and postgraduate levels. Data were collected during the first semester of the academic year 2024-2025, a period marked by the increasing use of generative AI tools in education, including GPT-4o, Gemini and QuillBot.

B. Survey Instrument

Two online questionnaires were developed: one for students and one for instructors. They were adapted from a semi-structured interview guide used in a previous study on generative AI tools in higher education [29], specifically focusing on

the use of LLMs to generate feedback on computing assessment tasks. The surveys included multiple-choice and Likert scale items covering LLM usage frequency, satisfaction, and perceived usefulness, along with four open-ended questions asking about experiences of using LLMs to generate feedback. To accommodate participants from both regions, the surveys were available in English and Arabic. Before deployment, a pilot study was conducted with seven participants (students and educators) to ensure clarity and usability.

C. Procedure

Participants were recruited through institutional mailing lists during the Fall 2024 semester and invited to complete an online survey. Participants were informed that participation was entirely voluntary and anonymous, with the option to withdraw at any time without any consequences. Ethical approval was granted by the Ethics Committee at the University of Glasgow and the Deanship of Scientific Research at Imam Abdulrahman Bin Faisal University in Saudi Arabia.

D. Data Analysis

Quantitative data were processed using IBM SPSS Statistics (version 29). Given the exploratory nature of the study, descriptive statistics such as frequencies and percentages were applied to summarize participant responses and identify key patterns. Qualitative data from open-ended questions were analyzed thematically following six-phase approach [36]. This process involved familiarizing with the data, generating initial codes, identifying and defining themes, reviewing and naming themes, and producing a final report. Coding was performed manually by the primary researcher, with independent review by a second researcher to enhance reliability. Disagreements were resolved through discussion.

V. RESULTS

A. Demographic

The study included 174 students and 77 Academics from KSA and the UK (see Tables II and I). In KSA, most of the students were female (78.5%), while in the UK, most were male (69.2%). A small number of UK students (7.7%) preferred not to say their gender. Most students were in advanced years, with juniors and seniors forming the largest groups (40.7% and 23.0% in KSA; 20.5% and 25.6% in the UK). Graduate students made up 8.1% in KSA and 15.4% in the UK.

Among instructors, women were the majority in KSA (78.1%), while men were more common in the UK (61.5%). Most of the instructors in KSA were lecturers (43.8%) or assistant professors (34.4%). In the UK, 38.5% were lecturers and 38.5% were graduate teaching assistants (GTA).

B. Quantitative Results (RQ1)

The findings reveal distinct patterns in the use, adoption, and intention to use LLMs for assessment feedback across students and academics in both the KSA and the UK, with notable

TABLE I
PARTICIPANT CHARACTERISTICS – ACADEMICS

Demographics	KSA (n=64)	UK (n=13)
Gender		
Male	14 (21.9%)	8 (61.5%)
Female	50 (78.1%)	5 (38.5%)
Prefer not to say	0 (0%)	0 (0%)
Teaching Role		
TA/GTA	7 (10.9%)	5 (38.5%)
Lecturer/Instructor	28 (43.8%)	5 (38.5%)
Sr. Lecturer/Asst. Prof	22 (34.4%)	2 (15.4%)
Assoc. Prof/Professor	7 (10.9%)	1 (7.7%)

TABLE II
PARTICIPANT CHARACTERISTICS – STUDENTS

Demographics	KSA (n=135)	UK (n=39)
Gender		
Male	29 (21.5%)	27 (69.2%)
Female	106 (78.5%)	9 (23.1%)
Prefer not to say	0 (0%)	3 (7.7%)
Year of Study		
Freshman	19 (14.1%)	3 (7.7%)
Sophomore	19 (14.1%)	12 (30.8%)
Junior	55 (40.7%)	8 (20.5%)
Senior	31 (23.0%)	10 (25.6%)
Graduate	11 (8.1%)	6 (15.4%)

variations between participant group (students and academics) and regional contexts (UK and KSA).

Usage Frequency

As illustrated in Figure 1, students in both countries reported more frequent use of LLMs than instructors, though usage patterns differed. In KSA, 94.1% of students had used LLMs, and nearly half used them frequently (daily or several times a week). Among instructors, half reported frequent use, while a quarter had never used them. In the UK, student use was also high (89.7%), with just over half using LLMs frequently. However, daily use was less common, and many students reported using it only occasionally. UK instructors showed the lowest overall engagement: while three-quarters had used LLMs, only a small fraction (16.7%) used them daily, and more than half used them rarely or not at all.

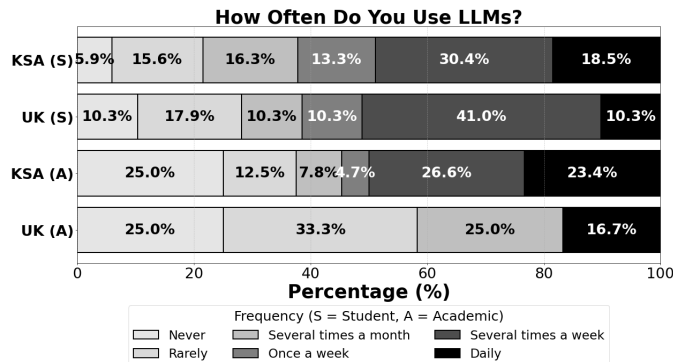


Fig. 1. Frequency of LLM usage by participant group and country

Tool Preference

As shown in Figure 2, ChatGPT was the most commonly used LLM among all participants. In KSA, 52.6% of students and 37.4% of instructors reported using it. In the UK, student usage was similar at 49.2%, while instructor usage was slightly higher at 38.5%. GitHub Copilot was slightly more used by UK students, while Microsoft Copilot was used the most by KSA students. Instructor use of these other tools was low and predominantly reported in KSA. Most instructors reported not using any LLM tool, especially in the UK, where 53.9% had never used one, compared to 26.4% in KSA. Among students, non-use of LLMs was low in both countries (KSA: 5.7%, UK: 10.2%).

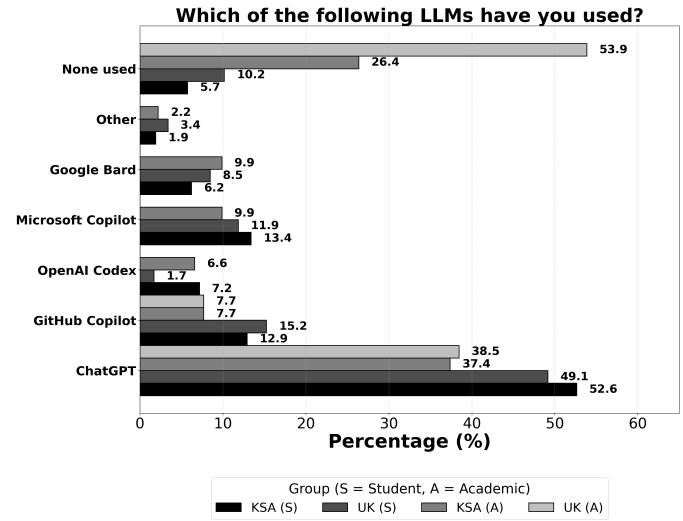


Fig. 2. LLM tool preference by participant group and country

Adoption of LLMs for Feedback Generation

As shown in Figure 3, students in both countries were more likely than instructors to use LLMs to generate feedback on assignments. In KSA, 66.7% of the students reported doing so, compared to 20.3% of the instructors. In the UK, more than half of the students had used LLMs for feedback, while none of the instructors reported any use. In KSA, 79.7% of the instructors had not used LLMs for feedback. Among these non-users, 50.9% were lecturers (26 out of 51), and 39.2% were senior lecturers or assistant professors (20 out of 51). Teaching assistants represented a smaller portion of non-users. On the student side, the adoption of LLM in KSA was highest among third-year students, with 72.7% (40 out of 55) reporting usage. The first and second year students had a combined adoption rate of 52.6%. These patterns highlight differences in adoption rates between roles and academic levels, with higher usage among students and lower usage among instructors, particularly those in mid- to senior-level positions.

Satisfaction levels with LLM-generated feedback

As shown in Figure 4, satisfaction levels varied between groups and countries. In KSA, most of the students reported positive experiences with LLM-generated feedback. A total of 62.1% said that they were very or extremely satisfied

Have you used LLMs to get feedback on your /your students' assignments?

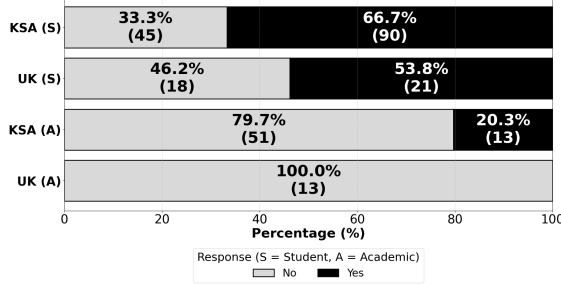


Fig. 3. Adoption of LLMs for feedback generation by participant group and country

and 34.5% were moderately satisfied. Only 3.4% reported low satisfaction. Overall satisfaction among UK students was lower, with 52.7% indicating they were very or extremely satisfied, and 36.8% reporting moderate satisfaction. Their dissatisfaction was higher compared to KSA, with 10.6% reporting low satisfaction.

Among KSA instructors, 75.0% were very or extremely satisfied (41.7% and 33.3%, respectively), and the remaining 25.0% were moderately satisfied. None reported dissatisfaction. In summary, satisfaction with LLM-generated feedback was higher in KSA than in the UK, especially among students and instructors who had used the tools. No satisfaction data was collected from UK instructors, as none reported using LLMs for feedback generation.

How satisfied are you with the feedback generated by LLMs?

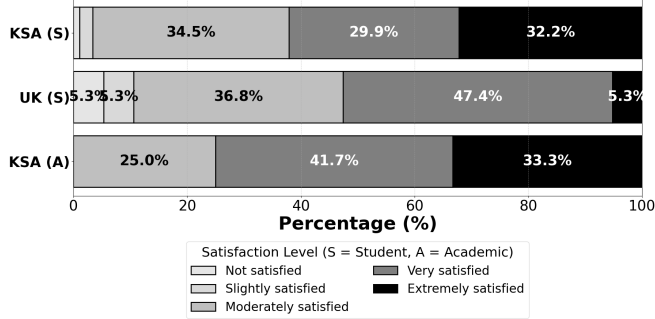


Fig. 4. Satisfaction with LLM-generated feedback by participant group and country

Helpfulness of LLM-generated feedback Participants rated the helpfulness of LLM-generated feedback. As shown in Figure 5, most students in KSA (73.6%) found the feedback very or extremely helpful. Only 1.1% considered it not helpful, while 10.3% rated it as slightly helpful and 14.9% as moderately helpful, indicating a generally high level of perceived usefulness. In contrast, responses from UK students were more varied. While 52.6% found the feedback very or extremely helpful, 36.8% rated it as moderately helpful, and 10.5% as slightly helpful. Notably, none considered it unhelpful. KSA instructors also reported favorable perceptions, with 75% rating the feedback as very or extremely helpful and 25% as moderately helpful. None rated it as unhelpful.

How helpful do you find the feedback generated by LLMs?

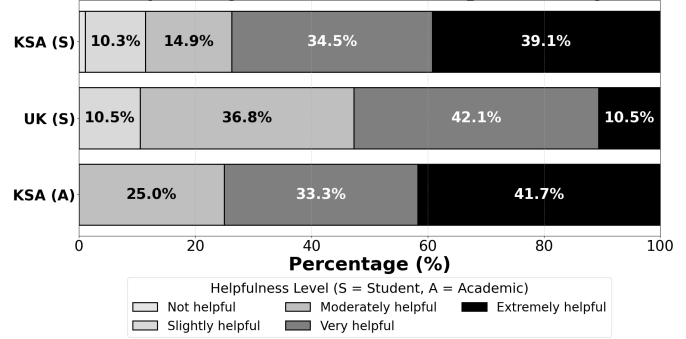


Fig. 5. Helpfulness of LLM-generated feedback by participant group and country

Intentions to Use LLMs for Feedback Generation As shown in Figure 6, intentions to use LLMs for feedback generation varied across groups and countries. In KSA, 64.4% of students expressed interest in future use, compared to only 5.6% in the UK. Among instructors, the pattern was more balanced, with 34.0% in KSA and 33.3% in the UK indicating interest. These results suggest a stronger openness to adopting LLMs among Saudi students in particular, whereas UK students appeared more resistant, with 94.4% reporting no intention to use such tools.

Do you plan to use LLMs to generate feedback on your /your students' assignments?

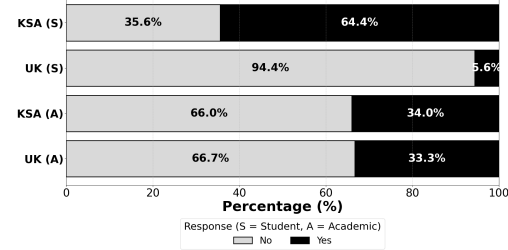


Fig. 6. Intentions to use LLMs for feedback generation by participant group and country

C. Qualitative Results (RQ2)

A thematic analysis of participants' qualitative responses revealed three overarching themes regarding LLM-generated feedback in computer science education across the UK and KSA: (a) "It saves effort", (b) "It is not as smart and genuine as a human", and (c) "Making it more like humans and less like a machine". These themes offer insight into how students and educators in both regions assess the role of LLMs in the feedback process, the concerns they have about its limitations, and the ways they envision enhancing its value in teaching and learning within computer science education.

1) "It Saves Effort": This theme reflects the perceived practical value of LLM-generated feedback, particularly its time-saving and academic support functions. It was the most prominent theme across both KSA and UK participants, who emphasized its usefulness, clarity, and efficiency. Several sub-themes highlighted specific aspects of this value.

a) *Efficiency and Speed*: One of the most frequently mentioned benefits was the ability of LLMs to provide instant feedback and save time. This feature was appreciated by students dealing with tight deadlines, as well as educators who manage heavy workloads. These responses illustrate a shared perception between regions and roles about the time-saving and efficiency-enhancing potential of LLM-generated feedback.

For example, a KSA teacher noted, *"Save time, can describe what I want in short and clear words quickly, save my effort for doing hard job."* Another KSA teacher emphasized the tone and delivery of the feedback as a key reason for using LLMs: *"It formulates and delivers feedback gently and focuses on the positive aspects, as some people do not accept criticism."* Similarly, a UK teacher shared, *"it saves my time and allows me to give detailed feedback quickly that helps students understand and fix their mistakes, improving their skills. It also frees up my time to focus on other important tasks in the course."*

The students also emphasized this benefit. A UK student commented that *"LLMs can act as instant tutors and often explain the whys of particular reasonings and conclusions."* and a student at KSA highlighted *"Getting sources faster by shortening the time spent searching online."*

b) *Detail and Clarity*: The participants reflect a common appreciation for the way LLM can enhance the depth and clarity of educational feedback. This was especially appreciated by students who found that the feedback from lecturers or teaching assistants was often too brief. As one UK student commented, *"They are more detailed than some of my lecturers or teaching assistants."* Educators also recognized this benefit in enhancing the quality of their feedback. A UK teacher stated *"I think I will use it to make my feedback more clear and without grammar mistakes."* A KSA teacher agreed with this view, saying, *"To analyze answers more thoroughly and get suggestions that could help develop the student."* Students in KSA similarly valued the level of detail and clarity provided by LLMs. One student noted, *"Due to the fact that it provides as many details as the person needs, it also provides images and examples to demonstrate, and very clear."*

c) *Error Spotting*: Student participants highlighted the role of LLMs in detecting errors, particularly in programming assignments. This feature was viewed as useful for self-assessment and revision. A KSA student explained, *"It helps me identify mistakes very quickly, for example, when I write code and there is an error, it helps me locate the mistake."* Similarly, a UK student noted, *"Can sometimes spot mistakes."*

d) *Improved Understanding*: LLMs were also perceived as effective tools to improve student understanding. Participants noted that these tools not only highlighted mistakes but also explained underlying concepts, helping learners identify areas for improvement. A UK student remarked, *"I learn much better when it explains it to me and tells me where I went wrong and how to improve."* A KSA student shared a similar view: *"To get an initial idea of the kind of feedback I might*

receive from the course instructor, and to see if there are things I can improve before submitting." Additionally, a KSA teacher emphasized the learning potential of LLMs, stating, *"I use it myself to look for suggestions to improve my work. I found it very helpful, so I plan to use it for the benefit of students."*

e) *Support for Coding and Projects*: Finally, LLMs were frequently described not only as a feedback tool but also as a supportive tool to shape and refine academic work. The students used them to brainstorm ideas and organize their thoughts, especially when struggling to express complex concepts. As a UK student shared, *"I use it to help me brainstorm ideas like discussing with another person instead of seeing it as a tool..."* A KSA student highlighted a similar benefit: *"To improve the quality of my work and identify missing parts of the project."* Educators also acknowledged this role. A KSA teacher stated, *"Helps me make necessary improvements, whether in student projects, research, or exams."*

2) ***"It is not as smart and genuine as a human"***: This theme reflects the concerns and perceived limitations of using LLMs for feedback. Although participants acknowledged the potential value of the system, they also raised concerns related to precision, depth, ethical use, and the human-like quality of the responses. Both students and educators expressed caution about relying too heavily on such tools.

a) *Accuracy and Clarity*: A shared concern in both regions was the reliability and clarity of the LLM-generated feedback, particularly in academic settings where precision is critical. This concern was especially emphasized by UK students who reported that responses were sometimes vague, misleading, or overly confident. One UK student noted, *"It doesn't always know the answer and can take you in the wrong direction."* Another added, *"The LLM could mark something as incorrect, but would not be able to report to the student where the flaw in their understanding is."* A KSA student also highlighted clarity issues, stating that *"uses complex and unusual words, and its style is unique."* Teachers expressed similar concerns. A KSA teacher remarked that *"It does not always provide correct answers."* A UK teacher commented on the use of LLM in research contexts, saying, *"Work with them for research – they are unreliable."*

b) *Prompting Difficulty*: Several participants reported that generating useful and accurate feedback from LLMs required highly specific and detailed instructions, which created a barrier to effective use. A student in the UK explained: *"I need to carefully craft the prompts because if you did not provide enough detail, often the answers are plain incorrect."* Another added, *"sometimes it takes too long to give it all the context."* Participants in KSA reported similar difficulties. A teacher shared, *"Challenge in trying to explain to the AI and help it understand my intent."* A student noted, *"It not comprehending the content I shared with it to make it give me complete and correct feedback."*

c) *Cost*: The financial barrier to accessing premium versions of LLM was another concern, especially for students. A UK participant stated, *"Paying for it."* Similarly, a KSA student similarly noted, *"Costly."*

d) *Institutional and Ethical Concerns*: The participants' responses about using LLMs for student work reflect a variety of ethical and institutional concerns, including academic integrity, data privacy, and discomfort with the broader AI industry. As one UK teacher explained, *"For ethical reasons I would not enter a student's submitted work in an LLM tool unless I know for sure that work won't be used by that LLM for training."* Another added, *"Auto-generating feedback seems like a betrayal of my duty as an educator."* Similarly, a KSA teacher commented, *"The feedback is about humans not machines, so I prefer to give feedback myself rather than get help from LLM."* Some students also expressed skepticism regarding the legitimacy or ethics of AI feedback. A UK student remarked, *"Morally opposed to LLM."* Another explained: *"I find the AI (LLM) industry icky and I don't want to use their tools, companies openly steal and take traffic on many websites to download their data without consent, and I can't imagine myself ever being comfortable supporting."* Similarly, a student of KSA stated *"I wish to use it, but my university considers it cheating."*

3) *Making It More Like Humans and Less Like a Machine*: A significant number of participants proposed improving LLMs to be more human-like, particularly in how they interpret, evaluate, and deliver feedback. These suggestions were often linked to dissatisfaction with responses that felt robotic or generic. A KSA student said, *"Feedback should be based on credible academic sources."* Another emphasized, *"Its responses should be reviewed by experts."* Similar concerns were raised by UK students, who called for greater reliability and logical accuracy: *"More factually correct feedback."* In addition, many participants believed that LLMs could be more effective if tailored to individual academic contexts, disciplines, or student needs. This included suggestions for personalized feedback, course-specific training, and better alignment with academic expectations. A UK student suggested, *"I think that LLMs should be trained in the specifics of a course."* Another noted, *"Have access to specialized LLMs that are trained for more specific purposes."* Similar requests were also raised in the KSA context. A student commented on *@Add customization options,* while a teacher proposed *"Adding more tools and more precise software to the AI."* The participants also shared practical recommendations for improving the overall user experience. For example, a KSA student stated *"Based on my experience LLM, tools, never give other tool name the machine just write theoretical book style answer without providing any name."*

VI. DISCUSSION

This study explored the use and perceptions of LLM-generated feedback in computer science education in two different academic and cultural contexts: the UK and KSA. The following discussion interprets the key findings and considers how cultural and institutional factors may have influenced the results.

The findings show that students in both regions reported more active use of LLMs than instructors. This may be due

to the regular use of AI-based generative tools by students for coding and problem solving, as also highlighted in [29]. This may also reflect the students' perceptions of these tools as effective and easy to use. As noted in [33], the acceptance of LLMs such as ChatGPT is strongly influenced by perceived efficiency and usability, consistent with the Technology Acceptance Model (TAM). In addition, students in technical fields often demonstrate a greater openness to digital tools [37], which may explain their willingness to engage with LLM as a reliable source of academic support. In contrast, instructors reported lower overall usage, although those in KSA showed greater engagement than their UK counterparts. This difference may be partially shaped by national initiatives such as Vision 2030, which actively promote educational innovation and the adoption of emerging technologies in Saudi Arabia [38].

ChatGPT was identified as the most frequently used LLM, largely due to its effectiveness in supporting coding tasks. Its capabilities, including code generation, natural language interaction with databases, and integration with programming tools such as Python, make it a practical and accessible resource for both students and instructors [5].

Students in both regions reported actively using LLMs to obtain feedback on their assignments. The open responses described that the students used these tools to check their code, identify errors, and improve their work prior to submission. This behavior may reflect their perception of LLMs as practical and accessible tools that support independent learning, particularly in programming contexts [29]. Previous research also highlights the ability of LLMs to provide immediate and targeted feedback on syntax and logic issues, making them especially valuable to CS students during task completion [16], [19], [20]. By contrast, instructors demonstrated a more cautious stance toward using LLMs for feedback. In the UK, no one reported using such tools, reflecting ongoing concerns about academic integrity and plagiarism [31]. In [29], instructors further expressed concerns that students might not fully grasp key concepts or skills if they relied too heavily on LLMs, and they also highlighted the lack of clear institutional policies as a major source of uncertainty regarding how to integrate these tools into their teaching. However, in Saudi Arabia, limited adoption among instructors may be related to growing national support. The Ministry of Education has issued official guidelines encouraging the regulated use of generative AI in education, including its application for feedback provision [39]. Another key finding relates to how participants evaluated the usefulness and helpfulness of LLM-generated feedback. UK instructors were excluded from this analysis, as none reported using LLMs for CS feedback. In KSA, overall satisfaction and perceived helpfulness were high, possibly influenced by greater familiarity with LLMs, openness to educational technology, and national support, such as the role of the Saudi Data and Artificial Intelligence Authority (SDAIA) [40]. These efforts, along with the Ministry of Education guidelines that promote responsible AI use, can enhance confidence and engagement [39]. In contrast, lower

satisfaction with LLM-generated feedback among UK students may be explained by limited experience with LLMs and uncertainty about how to use them effectively. Previous studies show that students with less familiarity tend to be more passive and less satisfied [28], and that satisfaction is influenced by ease of use and perceived efficiency [33]. Ethical concerns may also contribute to more cautious adoption in the UK [27].

The findings revealed a generally low intention to use LLM-based tools for academic assignments, particularly among students in the UK. This hesitation offers further insight into the institutional and cultural dynamics that may shape the attitudes of participants towards generative AI. Previous studies suggest that students in the UK often report uncertainty about what constitutes acceptable or ethical use of generative AI tools, such as ChatGPT [27], [28]. This uncertainty may be reinforced by the lack of clear institutional policies, training, or academic guidance, leaving students hesitant to apply such tools in formal coursework [28], [31]. As a result, even students who recognize the usefulness of LLM-generated feedback may avoid using it in graded tasks to prevent potential breaches of academic integrity or failure to comply with unclear institutional guidelines. This cautious approach contrasts with the greater willingness observed in contexts like KSA, where institutional openness and policy support have created a more encouraging environment for responsible and guided use of generative AI [38]–[40].

Thematic analysis of open-ended participants' responses revealed nuanced perspectives on LLM-generated feedback in CS education. The most prominent theme, *It saves effort*, reflects a widely shared appreciation of the speed, clarity, and usefulness of such tools, particularly in programming contexts. These findings align with previous research that highlights the ability of LLMs to improve efficiency, support creativity, and enhance student engagement through timely and detailed feedback [16], [19], [20], [23], [26], [33]. At the same time, participants raised concerns under the theme *It is not as smart and genuine as a human*, including issues related to reliability, vagueness, and ethical ambiguity. These concerns echo findings from earlier studies on the risks of academic misuse, the absence of human judgment, and the need for clearer institutional policies to guide responsible adoption [24], [24], [26]–[28], [31]. Finally, the theme *Making it more like humans and less like a machine* underscores the desire of participants for more personalized, context-sensitive and academically aligned feedback. This is consistent with the recommendations in the literature to improve accuracy, personalization, and consistency in LLM-driven feedback systems [16], [19], [20], [31].

VII. PEDAGOGICAL IMPLICATIONS AND CONCLUSION

This study investigated the usage, adoption, and perceptions of LLM-generated feedback among CS students and academics in the UK and KSA. The findings highlight the importance of institutional support, including clear ethical guidelines and training to ensure the effective and responsible use of these tools [28], [31]. Training should include practical

classroom examples to help users critically evaluate and reflect on LLM-generated feedback.

LLMs should be positioned as supportive learning tools rather than evaluative ones. To achieve this, educators must carefully craft prompts that guide the LLM to provide scaffolded feedback (e.g., step-by-step explanations) [41] and include gamified or interactive elements to improve engagement and reduce anxiety [14]. Prompts should also foster students' critical thinking and self-reflection, encouraging them to revise their work and engage more deeply with feedback [23]. However, if poorly constructed, such prompts may reduce student autonomy or hinder higher-order thinking [42]. Our recent mapping study [43] found limited pedagogical grounding in prompt design for LLM-generated feedback. It also revealed diverse and evolving usage patterns, often without consistent theoretical foundations. Furthermore, institutions should establish clear ethical guidelines to prevent misuse and ensure that AI-generated feedback aligns with academic integrity standards [24], [27].

Despite these implications, this study has limitations. Being exploratory and discipline-specific (Computer Science), results might not generalize. Additionally, voluntary participation could introduce bias. The findings reflect usage patterns specific to the data collection period (August–December 2024), and newer AI models may shift user behaviors.

Future research should explore how advancements in generative AI (e.g., GPT-4.5, DeepSeek) influence user perceptions, adoption patterns, and educational outcomes. Longitudinal or comparative studies across different disciplines are recommended.

REFERENCES

- [1] B. Min, H. Ross, E. Sulem, A. P. B. Veyseh, T. H. Nguyen, O. Sainz, E. Agirre, I. Heintz, and D. Roth, "Recent advances in natural language processing via large pre-trained language models: A survey," *ACM Comput. Surv.*, vol. 56, no. 2, Sep. 2023. [Online]. Available: <https://doi.org/10.1145/3605943>
- [2] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell *et al.*, "Language models are few-shot learners," *Advances in neural information processing systems*, vol. 33, pp. 1877–1901, 2020.
- [3] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, 2019, pp. 4171–4186.
- [4] D. M. Ziegler, N. Stiennon, J. Wu, T. B. Brown, A. Radford, D. Amodei, P. Christiano, and G. Irving, "Fine-tuning language models from human preferences," *arXiv preprint arXiv:1909.08593*, 2019.
- [5] I. Trummer, "Large language models: Principles and practice," in *2024 IEEE 40th International Conference on Data Engineering (ICDE)*, 2024, pp. 5354–5357.
- [6] S. Wang, T. Xu, H. Li, C. Zhang, J. Liang, J. Tang, P. S. Yu, and Q. Wen, "Large language models for education: A survey and outlook," *arXiv preprint arXiv:2403.18105*, 2024.
- [7] D. Boud and E. Molloy, "Rethinking models of feedback for learning: the challenge of design," *Assessment & Evaluation in higher education*, vol. 38, no. 6, pp. 698–712, 2013.
- [8] J. Hattie and H. Timperley, "The power of feedback," *Review of Educational Research*, vol. 77, no. 1, pp. 81–112, 2007.
- [9] D. Lohr, N. Kiesler, H. Keuning, and J. Jeuring, "'let them try to figure it out first' - reasons why experts (do not) provide feedback to novice programmers," in *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1*, ser. ITiCSE 2024.

- New York, NY, USA: Association for Computing Machinery, 2024, p. 38–44. [Online]. Available: <https://doi.org/10.1145/3649217.3653530>
- [10] C. Ott, A. Robins, and K. Shephard, “Translating principles of effective feedback for students into the cs1 context,” *ACM Transactions on Computing Education (TOCE)*, vol. 16, no. 1, pp. 1–27, 2016.
 - [11] D. P. Fisher, G. Brotto, I. Lim, and C. Southam, “The impact of timely formative feedback on university student motivation,” *Assessment & Evaluation in Higher Education*, pp. 1–10, 2025.
 - [12] E. Mulliner and M. Tucker, “Feedback on feedback practice: perceptions of students and academics,” *Assessment & Evaluation in Higher Education*, vol. 42, no. 2, pp. 266–288, 2017.
 - [13] A. Hellas, J. Leinonen, S. Sarsa, C. Koutcheme, L. Kujanpää, and J. Sorva, “Exploring the responses of large language models to beginner programmers’ help requests,” in *Proceedings of the 2023 ACM Conference on International Computing Education Research - Volume 1*, ser. ICER ’23. New York, NY, USA: Association for Computing Machinery, 2023, p. 93–105. [Online]. Available: <https://doi.org/10.1145/3568813.3600139>
 - [14] H. Nguyen and V. Allan, “Using gpt-4 to provide tiered, formative code feedback,” in *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1*, ser. SIGCSE 2024. New York, NY, USA: Association for Computing Machinery, 2024, p. 958–964. [Online]. Available: <https://doi.org/10.1145/3626252.3630960>
 - [15] L. Roest, H. Keuning, and J. Jeuring, “Next-step hint generation for introductory programming using large language models,” in *Proceedings of the 26th Australasian Computing Education Conference*, ser. ACE ’24. New York, NY, USA: Association for Computing Machinery, 2024, p. 144–153. [Online]. Available: <https://doi.org/10.1145/3636243.3636259>
 - [16] J. Woodrow, A. Malik, and C. Piech, “Ai teaches the art of elegant coding: Timely, fair, and helpful style feedback in a global course,” in *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1*, ser. SIGCSE 2024. New York, NY, USA: Association for Computing Machinery, 2024, p. 1442–1448. [Online]. Available: <https://doi.org/10.1145/3626252.3630773>
 - [17] C. Koutcheme, N. Dainese, S. Sarsa, A. Hellas, J. Leinonen, and P. Denny, “Open source language models can provide feedback: Evaluating llms’ ability to help students using gpt-4-as-a-judge,” in *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1*, ser. ITICSE 2024. New York, NY, USA: Association for Computing Machinery, 2024, p. 52–58. [Online]. Available: <https://doi.org/10.1145/3649217.3653612>
 - [18] I. Azaiz, N. Kiesler, and S. Strickroth, “Feedback-generation for programming exercises with gpt-4,” in *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1*, ser. ITICSE 2024. New York, NY, USA: Association for Computing Machinery, 2024, p. 31–37. [Online]. Available: <https://doi.org/10.1145/3649217.3653594>
 - [19] M. Pankiewicz and R. S. Baker, “Navigating compiler errors with ai assistance - a study of gpt hints in an introductory programming course,” in *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1*, ser. ITICSE 2024. New York, NY, USA: Association for Computing Machinery, 2024, p. 94–100. [Online]. Available: <https://doi.org/10.1145/3649217.3653608>
 - [20] Z. Zhang, Z. Dong, Y. Shi, T. Price, N. Matsuda, and D. Xu, “Students’ perceptions and preferences of generative artificial intelligence feedback for programming,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 21, 2024, pp. 23 250–23 258.
 - [21] T. Phung, J. Cambronero, S. Gulwani, T. Kohn, R. Majumdar, A. Singla, and G. Soares, “Generating high-precision feedback for programming syntax errors using large language models,” *arXiv preprint arXiv:2302.04662*, 2023.
 - [22] E. Smolić, M. Pavelić, B. Boras, I. Mekterović, and T. Jaguš, “Llm generative ai and students’ exam code evaluation: Qualitative and quantitative analysis,” in *2024 47th MIPRO ICT and Electronics Convention (MIPRO)*. IEEE, 2024, pp. 1261–1266.
 - [23] A. Almassaad, H. Alajlan, and R. Alebaikan, “Student perceptions of generative artificial intelligence: Investigating utilization, benefits, and challenges in higher education,” *Systems*, vol. 12, no. 10, p. 385, 2024.
 - [24] M. K. Alharbi, W. Syed, A. Innab, M. Basil A. Al-Rawi, A. Alsadoun, and A. Bashatah, “Healthcare students attitudes opinions perceptions and perceived obstacles regarding chatgpt in saudi arabia: a survey-based cross-sectional study,” *Scientific Reports*, vol. 14, no. 1, p. 22800, 2024.
 - [25] S. M. Alharthi, “Beyond traditional language learning: Efl student views on chatgpt in saudi arabia,” 2024.
 - [26] N. Alqahtani, “Benefits, challenges, and attitudes toward chatgpt in english writing courses at saudi universities,” *International Journal of Language and Literary Studies*, vol. 6, no. 2, pp. 396–413, 2024.
 - [27] M. Bikanga Ada, “It helps with crap lecturers and their low effort: Investigating computer science students’ perceptions of using chatgpt for learning,” *Education Sciences*, vol. 14, no. 10, p. 1106, 2024.
 - [28] R. English, R. Nash, and H. Mackenzie, “‘a rather stupid but always available brainstorming partner’: Use and understanding of generative ai by uk postgraduate researchers,” *Innovations in Education and Teaching International*, pp. 1–15, 2025.
 - [29] C. Zastudil, M. Rogalska, C. Kapp, J. Vaughn, and S. MacNeil, “Generative ai in computing education: Perspectives of students and instructors.(2023),” *arXiv preprint arXiv:2308.04309*, 2023.
 - [30] R. F. Kizilcec, E. Huber, E. C. Papanastasiou, A. Cram, C. A. Makridis, A. Smolansky, S. Zeivots, and C. Radulescu, “Perceived impact of generative ai on assessments: Comparing educator and student perspectives in australia, cyprus, and the united states,” *Computers and Education: Artificial Intelligence*, vol. 7, p. 100269, 2024.
 - [31] M. Bower, J. Torrington, J. W. Lai, P. Petocz, and M. Alfano, “How should we change teaching and assessment in response to increasingly powerful generative artificial intelligence? outcomes of the chatgpt teacher survey,” *Education and Information Technologies*, vol. 29, no. 12, pp. 15 403–15 439, 2024.
 - [32] Z. N. Khlaif, A. Ayyoub, B. Hamamra, E. Bensalem, M. A. Mitwally, A. Ayyoub, M. K. Hattab, and F. Shadid, “University teachers’ views on the adoption and integration of generative ai tools for student assessment in higher education,” *Education Sciences*, vol. 14, no. 10, p. 1090, 2024.
 - [33] C. Yu, J. Yan, and N. Cai, “Chatgpt in higher education: factors influencing chatgpt user satisfaction and continued use intention,” in *Frontiers in Education*, vol. 9. Frontiers Media SA, 2024, p. 1354929.
 - [34] J. W. Creswell and V. L. Plano Clark, *Designing and Conducting Mixed Methods Research*, 3rd ed. Thousand Oaks, CA: SAGE Publications, 2017.
 - [35] V. L. P. Clark and N. V. Ivankova, *Mixed methods research: A guide to the field*. Sage publications, 2015, vol. 3.
 - [36] V. Braun and V. Clarke, “Using thematic analysis in psychology,” *Qualitative research in psychology*, vol. 3, no. 2, pp. 77–101, 2006.
 - [37] D. Popa and I. R. Topală, “Students’ digital competencies, related attitudes and self directed learning,” *eLearning & Software for Education*, vol. 3, 2018.
 - [38] “Saudi vision 2030,” Kingdom of Saudi Arabia, 2016, accessed: 2025-04-02. [Online]. Available: <https://www.vision2030.gov.sa/en>
 - [39] Ministry of Education, Saudi Arabia, “A guideline for the use of generative artificial intelligence in public education in saudi arabia,” 2025, accessed: 29 March 2025. [Online]. Available: https://www.moe.gov.sa/ar/mediacenter/MOEnews/Pages/news1_24012025.aspx
 - [40] Saudi Data and Artificial Intelligence Authority (SDAIA), “Saudi data and artificial intelligence authority (sdaia),” 2024, accessed: April 2, 2025. [Online]. Available: <https://sdaia.gov.sa/en/default.aspx>
 - [41] D. Wood, J. S. Bruner, and G. Ross, “The role of tutoring in problem solving,” *Journal of child psychology and psychiatry*, vol. 17, no. 2, pp. 89–100, 1976.
 - [42] L. Xia, K. Shen, H. Sun, X. An, and Y. Dong, “Developing and validating the student learning agency scale in generative artificial intelligence (ai)-supported contexts,” *Education and Information Technologies*, pp. 1–23, 2025.
 - [43] E. Alyahyan, M. Bikanga Ada, and J. Lever, “A systematic mapping of large language models as feedback providers in higher education,” in *Proceedings of the 25th IEEE International Conference on Advanced Learning Technologies (ICALT 2025)*. Changhua, Taiwan: IEEE, July 14–17 2025, accepted for publication.

Alyahyan, Eyman, Bikanga Ada, Mireilla and Lever, Jake (2025) Adoption, Usage, and Perceptions of LLM-Generated Feedback in Computer Science Education: A Cross-Cultural Study of Students and Academics. In: 2025 Frontiers in Education Conference (FIE 2025), Nashville, Tennessee, 02-05 November 2025, ISBN 9798331501051 (doi: [10.1109/FIE63693.2025.11328530](https://doi.org/10.1109/FIE63693.2025.11328530))

[The University of Glasgow has an agreement with IEEE](#) which allows all UofG authors to self-archive accepted manuscripts submitted to any of the subscription-based (hybrid) IEEE journals, magazines, or conference proceedings. Authors can immediately self-archive accepted manuscripts in an institutional or subject based repository with a self-attributed CC BY licence. The agreement covers all original research and review articles.

This is the author accepted manuscript. There may be differences between this version and the published version. You are advised to consult the published version if you wish to cite from it: <https://doi.org/10.1109/FIE63693.2025.11328530>

© IEEE. Reproduced under a [Creative Commons Attribution 4.0 International Licence](#).



Enlighten record available at: <https://eprints.gla.ac.uk/357210/>

Deposited on 9 June 2025

<https://eprints.gla.ac.uk>